7.2E: Exercises for Section 7.2

Fill in the blank to make a true statement.

1) $\sin^2 x + _____ = 1$

Answer

$\cos^2 x$

2) $\sec^2 x - 1 = _____$

Answer

$\tan^2 x$

Use an identity to reduce the power of the trigonometric function to a trigonometric function raised to the first power.

3) $\sin^2 x = _____$

Answer

$\dfrac{1-\cos(2x)}{2}$

4) $\cos^2 x = _____$

Answer

$\dfrac{1+\cos(2x)}{2}$

Evaluate each of the following integrals by u-substitution.

5) $\int \sin^3 x \cos x \, dx$

Answer

$$\int \sin^3 x \cos x \, dx = \frac{\sin^4 x}{4} + C$$

6) $\int \sqrt{\cos x} \sin x \, dx$

7) $\int \tan^5(2x) \sec^2(2x) \, dx$

Answer

$$\int \tan^5(2x) \sec^2(2x) \, dx = \tfrac{1}{12}\tan^6(2x) + C$$

8) $\int \sin^7(2x) \cos(2x) \, dx$

9) $\int \tan(\frac{x}{2}) \sec^2(\frac{x}{2}) \, dx$

Answer

$$\int \tan(\frac{x}{2}) \sec^2(\frac{x}{2}) \, dx = \tan^2(\frac{x}{2}) + C$$

10) $\int \tan^2 x \sec^2 x \, dx$

Compute the following integrals using the guidelines for integrating powers of trigonometric functions. Use a CAS to check the solutions. (*Note*: Some of the problems may be done using techniques of integration learned previously.)

11) $\int \sin^3 x \, dx$

Answer

$$\int \sin^3 x\,dx \quad = -\cos x + \frac{\cos^3 x}{3} + C$$

12) $\int \cos^3 x\,dx$

13) $\int \sin x \cos x\,dx$

Answer

$$\int \sin x \cos x\,dx \quad = \quad -\tfrac{1}{2}\cos^2 x + C$$

14) $\int \cos^5 x\,dx$

15) $\int \sin^5 x \cos^2 x\,dx$

Answer

$$\int \sin^5 x \cos^2 x\,dx \quad = \quad -\frac{5\cos x}{64} - \tfrac{1}{192}\cos(3x) + \tfrac{3}{320}\cos(5x) - \tfrac{1}{448}\cos(7x) + C$$

16) $\int \sin^3 x \cos^3 x\,dx$

17) $\int \sqrt{\sin x}\cos x\,dx$

Answer

$$\int \sqrt{\sin x}\cos x\,dx \quad = \quad \tfrac{2}{3}(\sin x)^{3/2} + C$$

18) $\int \sqrt{\sin x}\cos^3 x\,dx$

19) $\int \sec x \tan x\,dx$

Answer

$$\int \sec x \tan x\,dx \quad = \quad \sec x + C$$

20) $\int \tan(5x)\,dx$

21) $\int \tan^2 x \sec x\,dx$

Answer

$$\int \tan^2 x \sec x\,dx \quad = \quad \tfrac{1}{2}\sec x \tan x - \tfrac{1}{2}\ln(\sec x + \tan x) + C$$

22) $\int \tan x \sec^3 x\,dx$

23) $\int \sec^4 x\,dx$

Answer

$$\int \sec^4 x\,dx \quad = \quad \frac{2\tan x}{3} + \tfrac{1}{3}\sec^2 x \tan x = \tan x + \frac{\tan^3 x}{3} + C$$

24) $\int \cot x\,dx$

25) $\int \csc x\,dx$

Answer

$$\int \csc x\,dx \quad = \quad -\ln|\cot x + \csc x| + C$$

26) $\int \frac{\tan^3 x}{\sqrt{\sec x}}\,dx$

For exercises 27 - 28, find a general formula for the integrals.

27) $\int \sin^2 ax \cos ax\,dx$

Answer

$$\int \sin^2 ax \cos ax\,dx \quad = \quad \frac{\sin^3(ax)}{3a} + C$$

28) $\int \sin ax \cos ax\,dx.$

Use the double-angle formulas to evaluate the integrals in exercises 29 - 34.

29) $\int_0^{\pi} \sin^2 x\,dx$

Answer

$$\int_0^{\pi} \sin^2 x\,dx \quad = \quad \frac{\pi}{2}$$

30) $\int_0^{\pi} \sin^4 x\,dx$

31) $\int \cos^2 3x\,dx$

Answer

$$\int \cos^2 3x\,dx \quad = \quad \frac{x}{2} + \tfrac{1}{12}\sin(6x) + C$$

32) $\int \sin^2 x \cos^2 x\,dx$

33) $\int \sin^2 x\,dx + \int \cos^2 x\,dx$

Answer

$$\int \sin^2 x\,dx + \int \cos^2 x\,dx \quad = \quad x + C$$

34) $\int \sin^2 x \cos^2(2x)\,dx$

For exercises 35 - 43, evaluate the definite integrals. Express answers in exact form whenever possible.

35) $\int_0^{2\pi} \cos x \sin 2x\,dx$

Answer

$$\int_0^{2\pi} \cos x \sin 2x\,dx \quad = \quad 0$$

36) $\int_0^{\pi} \sin 3x \sin 5x\,dx$

37) $\int_0^{\pi} \cos(99x)\sin(101x)\,dx$

Answer

$$\int_0^{\pi} \cos(99x)\sin(101x)\,dx \quad = \quad 0$$

38) $\int_{-\pi}^{\pi} \cos^2(3x)\,dx$

39) $\int_{0}^{2\pi} \sin x \sin(2x) \sin(3x)\,dx$

Answer

$$\int_{0}^{2\pi} \sin x \sin(2x) \sin(3x)\,dx \quad = \quad 0$$

40) $\int_{0}^{4\pi} \cos(x/2) \sin(x/2)\,dx$

41) $\int_{\pi/6}^{\pi/3} \frac{\cos^3 x}{\sqrt{\sin x}}\,dx$ (Round this answer to three decimal places.)

Answer

$$\int_{\pi/6}^{\pi/3} \frac{\cos^3 x}{\sqrt{\sin x}}\,dx \quad \approx \quad 0.239$$

42) $\int_{-\pi/3}^{\pi/3} \sqrt{\sec^2 x - 1}\,dx$

43) $\int_{0}^{\pi/2} \sqrt{1 - \cos(2x)}\,dx$

Answer

$$\int_{0}^{\pi/2} \sqrt{1 - \cos(2x)}\,dx \quad = \quad \sqrt{2}$$

44) Find the area of the region bounded by the graphs of the equations $y = \sin x,\ y = \sin^3 x,\ x = 0$, and $x = \frac{\pi}{2}$.

45) Find the area of the region bounded by the graphs of the equations $y = \cos^2 x,\ y = \sin^2 x,\ x = -\frac{\pi}{4}$, and $x = \frac{\pi}{4}$.

Answer

$A = 1 \text{ unit}^2$

46) A particle moves in a straight line with the velocity function $v(t) = \sin(\omega t) \cos^2(\omega t)$. Find its position function $x = f(t)$ if $f(0) = 0$.

47) Find the average value of the function $f(x) = \sin^2 x \cos^3 x$ over the interval $[-\pi, \pi]$.

Answer

0

For exercises 48 - 49, solve the differential equations.

48) $\frac{dy}{dx} = \sin^2 x$. The curve passes through point $(0, 0)$.

49) $\frac{dy}{d\theta} = \sin^4(\pi\theta)$

Answer

$$f(x) = \frac{3\theta}{8} - \frac{1}{4\pi}\sin(2\pi\theta) + \frac{1}{32\pi}\sin(4\pi\theta) + C$$

50) Find the length of the curve $y = \ln(\csc x),\ \text{for } \frac{\pi}{4} \le x \le \frac{\pi}{2}$.

51) Find the length of the curve $y = \ln(\sin x),\ \text{for } \frac{\pi}{3} \le x \le \frac{\pi}{2}$.

Answer

$s = \ln(\sqrt{3})$

52) Find the volume generated by revolving the curve $y = \cos(3x)$ about the x-axis, for $0 \le x \le \frac{\pi}{36}$.

For exercises 53 - 54, use this information: The inner product of two functions f and g over $[a, b]$ is defined by $f(x) \cdot g(x) = \langle f, g \rangle = \int_a^b f \cdot g \, dx$. Two distinct functions f and g are said to be orthogonal if $\langle f, g \rangle = 0$.

53) Show that $\sin(2x)$, $\cos(3x)$ are orthogonal over the interval $[-\pi, \pi]$.

Answer

$$\int_{-\pi}^{\pi} \sin(2x)\cos(3x)\,dx = 0$$

54) Evaluate $\int_{-\pi}^{\pi} \sin(mx)\cos(nx)\,dx$.

55) Integrate $y' = \sqrt{\tan x}\sec^4 x$.

Answer

$$y = \int \sqrt{\tan x}\sec^4 x\,dx \quad = \quad \tfrac{2}{3}(\tan x)^{3/2} + \tfrac{2}{7}(\tan x)^{7/2} + C = \tfrac{2}{21}(\tan x)^{3/2}\left[7 + 3\tan^2 x\right] + C$$

For each pair of integrals in exercises 56 - 57, determine which one is more difficult to evaluate. Explain your reasoning.

56) $\int \sin^{456} x \cos x\,dx$ or $\int \sin^2 x \cos^2 x\,dx$

57) $\int \tan^{350} x \sec^2 x\,dx$ or $\int \tan^{350} x \sec x\,dx$

Answer

The second integral is more difficult because the first integral is simply a u-substitution type.

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7.3: Trigonometric Substitution

Learning Objectives

- Solve integration problems involving the square root of a sum or difference of two squares.

In this section, we explore integrals containing expressions of the form $\sqrt{a^2-x^2}$, $\sqrt{a^2+x^2}$, and $\sqrt{x^2-a^2}$, where the values of a are positive. We have already encountered and evaluated integrals containing some expressions of this type, but many still remain inaccessible. The technique of *trigonometric substitution* comes in very handy when evaluating these integrals. This technique uses substitution to rewrite these integrals as trigonometric integrals.

Integrals Involving $\sqrt{a^2-x^2}$

Before developing a general strategy for integrals containing $\sqrt{a^2-x^2}$, consider the integral $\int \sqrt{9-x^2}dx$. This integral cannot be evaluated using any of the techniques we have discussed so far. However, if we make the substitution $x=3\sin\theta$, we have $dx=3\cos\theta\,d\theta$. After substituting into the integral, we have

$$\int \sqrt{9-x^2}\,dx = \int \sqrt{9-(3\sin\theta)^2}\cdot 3\cos\theta\,d\theta.$$

After simplifying, we have

$$\int \sqrt{9-x^2}\,dx = \int 9\sqrt{1-\sin^2\theta}\cdot\cos\theta\,d\theta.$$

Letting $1-\sin^2\theta=\cos^2\theta$, we now have

$$\int \sqrt{9-x^2}\,dx = \int 9\sqrt{\cos^2\theta}\cos\theta\,d\theta.$$

Assuming that $\cos\theta\geq 0$, we have

$$\int \sqrt{9-x^2}\,dx = \int 9\cos^2\theta\,d\theta.$$

At this point, we can evaluate the integral using the techniques developed for integrating powers and products of trigonometric functions. Before completing this example, let's take a look at the general theory behind this idea.

To evaluate integrals involving $\sqrt{a^2-x^2}$, we make the substitution $x=a\sin\theta$ and $dx=a\cos\theta$. To see that this actually makes sense, consider the following argument: The domain of $\sqrt{a^2-x^2}$ is $[-a,a]$. Thus,

$$-a\leq x\leq a.$$

Consequently,

$$-1\leq \frac{x}{a}\leq 1.$$

Since the range of $\sin x$ over $[-(\pi/2),\pi/2]$ is $[-1,1]$, there is a unique angle θ satisfying $-(\pi/2)\leq\theta\leq\pi/2$ so that $\sin\theta=x/a$, or equivalently, so that $x=a\sin\theta$. If we substitute $x=a\sin\theta$ into $\sqrt{a^2-x^2}$, we get

$$\begin{aligned}
\sqrt{a^2-x^2} &= \sqrt{a^2-(a\sin\theta)^2} && \text{Let } x=a\sin\theta \text{ where } -\frac{\pi}{2}\leq\theta\leq\frac{\pi}{2}.\\
&&& \text{Simplify.}\\
&= \sqrt{a^2-a^2\sin^2\theta} && \text{Factor out } a^2.\\
&= \sqrt{a^2(1-\sin^2\theta)} && \text{Substitute } 1-\sin^2 x=\cos^2 x.\\
&= \sqrt{a^2\cos^2\theta} && \text{Take the square root.}\\
&= |a\cos\theta|\\
&= a\cos\theta
\end{aligned}$$

Since $\cos x \geq 0$ on $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$ and $a > 0, |a\cos\theta| = a\cos\theta$. We can see, from this discussion, that by making the substitution $x = a\sin\theta$, we are able to convert an integral involving a radical into an integral involving trigonometric functions. After we evaluate the integral, we can convert the solution back to an expression involving x. To see how to do this, let's begin by assuming that $0 < x < a$. In this case, $0 < \theta < \frac{\pi}{2}$. Since $\sin\theta = \frac{x}{a}$, we can draw the reference triangle in Figure 7.3.1 to assist in expressing the values of $\cos\theta$, $\tan\theta$, and the remaining trigonometric functions in terms of x. It can be shown that this triangle actually produces the correct values of the trigonometric functions evaluated at θ for all θ satisfying $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$. It is useful to observe that the expression $\sqrt{a^2 - x^2}$ actually appears as the length of one side of the triangle. Last, should θ appear by itself, we use $\theta = \sin^{-1}\left(\frac{x}{a}\right)$.

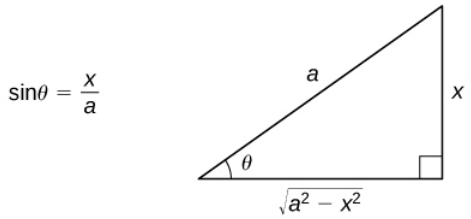


Figure 7.3.1: A reference triangle can help express the trigonometric functions evaluated at θ in terms of x.

The essential part of this discussion is summarized in the following problem-solving strategy.

Problem-Solving Strategy: Integrating Expressions Involving $\sqrt{a^2 - x^2}$

1. It is a good idea to make sure the integral cannot be evaluated easily in another way. For example, although this method can be applied to integrals of the form $\int \frac{1}{\sqrt{a^2 - x^2}}dx$, $\int \frac{x}{\sqrt{a^2 - x^2}}dx$, and $\int x\sqrt{a^2 - x^2}\,dx$, they can each be integrated directly either by formula or by a simple u-substitution.
2. Make the substitution $x = a\sin\theta$ and $dx = a\cos\theta\,d\theta$. Note: This substitution yields $\sqrt{a^2 - x^2} = a\cos\theta$.
3. Simplify the expression.
4. Evaluate the integral using techniques from the section on trigonometric integrals.
5. Use the reference triangle from Figure 1 to rewrite the result in terms of x. You may also need to use some trigonometric identities and the relationship $\theta = \sin^{-1}\left(\frac{x}{a}\right)$.

The following example demonstrates the application of this problem-solving strategy.

Example 7.3.1: Integrating an Expression Involving $\sqrt{a^2 - x^2}$

Evaluate

$$\int \sqrt{9 - x^2}dx.$$

Solution

Begin by making the substitutions $x = 3\sin\theta$ and $dx = 3\cos\theta\,d\theta$. Since $\sin\theta = \frac{x}{3}$, we can construct the reference triangle shown in Figure 2.

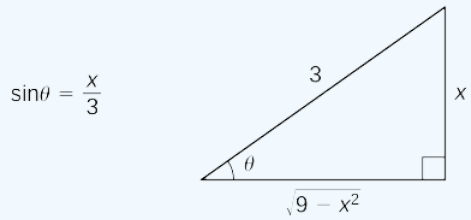


Figure 7.3.2: A reference triangle can be constructed for Example 7.3.1.

Thus,

$\int \sqrt{9 - x^2}\,dx = \int \sqrt{9 - (3\sin\theta)^2}3\cos\theta\,d\theta$ Substitute $x = 3\sin\theta$ and $dx = 3\cos\theta\,d\theta$.

$= \int \sqrt{9(1-\sin^2\theta)} \cdot 3\cos\theta\, d\theta$ Simplify.

$= \int \sqrt{9\cos^2\theta} \cdot 3\cos\theta\, d\theta$ Substitute $\cos^2\theta = 1-\sin^2\theta$.

$= \int 3|\cos\theta| 3\cos\theta\, d\theta$ Take the square root.

$= \int 9\cos^2\theta\, d\theta$ Simplify. Since $-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}, \cos\theta \ge 0$ and $|\cos\theta| = \cos\theta$.

$= \int 9\left(\frac{1}{2} + \frac{1}{2}\cos(2\theta)\right) d\theta$ Use the strategy for integrating an even power of $\cos\theta$.

$= \frac{9}{2}\theta + \frac{9}{4}\sin(2\theta) + C$ Evaluate the integral.

$= \frac{9}{2}\theta + \frac{9}{4}(2\sin\theta\cos\theta) + C$ Substitute $\sin(2\theta) = 2\sin\theta\cos\theta$.

$= \frac{9}{2}\sin^{-1}\left(\frac{x}{3}\right) + \frac{9}{2}\cdot\frac{x}{3}\cdot\frac{\sqrt{9-x^2}}{3} + C$ Substitute $\sin^{-1}\left(\frac{x}{3}\right) = \theta$ and $\sin\theta = \frac{x}{3}$.

Use the reference triangle to see that $\cos\theta = \frac{\sqrt{9-x^2}}{3}$ and make this substitution. Simplify.

$= \frac{9}{2}\sin^{-1}\left(\frac{x}{3}\right) + \frac{x\sqrt{9-x^2}}{2} + C.$ Simplify.

Example 7.3.2: Integrating an Expression Involving $\sqrt{a^2-x^2}$

Evaluate

$$\int \frac{\sqrt{4-x^2}}{x} dx.$$

Solution

First make the substitutions $x = 2\sin\theta$ and $dx = 2\cos\theta\, d\theta$. Since $\sin\theta = \frac{x}{2}$, we can construct the reference triangle shown in Figure 7.3.3.

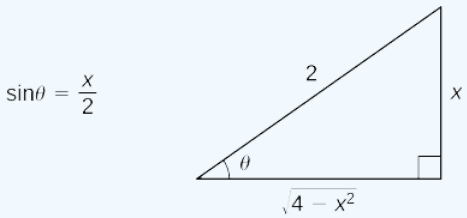


Figure 7.3.3: A reference triangle can be constructed for Example 7.3.2.

Thus,

$\int \frac{\sqrt{4-x^2}}{x} dx = \int \frac{\sqrt{4-(2\sin\theta)^2}}{2\sin\theta} 2\cos\theta\, d\theta$ Substitute $x = 2\sin\theta$ and $dx = 2\cos\theta\, d\theta$.

$= \int \frac{2\cos^2\theta}{\sin\theta} d\theta$ Substitute $\cos^2\theta = 1-\sin^2\theta$ and simplify.

$= \int \frac{2(1-\sin^2\theta)}{\sin\theta} d\theta$ Substitute $\cos^2\theta = 1-\sin^2\theta$.

$= \int (2\csc\theta - 2\sin\theta)\, d\theta$ Separate the numerator, simplify, and use $\csc\theta = \frac{1}{\sin\theta}$.

$= 2\ln|\csc\theta - \cot\theta| + 2\cos\theta + C$ Evaluate the integral.

$= 2\ln\left|\frac{2}{x} - \frac{\sqrt{4-x^2}}{x}\right| + \sqrt{4-x^2} + C.$ Use the reference triangle to rewrite the expression in terms of x and simplify.

In the next example, we see that we sometimes have a choice of methods.

Example 7.3.3: Integrating an Expression Involving $\sqrt{a^2 - x^2}$ Two Ways

Evaluate $\int x^3\sqrt{1-x^2}dx$ two ways: first by using the substitution $u = 1 - x^2$ and then by using a trigonometric substitution.

Method 1

Let $u = 1 - x^2$ and hence $x^2 = 1 - u$. Thus, $du = -2x\,dx$. In this case, the integral becomes

$$\begin{aligned}\int x^3\sqrt{1-x^2}\,dx &= -\frac{1}{2}\int x^2\sqrt{1-x^2}(-2x\,dx) \text{ Make the substitution.}\\ &= -\frac{1}{2}\int(1-u)\sqrt{u}\,du \text{ Expand the expression.}\\ &= -\frac{1}{2}\int(u^{1/2}-u^{3/2})\,du \text{ Evaluate the integral.}\\ &= -\frac{1}{2}(\frac{2}{3}u^{3/2}-\frac{2}{5}u^{5/2})+C \text{ Rewrite in terms of x.}\\ &= -\frac{1}{3}(1-x^2)^{3/2}+\frac{1}{5}(1-x^2)^{5/2}+C.\end{aligned}$$

Method 2

Let $x = \sin\theta$. In this case, $dx = \cos\theta\,d\theta$. Using this substitution, we have

$$\begin{aligned}\int x^3\sqrt{1-x^2}dx &= \int \sin^3\theta\cos^2\theta\,d\theta\\ &= \int(1-\cos^2\theta)\cos^2\theta\sin\theta\,d\theta \text{ Let } u=\cos\theta. \text{ Thus,} du=-\sin\theta\,d\theta.\\ &= \int(u^4-u^2)\,du\\ &= \frac{1}{5}u^5-\frac{1}{3}u^3+C \text{ Substitute } \cos\theta=u.\\ &= \frac{1}{5}\cos^5\theta-\frac{1}{3}\cos^3\theta+C \text{ Use a reference triangle to see that } \cos\theta=\sqrt{1-x^2}.\\ &= \frac{1}{5}(1-x^2)^{5/2}-\frac{1}{3}(1-x^2)^{3/2}+C.\end{aligned}$$

Exercise 7.3.1

Rewrite the integral $\displaystyle\int\frac{x^3}{\sqrt{25-x^2}}\,dx$ using the appropriate trigonometric substitution (do not evaluate the integral).

Hint

Substitute $x = 5\sin\theta$ and $dx = 5\cos\theta\,d\theta$.

Answer

$$\int 125\sin^3\theta\,d\theta$$

Integrating Expressions Involving $\sqrt{a^2 + x^2}$

For integrals containing $\sqrt{a^2+x^2}$,let's first consider the domain of this expression. Since $\sqrt{a^2+x^2}$ is defined for all real values of x, we restrict our choice to those trigonometric functions that have a range of all real numbers. Thus, our choice is restricted to selecting either $x = a\tan\theta$ or $x = a\cot\theta$. Either of these substitutions would actually work, but the standard substitution is $x = a\tan\theta$ or, equivalently, $\tan\theta = x/a$. With this substitution, we make the assumption that $-(\pi/2) < \theta < \pi/2$, so that we also have $\theta = \tan^{-1}(x/a)$. The procedure for using this substitution is outlined in the following problem-solving strategy.

Problem-Solving Strategy: Integrating Expressions Involving $\sqrt{a^2 + x^2}$

1. Check to see whether the integral can be evaluated easily by using another method. In some cases, it is more convenient to use an alternative method.
2. Substitute $x = a\tan\theta$ and $dx = a\sec^2\theta\,d\theta$. This substitution yields $\sqrt{a^2+x^2} = \sqrt{a^2+(a\tan\theta)^2} = \sqrt{a^2(1+\tan^2\theta)} = \sqrt{a^2sec^2\theta}$ $= |a\sec\theta| = a\sec\theta$. (Since $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$ and $\sec\theta > 0$ over this interval, $|a\sec\theta| = a\sec\theta$.)

3. Simplify the expression.
4. Evaluate the integral using techniques from the section on trigonometric integrals.
5. Use the reference triangle from Figure 7.3.4 to rewrite the result in terms of x. You may also need to use some trigonometric identities and the relationship $\theta = \tan^{-1}\left(\frac{x}{a}\right)$. (Note: The reference triangle is based on the assumption that $x > 0$; however, the trigonometric ratios produced from the reference triangle are the same as the ratios for which $x \leq 0$.)

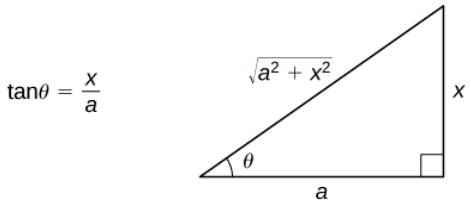


Figure 7.3.4: A reference triangle can be constructed to express the trigonometric functions evaluated at θ in terms of x.

Example 7.3.4: Integrating an Expression Involving $\sqrt{a^2 + x^2}$

Evaluate $\int \frac{dx}{\sqrt{1+x^2}}$ and check the solution by differentiating.

Solution

Begin with the substitution $x = \tan\theta$ and $dx = sec^2\theta\, d\theta$. Since $\tan\theta = x$, draw the reference triangle in Figure 7.3.5.

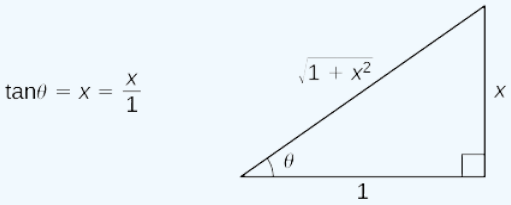


Figure 7.3.5: The reference triangle for Example 7.3.4.

Thus,

$$\int \frac{dx}{\sqrt{1+x^2}} = \int \frac{\sec^2\theta}{\sec\theta} d\theta \qquad \text{Substitute } x = \tan\theta \text{ and } dx = \sec^2\theta\, d\theta.$$

$$\text{This substitution makes } \sqrt{1+x^2} = \sec\theta. \text{ Simplify.}$$

$$= \int \sec\theta\, d\theta \qquad \text{Evaluate the integral.}$$

$$= \ln|\sec\theta + \tan\theta| + C \qquad \text{Use the reference triangle to express the result in terms of } x.$$

$$= \ln|\sqrt{1+x^2} + x| + C$$

To check the solution, differentiate:

$$\frac{d}{dx}\left(\ln|\sqrt{1+x^2}+x|\right) = \frac{1}{\sqrt{1+x^2}+x} \cdot \left(\frac{x}{\sqrt{1+x^2}} + 1\right) = \frac{1}{\sqrt{1+x^2}+x} \cdot \frac{x+\sqrt{1+x^2}}{\sqrt{1+x^2}} = \frac{1}{\sqrt{1+x^2}}.$$

Since $\sqrt{1+x^2} + x > 0$ for all values of x, we could rewrite $\ln|\sqrt{1+x^2}+x| + C = \ln(\sqrt{1+x^2}+x) + C$, if desired.

Example 7.3.5: Evaluating $\int \frac{dx}{\sqrt{1+x^2}}$ Using a Different Substitution

Use the substitution $x = \sinh\theta$ to evaluate $\int \frac{dx}{\sqrt{1+x^2}}$.

Solution

Because $\sinh\theta$ has a range of all real numbers, and $1 + \sinh^2\theta = \cosh^2\theta$, we may also use the substitution $x = \sinh\theta$ to evaluate this integral. In this case, $dx = \cosh\theta\, d\theta$. Consequently,

$$\int \frac{dx}{\sqrt{1+x^2}} = \int \frac{\cosh\theta}{\sqrt{1+\sinh^2\theta}}d\theta \qquad \text{Substitute } x=\sinh\theta \text{ and } dx=\cosh\theta\,d\theta.$$

$$\text{Substitute } 1+\sinh^2\theta=\cosh^2\theta.$$

$$= \int \frac{\cosh\theta}{\sqrt{\cosh^2\theta}}d\theta \qquad \text{Since } \sqrt{\cosh^2\theta}=|\cosh\theta|$$

$$= \int \frac{\cosh\theta}{|\cosh\theta|}d\theta \qquad |\cosh\theta|=\cosh\theta \text{ since } \cosh\theta>0 \text{ for all } \theta.$$

$$= \int \frac{\cosh\theta}{\cosh\theta}d\theta \qquad \text{Simplify.}$$

$$= \int 1\,d\theta \qquad \text{Evaluate the integral.}$$

$$= \theta + C \qquad \text{Since } x=\sinh\theta, \text{ we know } \theta=\sinh^{-1}x.$$

$$= \sinh^{-1}x + C.$$

Analysis

This answer looks quite different from the answer obtained using the substitution $x=\tan\theta$. To see that the solutions are the same, set $y=\sinh^{-1}x$. Thus, $\sinh y = x$. From this equation we obtain:

$$\frac{e^y - e^{-y}}{2} = x.$$

After multiplying both sides by $2e^y$ and rewriting, this equation becomes:

$$e^{2y} - 2xe^y - 1 = 0.$$

Use the quadratic equation to solve for e^y:

$$e^y = \frac{2x \pm \sqrt{4x^2+4}}{2}.$$

Simplifying, we have:

$$e^y = x \pm \sqrt{x^2+1}.$$

Since $x-\sqrt{x^2+1}<0$, it must be the case that $e^y = x+\sqrt{x^2+1}$. Thus,

$$y = \ln(x+\sqrt{x^2+1}).$$

Last, we obtain

$$\sinh^{-1}x = \ln(x+\sqrt{x^2+1}).$$

After we make the final observation that, since $x+\sqrt{x^2+1}>0$,

$$\ln(x+\sqrt{x^2+1}) = \ln|\sqrt{1+x^2}+x|,$$

we see that the two different methods produced equivalent solutions.

Example 7.3.6: Finding an Arc Length

Find the length of the curve $y=x^2$ over the interval $[0,\frac{1}{2}]$.

Solution

Because $\frac{dy}{dx}=2x$, the arc length is given by

$$\int_0^{1/2}\sqrt{1+(2x)^2}dx = \int_0^{1/2}\sqrt{1+4x^2}dx.$$

To evaluate this integral, use the substitution $x = \frac{1}{2}\tan\theta$ and $dx = \frac{1}{2}\sec^2\theta\, d\theta$. We also need to change the limits of integration. If $x = 0$, then $\theta = 0$ and if $x = \frac{1}{2}$, then $\theta = \frac{\pi}{4}$. Thus,

$\int_0^{1/2} \sqrt{1+4x^2}dx = \int_0^{\pi/4} \sqrt{1+\tan^2\theta}\cdot \frac{1}{2}\sec^2\theta\, d\theta$ After substitution, $\sqrt{1+4x^2} = \sec\theta$. (Substitute $1+\tan^2\theta = \sec^2\theta$ and simplify.)

$= \frac{1}{2}\int_0^{\pi/4} \sec^3\theta\, d\theta$ We derived this integral in the previous section.

$= \frac{1}{2}(\frac{1}{2}\sec\theta\tan\theta + \frac{1}{2}\ln|\sec\theta+\tan\theta|)\,|_0^{\pi/4}$ Evaluate and simplify.

$= \frac{1}{4}(\sqrt{2}+\ln(\sqrt{2}+1))$.

Exercise 7.3.2

Rewrite $\int x^3\sqrt{x^2+4}dx$ by using a substitution involving $\tan\theta$.

Hint

Use $x = 2\tan\theta$ and $dx = 2\sec^2\theta\, d\theta$.

Answer

$$\int 32\tan^3\theta\sec^3\theta\, d\theta$$

Integrating Expressions Involving $\sqrt{x^2-a^2}$

The domain of the expression $\sqrt{x^2-a^2}$ is $(-\infty,-a]\cup[a,+\infty)$. Thus, either $x \le -a$ or $x \ge a$. Hence, $\frac{x}{a} \le -1$ or $\frac{x}{a} \ge 1$. Since these intervals correspond to the range of $\sec\theta$ on the set $[0,\frac{\pi}{2})\cup(\frac{\pi}{2},\pi]$, it makes sense to use the substitution $\sec\theta = \frac{x}{a}$ or, equivalently, $x = a\sec\theta$, where $0 \le \theta < \frac{\pi}{2}$ or $\frac{\pi}{2} < \theta \le \pi$. The corresponding substitution for dx is $dx = a\sec\theta\tan\theta\, d\theta$. The procedure for using this substitution is outlined in the following problem-solving strategy.

Problem-Solving Strategy: Integrals Involving $\sqrt{x^2-a^2}$

1. Check to see whether the integral cannot be evaluated using another method. If so, we may wish to consider applying an alternative technique.
2. Substitute $x = a\sec\theta$ and $dx = a\sec\theta\tan\theta\, d\theta$. This substitution yields

$$\sqrt{x^2-a^2} = \sqrt{(a\sec\theta)^2-a^2} = \sqrt{a^2(\sec^2\theta-1)} = \sqrt{a^2\tan^2\theta} = |a\tan\theta|.$$

   For $x \ge a, |a\tan\theta| = a\tan\theta$ and for $x \le -a, |a\tan\theta| = -a\tan\theta$.
3. Simplify the expression.
4. Evaluate the integral using techniques from the section on trigonometric integrals.
5. Use the reference triangles from Figure 7.3.6 to rewrite the result in terms of x.
6. You may also need to use some trigonometric identities and the relationship $\theta = \sec^{-1}\left(\frac{x}{a}\right)$. (Note: We need both reference triangles, since the values of some of the trigonometric ratios are different depending on whether $x > a$ or $x < -a$.)

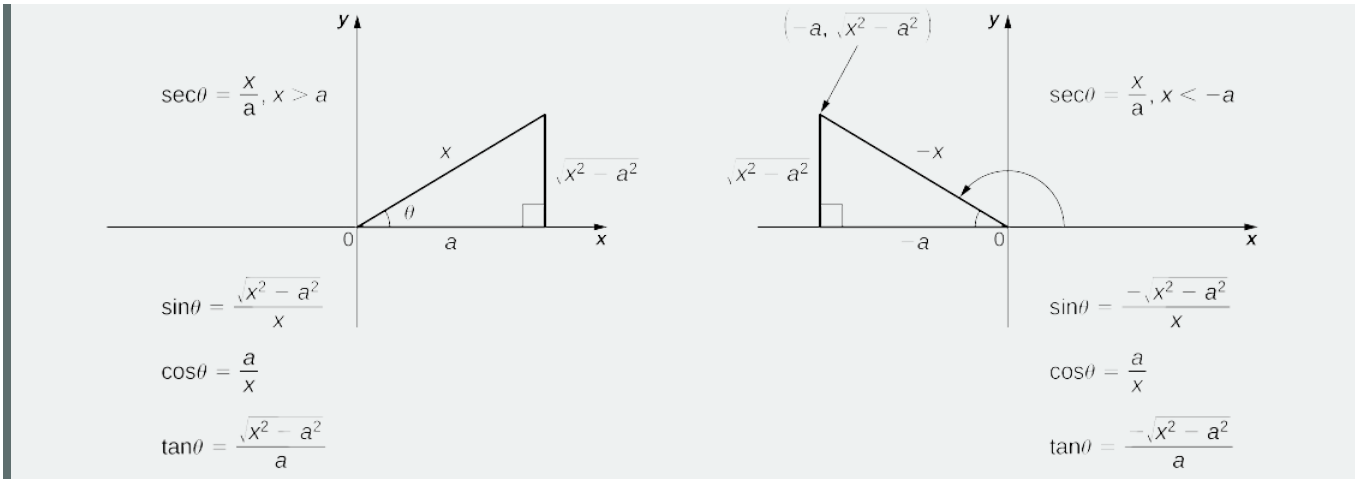


Figure 7.3.6: Use the appropriate reference triangle to express the trigonometric functions evaluated at θ in terms of x.

Example 7.3.7: Finding the Area of a Region

Find the area of the region between the graph of $f(x) = \sqrt{x^2 - 9}$ and the x-axis over the interval $[3, 5]$.

Solution

First, sketch a rough graph of the region described in the problem, as shown in the following figure.

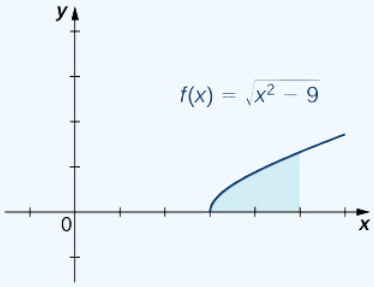


Figure 7.3.7: Calculating the area of the shaded region requires evaluating an integral with a trigonometric substitution.

We can see that the area is $A = \int_3^5 \sqrt{x^2 - 9}dx$. To evaluate this definite integral, substitute $x = 3\sec\theta$ and $dx = 3\sec\theta\tan\theta\, d\theta$. We must also change the limits of integration. If $x = 3$, then $3 = 3\sec\theta$ and hence $\theta = 0$. If $x = 5$, then $\theta = \sec^{-1}(\frac{5}{3})$. After making these substitutions and simplifying, we have

$$\begin{aligned}
\textbf{Area} &= \int_3^5 \sqrt{x^2 - 9}dx \\
&= \int_0^{\sec^{-1}(5/3)} 9\tan^2\theta\sec\theta\, d\theta \text{ Use } \tan^2\theta = \sec^2\theta - 1. \\
&= \int_0^{\sec^{-1}(5/3)} 9(\sec^2\theta - 1)\sec\theta\, d\theta \text{ Expand.} \\
&= \int_0^{\sec^{-1}(5/3)} 9(\sec^3\theta - \sec\theta)\, d\theta \text{ Evaluate the integral.} \\
&= (\frac{9}{2}\ln|\sec\theta + \tan\theta| + \frac{9}{2}\sec\theta\tan\theta) - 9\ln|\sec\theta + \tan\theta|\,|_0^{\sec^{-1}(5/3)} \text{ Simplify.} \\
&= \frac{9}{2}\sec\theta\tan\theta - \frac{9}{2}\ln|\sec\theta + \tan\theta|\,|_0^{\sec^{-1}(5/3)} \text{ Evaluate. Use } \sec(\sec^{-1}\frac{5}{3}) = \frac{5}{3} \text{ and } \tan(\sec^{-1}\frac{5}{3}) = \frac{4}{3}. \\
&= \frac{9}{2}\cdot\frac{5}{3}\cdot\frac{4}{3} - \frac{9}{2}\ln|\frac{5}{3} + \frac{4}{3}| - (\frac{9}{2}\cdot 1\cdot 0 - \frac{9}{2}\ln|1 + 0|) \\
&= 10 - \frac{9}{2}\ln 3
\end{aligned}$$

Exercise 7.3.3

Evaluate

$$\int \frac{dx}{\sqrt{x^2-4}}.$$

Assume that $x > 2$.

Hint

Substitute $x = 2\sec\theta$ and $dx = 2\sec\theta\tan\theta\, d\theta$.

Answer

$$\ln\left|\frac{x}{2} + \frac{\sqrt{x^2-4}}{2}\right| + C$$

Key Concepts

- For integrals involving $\sqrt{a^2-x^2}$, use the substitution $x = a\sin\theta$ and $dx = a\cos\theta\, d\theta$.
- For integrals involving $\sqrt{a^2+x^2}$, use the substitution $x = a\tan\theta$ and $dx = a\sec^2\theta\, d\theta$.
- For integrals involving $\sqrt{x^2-a^2}$, substitute $x = a\sec\theta$ and $dx = a\sec\theta\tan\theta\, d\theta$.

Glossary

trigonometric substitution

an integration technique that converts an algebraic integral containing expressions of the form $\sqrt{a^2-x^2}$, $\sqrt{a^2+x^2}$, or $\sqrt{x^2-a^2}$ into a trigonometric integral

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7.3E: Exercises for Section 7.3

Simplify the expressions in exercises 1 - 5 by writing each one using a single trigonometric function.

1) $4 - 4\sin^2\theta$

2) $9\sec^2\theta - 9$

Answer

$$9\sec^2\theta - 9 \quad = \quad 9\tan^2\theta$$

3) $a^2 + a^2\tan^2\theta$

4) $a^2 + a^2\sinh^2\theta$

Answer

$$a^2 + a^2\sinh^2\theta \quad = \quad a^2\cosh^2\theta$$

5) $16\cosh^2\theta - 16$

Use the technique of completing the square to express each trinomial in exercises 6 - 8 as the square of a binomial.

6) $4x^2 - 4x + 1$

Answer

$4(x - \frac{1}{2})^2$

7) $2x^2 - 8x + 3$

8) $-x^2 - 2x + 4$

Answer

$-(x+1)^2 + 5$

In exercises 9 - 28, integrate using the method of trigonometric substitution. Express the final answer in terms of the original variable.

9) $\int \frac{dx}{\sqrt{4 - x^2}}$

10) $\int \frac{dx}{\sqrt{x^2 - a^2}}$

Answer

$$\int \frac{dx}{\sqrt{x^2 - a^2}} \quad = \quad \ln\left| x + \sqrt{-a^2 + x^2}\right| + C$$

11) $\int \sqrt{4 - x^2}\, dx$

12) $\int \frac{dx}{\sqrt{1 + 9x^2}}$

Answer

$$\int \frac{dx}{\sqrt{1 + 9x^2}} \quad = \quad \tfrac{1}{3}\ln\left| \sqrt{9x^2 + 1} + 3x\right| + C$$

13) $\int \frac{x^2\, dx}{\sqrt{1 - x^2}}$

14) $\int \frac{dx}{x^2\sqrt{1 - x^2}}$

Answer

$$\int \frac{dx}{x^2\sqrt{1-x^2}} = -\frac{\sqrt{1-x^2}}{x} + C$$

15) $\int \frac{dx}{(1+x^2)^2}$

16) $\int \sqrt{x^2+9}\,dx$

Answer

$$\int \sqrt{x^2+9}\,dx = 9\left[\frac{x\sqrt{x^2+9}}{18} + \tfrac{1}{2}\ln\left|\frac{\sqrt{x^2+9}}{3} + \frac{x}{3}\right|\right] + C$$

17) $\int \frac{\sqrt{x^2-25}}{x}\,dx$

18) $\int \frac{\theta^3}{\sqrt{9-\theta^2}}\,d\theta$

Answer

$$\int \frac{\theta^3 d\theta}{\sqrt{9-\theta^2}}\,d\theta = -\tfrac{1}{3}\sqrt{9-\theta^2}(18+\theta^2) + C$$

19) $\int \frac{dx}{\sqrt{x^6-x^2}}$

20) $\int \sqrt{x^6-x^8}\,dx$

Answer

$$\int \sqrt{x^6-x^8}\,dx = \frac{(-1+x^2)(2+3x^2)\sqrt{x^6-x^8}}{15x^3} + C$$

21) $\int \frac{dx}{(1+x^2)^{3/2}}$

22) $\int \frac{dx}{(x^2-9)^{3/2}}$

Answer

$$\int \frac{dx}{(x^2-9)^{3/2}} = -\frac{x}{9\sqrt{x^2-9}} + C$$

23) $\int \frac{\sqrt{1+x^2}}{x}\,dx$

24) $\int \frac{x^2}{\sqrt{x^2-1}}\,dx$

Answer

$$\int \frac{x^2}{\sqrt{x^2-1}}\,dx = \tfrac{1}{2}(\ln\mid x+\sqrt{x^2-1}\mid + x\sqrt{x^2-1}) + C$$

25) $\int \frac{x^2}{x^2+4}\,dx$

26) $\int \frac{dx}{x^2\sqrt{x^2+1}}$

Answer

$$\int \frac{dx}{x^2\sqrt{x^2+1}} = -\frac{\sqrt{1+x^2}}{x} + C$$

27) $\int \frac{x^2}{\sqrt{1+x^2}}\,dx$

28) $\int_{-1}^{1}(1-x^2)^{3/2}\,dx$

Answer

$$\int_{-1}^{1}(1-x^2)^{3/2}\,dx \quad = \quad \tfrac{1}{8}\left(x(5-2x^2)\sqrt{1-x^2}+3\arcsin x\right)+C$$

In exercises 29 - 34, use the substitutions $x=\sinh\theta$, $\cosh\theta$, or $\tanh\theta$. Express the final answers in terms of the variable x.

29) $\int \dfrac{dx}{\sqrt{x^2-1}}$

30) $\int \dfrac{dx}{x\sqrt{1-x^2}}$

Answer

$$\int \frac{dx}{x\sqrt{1-x^2}} \quad = \quad \ln x - \ln\left|\,1+\sqrt{1-x^2}\,\right| + C$$

31) $\int \sqrt{x^2-1}\,dx$

32) $\int \dfrac{\sqrt{x^2-1}}{x^2}\,dx$

Answer

$$\int \frac{\sqrt{x^2-1}}{x^2}\,dx \quad = \quad -\frac{\sqrt{-1+x^2}}{x}+\ln\left|x+\sqrt{-1+x^2}\right|+C$$

33) $\int \dfrac{dx}{1-x^2}$

34) $\int \dfrac{\sqrt{1+x^2}}{x^2}\,dx$

Answer

$$\int \frac{\sqrt{1+x^2}}{x^2}\,dx \quad = \quad -\frac{\sqrt{1+x^2}}{x}+\operatorname{arcsinh} x+C$$

Use the technique of completing the square to evaluate the integrals in exercises 35 - 39.

35) $\int \dfrac{1}{x^2-6x}\,dx$

36) $\int \dfrac{1}{x^2+2x+1}\,dx$

Answer

$$\int \frac{1}{x^2+2x+1}\,dx \quad = \quad -\frac{1}{1+x}+C$$

37) $\int \dfrac{1}{\sqrt{-x^2+2x+8}}\,dx$

38) $\int \dfrac{1}{\sqrt{-x^2+10x}}\,dx$

Answer

$$\int \frac{1}{\sqrt{-x^2+10x}}\,dx \quad = \quad \arcsin\left(\frac{x-5}{5}\right)+C$$

39) $\int \dfrac{1}{\sqrt{x^2+4x-12}}\,dx$

40) Evaluate the integral without using calculus: $\int_{-3}^{3}\sqrt{9-x^2}\,dx$.

Answer

$\int_{-3}^{3} \sqrt{9-x^2}\,dx = \frac{9\pi}{2}$; area of a semicircle with radius 3

41) Find the area enclosed by the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$.

42) Evaluate the integral $\int \frac{dx}{\sqrt{1-x^2}}$ using two different substitutions. First, let $x = \cos\theta$ and evaluate using trigonometric substitution. Second, let $x = \sin\theta$ and use trigonometric substitution. Are the answers the same?

Answer

$\int \frac{dx}{\sqrt{1-x^2}} = \arcsin(x) + C$ is the common answer.

43) Evaluate the integral $\int \frac{dx}{x\sqrt{x^2-1}}$ using the substitution $x = \sec\theta$. Next, evaluate the same integral using the substitution $x = \csc\theta$. Show that the results are equivalent.

44) Evaluate the integral $\int \frac{x}{x^2+1}\,dx$ using the form $\int \frac{1}{u}\,du$. Next, evaluate the same integral using $x = \tan\theta$. Are the results the same?

Answer

$\int \frac{x}{x^2+1}\,dx = \frac{1}{2}\ln(1+x^2) + C$ is the result using either method.

45) State the method of integration you would use to evaluate the integral $\int x\sqrt{x^2+1}\,dx$. Why did you choose this method?

46) State the method of integration you would use to evaluate the integral $\int x^2\sqrt{x^2-1}\,dx$. Why did you choose this method?

Answer

Use trigonometric substitution. Let $x = \sec(\theta)$.

47) Evaluate $\int_{-1}^{1} \frac{x}{x^2+1}\,dx$

48) Find the length of the arc of the curve over the specified interval: $y = \ln x, \quad [1, 5]$. Round the answer to three decimal places.

Answer

$s = 4.367$ units

49) Find the surface area of the solid generated by revolving the region bounded by the graphs of $y = x^2$, $y = 0$, $x = 0$, and $x = \sqrt{2}$ about the x-axis. (Round the answer to three decimal places).

50) The region bounded by the graph of $f(x) = \frac{1}{1+x^2}$ and the x-axis between $x = 0$ and $x = 1$ is revolved about the x-axis. Find the volume of the solid that is generated.

Answer

$V = \left(\frac{\pi^2}{8} + \frac{\pi}{4}\right)$ units^3

In exercises 51 - 52, solve the initial-value problem for y as a function of x.

51) $(x^2+36)\frac{dy}{dx} = 1, \quad y(6) = 0$

52) $(64-x^2)\frac{dy}{dx} = 1, \quad y(0) = 3$

Answer

$y = \frac{1}{16}\ln\left|\frac{x+8}{x-8}\right| + 3$

53) Find the area bounded by $y = \dfrac{2}{\sqrt{64 - 4x^2}}$, $x = 0$, $y = 0$, and $x = 2$.

54) An oil storage tank can be described as the volume generated by revolving the area bounded by $y = \dfrac{16}{\sqrt{64 + x^2}}$, $x = 0$, $y = 0$, $x = 2$ about the x-axis. Find the volume of the tank (in cubic meters).

Answer

$V = 24.6 \text{ m}^3$

55) During each cycle, the velocity v (in feet per second) of a robotic welding device is given by $v = 2t - \dfrac{14}{4 + t^2}$, where t is time in seconds. Find the expression for the displacement s (in feet) as a function of t if $s = 0$ when $t = 0$.

56) Find the length of the curve $y = \sqrt{16 - x^2}$ between $x = 0$ and $x = 2$.

Answer

$s = \frac{2\pi}{3}$ units

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7.4: Partial Fractions

Learning Objectives

- Integrate a rational function using the method of partial fractions.
- Recognize simple linear factors in a rational function.
- Recognize repeated linear factors in a rational function.
- Recognize quadratic factors in a rational function.

We have seen some techniques that allow us to integrate specific rational functions. For example, we know that

$$\int \frac{du}{u} = \ln|u| + C$$

and

$$\int \frac{du}{u^2 + a^2} = \frac{1}{a}\tan^{-1}\left(\frac{u}{a}\right) + C.$$

However, we do not yet have a technique that allows us to tackle arbitrary quotients of this type. Thus, it is not immediately obvious how to go about evaluating

$$\int \frac{3x}{x^2 - x - 2}\, dx.$$

However, we know from material previously developed that

$$\int \left(\frac{1}{x+1} + \frac{2}{x-2}\right) dx = \ln|x+1| + 2\ln|x-2| + C.$$

In fact, by getting a common denominator, we see that

$$\frac{1}{x+1} + \frac{2}{x-2} = \frac{3x}{x^2 - x - 2}.$$

Consequently,

$$\int \frac{3x}{x^2 - x - 2}\, dx = \int \left(\frac{1}{x+1} + \frac{2}{x-2}\right) dx.$$

In this section, we examine the method of **partial fraction decomposition**, which allows us to decompose **rational functions** into sums of simpler, more easily integrated rational functions. Using this method, we can rewrite an expression such as:

$$\frac{3x}{x^2 - x - 2}$$

as an expression such as

$$\frac{1}{x+1} + \frac{2}{x-2}.$$

The key to the method of partial fraction decomposition is being able to anticipate the form that the decomposition of a rational function will take. As we shall see, this form is both predictable and highly dependent on the factorization of the denominator of the rational function. It is also extremely important to keep in mind that partial fraction decomposition can be applied to a rational function $\frac{P(x)}{Q(x)}$ only if $deg(P(x)) < deg(Q(x))$. In the case when $deg(P(x)) \geq deg(Q(x))$, we must first perform long division to rewrite the quotient $\frac{P(x)}{Q(x)}$ in the form $A(x) + \frac{R(x)}{Q(x)}$, where $deg(R(x)) < deg(Q(x))$. We then do a partial fraction decomposition on $\frac{R(x)}{Q(x)}$. The following example, although not requiring partial fraction decomposition, illustrates our approach to integrals of rational functions of the form $\int \frac{P(x)}{Q(x)}\, dx$, where $deg(P(x)) \geq deg(Q(x))$.

Example 7.4.1: Integrating $\int \frac{P(x)}{Q(x)}\,dx$, where $deg(P(x)) \geq deg(Q(x))$

Evaluate

$$\int \frac{x^2+3x+5}{x+1}\,dx.$$

Solution

Since $deg(x^2+3x+5) \geq deg(x+1)$, we perform long division to obtain

$$\frac{x^2+3x+5}{x+1} = x+2+\frac{3}{x+1}.$$

Thus,

$$\int \frac{x^2+3x+5}{x+1}\,dx = \int \left(x+2+\frac{3}{x+1}\right)\,dx = \frac{1}{2}x^2+2x+3\ln|x+1|+C.$$

Visit this website for a review of long division of polynomials.

Exercise 7.4.1

Evaluate

$$\int \frac{x-3}{x+2}\,dx.$$

Hint

Use long division to obtain $\frac{x-3}{x+2} = 1-\frac{5}{x+2}$.

Answer

$$x-5\ln|x+2|+C$$

To integrate $\int \frac{P(x)}{Q(x)}\,dx$, where $deg(P(x)) < deg(Q(x))$, we must begin by factoring $Q(x)$.

Nonrepeated Linear Factors

If $Q(x)$ can be factored as $(a_1x+b_1)(a_2x+b_2)\ldots(a_nx+b_n)$, where each linear factor is distinct, then it is possible to find constants $A_1, A_2, \ldots A_n$ satisfying

$$\frac{P(x)}{Q(x)} = \frac{A_1}{a_1x+b_1}+\frac{A_2}{a_2x+b_2}+\cdots+\frac{A_n}{a_nx+b_n}. \tag{7.4.1}$$

The proof that such constants exist is beyond the scope of this course.

In this next example, we see how to use partial fractions to integrate a rational function of this type.

Example 7.4.2: Partial Fractions with Nonrepeated Linear Factors

Evaluate $\int \frac{3x+2}{x^3-x^2-2x}\,dx$.

Solution

Since $deg(3x+2) < deg(x^3-x^2-2x)$, we begin by factoring the denominator of $\frac{3x+2}{x^3-x^2-2x}$. We can see that $x^3-x^2-2x = x(x-2)(x+1)$. Thus, there are constants A, B, and C satisfying Equation 7.4.1 such that

$$\frac{3x+2}{x(x-2)(x+1)} = \frac{A}{x} + \frac{B}{x-2} + \frac{C}{x+1}.$$

We must now find these constants. To do so, we begin by getting a common denominator on the right. Thus,

$$\frac{3x+2}{x(x-2)(x+1)} = \frac{A(x-2)(x+1) + Bx(x+1) + Cx(x-2)}{x(x-2)(x+1)}.$$

Now, we set the numerators equal to each other, obtaining

$$3x+2 = A(x-2)(x+1) + Bx(x+1) + Cx(x-2). \tag{7.4.2}$$

There are two different strategies for finding the coefficients A, B, and C. We refer to these as the *method of equating coefficients* and the *method of strategic substitution*.

Strategy one: Method of Equating Coefficients

Rewrite Equation 7.4.2 in the form

$$3x+2 = (A+B+C)x^2 + (-A+B-2C)x + (-2A).$$

Equating coefficients produces the system of equations

$$\begin{aligned} A+B+C &= 0 \\ -A+B-2C &= 3 \\ -2A &= 2. \end{aligned}$$

To solve this system, we first observe that $-2A = 2 \Rightarrow A = -1$. Substituting this value into the first two equations gives us the system

$$B+C = 1$$

$$B-2C = 2.$$

Multiplying the second equation by -1 and adding the resulting equation to the first produces

$$-3C = 1,$$

which in turn implies that $C = -\frac{1}{3}$. Substituting this value into the equation $B+C=1$ yields $B = \frac{4}{3}$. Thus, solving these equations yields $A = -1, B = \frac{4}{3}$, and $C = -\frac{1}{3}$.

It is important to note that the system produced by this method is consistent if and only if we have set up the decomposition correctly. If the system is inconsistent, there is an error in our decomposition.

Strategy two: Method of Strategic Substitution

The method of strategic substitution is based on the assumption that we have set up the decomposition correctly. If the decomposition is set up correctly, then there must be values of A, B, and C that satisfy Equation 7.4.2 for all values of x. That is, this equation must be true for any value of x we care to substitute into it. Therefore, by choosing values of x carefully and substituting them into the equation, we may find A, B, and C easily. For example, if we substitute $x = 0$, the equation reduces to $2 = A(-2)(1)$. Solving for A yields $A = -1$. Next, by substituting $x = 2$, the equation reduces to $8 = B(2)(3)$, or equivalently $B = 4/3$. Last, we substitute $x = -1$ into the equation and obtain $-1 = C(-1)(-3)$. Solving, we have $C = -\frac{1}{3}$.

It is important to keep in mind that if we attempt to use this method with a decomposition that has not been set up correctly, we are still able to find values for the constants, but these constants are meaningless. If we do opt to use the method of strategic substitution, then it is a good idea to check the result by recombining the terms algebraically.

Now that we have the values of A, B, and C, we rewrite the original integral:

$$\int \frac{3x+2}{x^3-x^2-2x}\,dx = \int \left(-\frac{1}{x} + \frac{4}{3}\cdot\frac{1}{x-2} - \frac{1}{3}\cdot\frac{1}{x+1}\right)\,dx.$$

Evaluating the integral gives us

$$\int \frac{3x+2}{x^3-x^2-2x}\,dx = -\ln|x| + \frac{4}{3}\ln|x-2| - \frac{1}{3}\ln|x+1| + C.$$

In the next example, we integrate a rational function in which the degree of the numerator is not less than the degree of the denominator.

Example 7.4.3: Dividing before Applying Partial Fractions

Evaluate $\int \frac{x^2+3x+1}{x^2-4}\,dx$.

Solution

Since $deg(x^2+3x+1) \geq deg(x^2-4)$, we must perform long division of polynomials. This results in

$$\frac{x^2+3x+1}{x^2-4} = 1 + \frac{3x+5}{x^2-4}$$

Next, we perform partial fraction decomposition on $\frac{3x+5}{x^2-4} = \frac{3x+5}{(x+2)(x-2)}$. We have

$$\frac{3x+5}{(x-2)(x+2)} = \frac{A}{x-2} + \frac{B}{x+2}.$$

Thus,

$$3x+5 = A(x+2) + B(x-2).$$

Solving for A and B using either method, we obtain $A = 11/4$ and $B = 1/4$.

Rewriting the original integral, we have

$$\int \frac{x^2+3x+1}{x^2-4}\,dx = \int \left(1 + \frac{11}{4}\cdot\frac{1}{x-2} + \frac{1}{4}\cdot\frac{1}{x+2}\right)\,dx.$$

Evaluating the integral produces

$$\int \frac{x^2+3x+1}{x^2-4}\,dx = x + \frac{11}{4}\ln|x-2| + \frac{1}{4}\ln|x+2| + C.$$

As we see in the next example, it may be possible to apply the technique of partial fraction decomposition to a nonrational function. The trick is to convert the nonrational function to a rational function through a substitution.

Example 7.4.4: Applying Partial Fractions after a Substitution

Evaluate $\int \frac{\cos x}{\sin^2 x - \sin x}\,dx$.

Solution

Let's begin by letting $u = \sin x$. Consequently, $du = \cos x\,dx$. After making these substitutions, we have

$$\int \frac{\cos x}{\sin^2 x - \sin x}\,dx = \int \frac{du}{u^2-u} = \int \frac{du}{u(u-1)}.$$

Applying partial fraction decomposition to $\frac{1}{u(u-1)}$ gives $\frac{1}{u(u-1)} = -\frac{1}{u} + \frac{1}{u-1}$.

Thus,

$$\int \frac{\cos x}{\sin^2 x - \sin x}\,dx = -\ln|u| + \ln|u-1| + C = -\ln|\sin x| + \ln|\sin x - 1| + C.$$

Exercise 7.4.2

Evaluate $\int \frac{x+1}{(x+3)(x-2)}\,dx$.

Hint

$$\frac{x+1}{(x+3)(x-2)} = \frac{A}{x+3} + \frac{B}{x-2}$$

Answer

$$\frac{2}{5}\ln|x+3| + \frac{3}{5}\ln|x-2| + C$$

Repeated Linear Factors

For some applications, we need to integrate rational expressions that have denominators with repeated linear factors—that is, rational functions with at least one factor of the form $(ax+b)^n$, where n is a positive integer greater than or equal to 2. If the denominator contains the repeated linear factor $(ax+b)^n$, then the decomposition must contain

$$\frac{A_1}{ax+b} + \frac{A_2}{(ax+b)^2} + \cdots + \frac{A_n}{(ax+b)^n}. \tag{7.4.3}$$

As we see in our next example, the basic technique used for solving for the coefficients is the same, but it requires more algebra to determine the numerators of the partial fractions.

Example 7.4.5: Partial Fractions with Repeated Linear Factors

Evaluate $\int \frac{x-2}{(2x-1)^2(x-1)}\,dx$.

Solution

We have $deg(x-2) < deg((2x-1)^2(x-1))$, so we can proceed with the decomposition. Since $(2x-1)^2$ is a repeated linear factor, include

$$\frac{A}{2x-1} + \frac{B}{(2x-1)^2}$$

in the decomposition in Equation 7.4.3. Thus,

$$\frac{x-2}{(2x-1)^2(x-1)} = \frac{A}{2x-1} + \frac{B}{(2x-1)^2} + \frac{C}{x-1}.$$

After getting a common denominator and equating the numerators, we have

$$x-2 = A(2x-1)(x-1) + B(x-1) + C(2x-1)^2. \tag{7.4.4}$$

We then use the method of equating coefficients to find the values of A, B, and C.

$$x-2 = (2A+4C)x^2 + (-3A+B-4C)x + (A-B+C).$$

Equating coefficients yields $2A+4C=0$, $-3A+B-4C=1$, and $A-B+C=-2$. Solving this system yields $A=2, B=3$, and $C=-1$.

Alternatively, we can use the method of strategic substitution. In this case, substituting $x=1$ and $x=1/2$ into Equation 7.4.4 easily produces the values $B=3$ and $C=-1$. At this point, it may seem that we have run out of good choices for x, however, since we already have values for B and C, we can substitute in these values and choose any value for x not previously used. The value $x=0$ is a good option. In this case, we obtain the equation $-2 = A(-1)(-1) + 3(-1) + (-1)(-1)^2$ or, equivalently, $A=2$.

Now that we have the values for A, B, and C, we rewrite the original integral and evaluate it:

$$\int \frac{x-2}{(2x-1)^2(x-1)}\,dx = \int \left(\frac{2}{2x-1} + \frac{3}{(2x-1)^2} - \frac{1}{x-1}\right) dx$$
$$= \ln|2x-1| - \frac{3}{2(2x-1)} - \ln|x-1| + C.$$

? Exercise 7.4.3

Set up the partial fraction decomposition for

$$\int \frac{x+2}{(x+3)^3(x-4)^2}\,dx.$$

(Do not solve for the coefficients or complete the integration.)

Hint

Use the problem-solving method of Example 7.4.5 for guidance.

Answer

$$\frac{x+2}{(x+3)^3(x-4)^2} = \frac{A}{x+3} + \frac{B}{(x+3)^2} + \frac{C}{(x+3)^3} + \frac{D}{(x-4)} + \frac{E}{(x-4)^2}$$

The General Method

Now that we are beginning to get the idea of how the technique of partial fraction decomposition works, let's outline the basic method in the following problem-solving strategy.

Problem-Solving Strategy: Partial Fraction Decomposition

To decompose the rational function $P(x)/Q(x)$, use the following steps:

1. Make sure that $deg(P(x)) < deg(Q(x))$. If not, perform long division of polynomials.
2. Factor $Q(x)$ into the product of linear and irreducible quadratic factors. An irreducible quadratic is a quadratic that has no real zeros.
3. Assuming that $deg(P(x)) < deg(Q(x)$, the factors of $Q(x)$ determine the form of the decomposition of $P(x)/Q(x)$.
   a. If $Q(x)$ can be factored as $(a_1x+b_1)(a_2x+b_2)\ldots(a_nx+b_n)$, where each linear factor is distinct, then it is possible to find constants $A_1, A_2, \ldots A_n$ satisfying
   $$\frac{P(x)}{Q(x)} = \frac{A_1}{a_1x+b_1} + \frac{A_2}{a_2x+b_2} + \cdots + \frac{A_n}{a_nx+b_n}.$$
   b. If $Q(x)$ contains the repeated linear factor $(ax+b)^n$, then the decomposition must contain
   $$\frac{A_1}{ax+b} + \frac{A_2}{(ax+b)^2} + \cdots + \frac{A_n}{(ax+b)^n}.$$
   c. For each irreducible quadratic factor ax^2+bx+c that $Q(x)$ contains, the decomposition must include
   $$\frac{Ax+B}{ax^2+bx+c}.$$
   d. For each repeated irreducible quadratic factor $(ax^2+bx+c)^n$, the decomposition must include
   $$\frac{A_1x+B_1}{ax^2+bx+c} + \frac{A_2x+B_2}{(ax^2+bx+c)^2} + \cdots + \frac{A_nx+B_n}{(ax^2+bx+c)^n}.$$
   e. After the appropriate decomposition is determined, solve for the constants.
   f. Last, rewrite the integral in its decomposed form and evaluate it using previously developed techniques or integration formulas.

Simple Quadratic Factors

Now let's look at integrating a rational expression in which the denominator contains an irreducible quadratic factor. Recall that the quadratic ax^2+bx+c is irreducible if $ax^2+bx+c=0$ has no real zeros—that is, if $b^2-4ac<0$.

Example 7.4.6: Rational Expressions with an Irreducible Quadratic Factor

Evaluate

$$\int \frac{2x-3}{x^3+x}\,dx.$$

Solution

Since $deg(2x-3)<deg(x^3+x)$, factor the denominator and proceed with partial fraction decomposition. Since $x^3+x=x(x^2+1)$ contains the irreducible quadratic factor x^2+1, include $\frac{Ax+B}{x^2+1}$ as part of the decomposition, along with $\frac{C}{x}$ for the linear term x. Thus, the decomposition has the form

$$\frac{2x-3}{x(x^2+1)}=\frac{Ax+B}{x^2+1}+\frac{C}{x}.$$

After getting a common denominator and equating the numerators, we obtain the equation

$$2x-3=(Ax+B)x+C(x^2+1).$$

Solving for A, B, and C, we get $A=3$, $B=2$, and $C=-3$.

Thus,

$$\frac{2x-3}{x^3+x}=\frac{3x+2}{x^2+1}-\frac{3}{x}.$$

Substituting back into the integral, we obtain

$$\begin{aligned}\int \frac{2x-3}{x^3+x}\,dx &= \int\left(\frac{3x+2}{x^2+1}-\frac{3}{x}\right)dx \\ &= 3\int\frac{x}{x^2+1}\,dx+2\int\frac{1}{x^2+1}\,dx-3\int\frac{1}{x}\,dx && \text{Split up the integral} \\ &= \frac{3}{2}\ln\mid x^2+1\mid+2\tan^{-1}x-3\ln|x|+C. && \text{Evaluate each integral}\end{aligned}$$

Note: We may rewrite $\ln\mid x^2+1\mid=\ln(x^2+1)$, if we wish to do so, since $x^2+1>0$.

Example 7.4.7: Partial Fractions with an Irreducible Quadratic Factor

Evaluate $\int\frac{dx}{x^3-8}$.

Solution: We can start by factoring $x^3-8=(x-2)(x^2+2x+4)$. We see that the quadratic factor x^2+2x+4 is irreducible since $2^2-4(1)(4)=-12<0$. Using the decomposition described in the problem-solving strategy, we get

$$\frac{1}{(x-2)(x^2+2x+4)}=\frac{A}{x-2}+\frac{Bx+C}{x^2+2x+4}.$$

After obtaining a common denominator and equating the numerators, this becomes

$$1=A(x^2+2x+4)+(Bx+C)(x-2).$$

Applying either method, we get $A=\frac{1}{12}$, $B=-\frac{1}{12}$, and $C=-\frac{1}{3}$.

Rewriting $\int\frac{dx}{x^3-8}$, we have

$$\int\frac{dx}{x^3-8}=\frac{1}{12}\int\frac{1}{x-2}\,dx-\frac{1}{12}\int\frac{x+4}{x^2+2x+4}\,dx.$$

We can see that

$$\int \frac{1}{x-2}\,dx = \ln|x-2| + C,$$

but

$$\int \frac{x+4}{x^2+2x+4}\,dx$$

requires a bit more effort. Let's begin by **completing the square** on x^2+2x+4 to obtain

$$x^2+2x+4 = (x+1)^2+3.$$

By letting $u = x+1$ and consequently $du = dx$, we see that

$$\begin{aligned}
\int \frac{x+4}{x^2+2x+4}\,dx &= \int \frac{x+4}{(x+1)^2+3}\,dx && \text{Complete the square on the denominator}\\
&= \int \frac{u+3}{u^2+3}\,du && \text{Substitute } u = x+1,\ x = u-1, \text{ and } du = dx\\
&= \int \frac{u}{u^2+3}du + \int \frac{3}{u^2+3}du && \text{Split the numerator apart}\\
&= \frac{1}{2}\ln|u^2+3| && \text{Evaluate each integral}\\
&\quad + \frac{3}{\sqrt{3}}\tan^{-1}\frac{u}{\sqrt{3}} + C\\
&= \frac{1}{2}\ln|x^2+2x+4| && \text{Rewrite in terms of } x \text{ and simplify}\\
&\quad + \sqrt{3}\tan^{-1}\left(\frac{x+1}{\sqrt{3}}\right) + C
\end{aligned}$$

Substituting back into the original integral and simplifying gives

$$\int \frac{dx}{x^3-8} = \frac{1}{12}\ln|x-2| - \frac{1}{24}\ln|x^2+2x+4| - \frac{\sqrt{3}}{12}\tan^{-1}\left(\frac{x+1}{\sqrt{3}}\right) + C.$$

Here again, we can drop the absolute value if we wish to do so, since $x^2+2x+4 > 0$ for all x.

Example 7.4.8: Finding a Volume

Find the volume of the solid of revolution obtained by revolving the region enclosed by the graph of $f(x) = \dfrac{x^2}{(x^2+1)^2}$ and the x-axis over the interval $[0,1]$ about the y-axis.

Solution

Let's begin by sketching the region to be revolved (see Figure 7.4.1). From the sketch, we see that the shell method is a good choice for solving this problem.

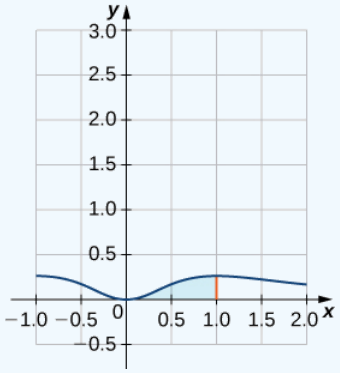


Figure 7.4.1: We can use the shell method to find the volume of revolution obtained by revolving the region shown about the y-axis.

The volume is given by

$$V = 2\pi \int_0^1 x \cdot \frac{x^2}{(x^2+1)^2}\, dx = 2\pi \int_0^1 \frac{x^3}{(x^2+1)^2}\, dx.$$

Since $deg((x^2+1)^2) = 4 > 3 = deg(x^3)$, we can proceed with partial fraction decomposition. Note that $(x^2+1)^2$ is a repeated irreducible quadratic. Using the decomposition described in the problem-solving strategy, we get

$$\frac{x^3}{(x^2+1)^2} = \frac{Ax+B}{x^2+1} + \frac{Cx+D}{(x^2+1)^2}.$$

Finding a common denominator and equating the numerators gives

$$x^3 = (Ax+B)(x^2+1) + Cx + D.$$

Solving, we obtain $A = 1, B = 0, C = -1$, and $D = 0$. Substituting back into the integral, we have

$$\begin{aligned} V &= 2\pi \int_0^1 \frac{x^3}{(x^2+1)^2}\, dx = 2\pi \int_0^1 \left(\frac{x}{x^2+1} - \frac{x}{(x^2+1)^2} \right) dx \\ &= 2\pi \left(\frac{1}{2}\ln(x^2+1) + \frac{1}{2} \cdot \frac{1}{x^2+1} \right) \Big|_0^1 = \pi \left(\ln 2 - \tfrac{1}{2} \right). \end{aligned}$$

? Exercise 7.4.4

Set up the partial fraction decomposition for

$$\int \frac{x^2+3x+1}{(x+2)(x-3)^2(x^2+4)^2}\, dx.$$

Hint

Use the problem-solving strategy.

Answer

$$\frac{x^2+3x+1}{(x+2)(x-3)^2(x^2+4)^2} = \frac{A}{x+2} + \frac{B}{x-3} + \frac{C}{(x-3)^2} + \frac{Dx+E}{x^2+4} + \frac{Fx+G}{(x^2+4)^2}$$

Key Concepts

- Partial fraction decomposition is a technique used to break down a rational function into a sum of simple rational functions that can be integrated using previously learned techniques.
- When applying partial fraction decomposition, we must make sure that the degree of the numerator is less than the degree of the denominator. If not, we need to perform long division before attempting partial fraction decomposition.
- The form the decomposition takes depends on the type of factors in the denominator. The types of factors include nonrepeated linear factors, repeated linear factors, nonrepeated irreducible quadratic factors, and repeated irreducible quadratic factors.

Glossary

partial fraction decomposition

a technique used to break down a rational function into the sum of simple rational functions

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7.4E: Exercises for Section 7.4

Use partial fraction decomposition (or a simpler technique) to express the rational function as a sum or difference of two or more simpler rational expressions.

1) $\dfrac{1}{(x-3)(x-2)}$

2) $\dfrac{x^2+1}{x(x+1)(x+2)}$

Answer

$$\frac{x^2+1}{x(x+1)(x+2)} = -\frac{2}{x+1}+\frac{5}{2(x+2)}+\frac{1}{2x}$$

3) $\dfrac{1}{x^3-x}$

4) $\dfrac{3x+1}{x^2}$

Answer

$$\frac{3x+1}{x^2} = \frac{1}{x^2}+\frac{3}{x}$$

5) $\dfrac{3x^2}{x^2+1}$ (Hint: Use long division first.)

6) $\dfrac{2x^4}{x^2-2x}$

Answer

$$\frac{2x^4}{x^2-2x} = 2x^2+4x+8+\frac{16}{x-2}$$

7) $\dfrac{1}{(x-1)(x^2+1)}$

8) $\dfrac{1}{x^2(x-1)}$

Answer

$$\frac{1}{x^2(x-1)} = -\frac{1}{x^2}-\frac{1}{x}+\frac{1}{x-1}$$

9) $\dfrac{x}{x^2-4}$

10) $\dfrac{1}{x(x-1)(x-2)(x-3)}$

Answer

$$\frac{1}{x(x-1)(x-2)(x-3)} = -\frac{1}{2(x-2)}+\frac{1}{2(x-1)}-\frac{1}{6x}+\frac{1}{6(x-3)}$$

11) $\dfrac{1}{x^4-1} = \dfrac{1}{(x+1)(x-1)(x^2+1)}$

12) $\dfrac{3x^2}{x^3-1} = \dfrac{3x^2}{(x-1)(x^2+x+1)}$

Answer

$$\frac{3x^2}{x^3-1} = \frac{1}{x-1}+\frac{2x+1}{x^2+x+1}$$

13) $\dfrac{2x}{(x+2)^2}$

14) $\dfrac{3x^4+x^3+20x^2+3x+31}{(x+1)(x^2+4)^2}$

Answer

$$\frac{3x^4+x^3+20x^2+3x+31}{(x+1)(x^2+4)^2} = \frac{2}{x+1}+\frac{x}{x^2+4}-\frac{1}{(x^2+4)^2}$$

In exercises 15 - 25, use the method of partial fractions to evaluate each of the following integrals.

15) $\displaystyle\int \frac{dx}{(x-3)(x-2)}$

16) $\displaystyle\int \frac{3x}{x^2+2x-8}\,dx$

Answer

$$\int \frac{3x}{x^2+2x-8}\,dx = 2\ln|x+4|+\ln|x-2|+C = \ln\left|(x+4)^2(x-2)\right|+C$$

17) $\displaystyle\int \frac{dx}{x^3-x}$

18) $\displaystyle\int \frac{x}{x^2-4}\,dx$

Answer

Note that you don't need Partial Fractions here. We use a simple u-substitution.

$$\int \frac{x}{x^2-4}\,dx = \tfrac{1}{2}\ln|4-x^2|+C$$

19) $\displaystyle\int \frac{dx}{x(x-1)(x-2)(x-3)}$

20) $\displaystyle\int \frac{2x^2+4x+22}{x^2+2x+10}\,dx$

Answer

Note that since the degree of the numerator is equal to the degree of the denominator, we need to start with long division. Then note that we will need to use completing the square to continue since we cannot factor the trinomial in the denominator.

$$\int \frac{2x^2+4x+22}{x^2+2x+10}\,dx = 2\left(x+\tfrac{1}{3}\arctan\left(\frac{1+x}{3}\right)\right)+C$$

21) $\displaystyle\int \frac{dx}{x^2-5x+6}$

22) $\displaystyle\int \frac{2-x}{x^2+x}\,dx$

Answer

$$\int \frac{2-x}{x^2+x}\,dx = 2\ln|x|-3\ln|1+x|+C = \ln\left|\frac{x^2}{(1+x)^3}\right|+C$$

23) $\displaystyle\int \frac{2}{x^2-x-6}\,dx$

24) $\displaystyle\int \frac{dx}{x^3-2x^2-4x+8}$

Answer

$$\int \frac{dx}{x^3-2x^2-4x+8} = \tfrac{1}{16}\left(-\frac{4}{-2+x}-\ln|-2+x|+\ln|2+x|\right)+C = \tfrac{1}{16}\left(-\frac{4}{-2+x}+\ln\left|\frac{x+2}{x-2}\right|\right)+C$$

25) $\displaystyle\int \frac{dx}{x^4-10x^2+9}$

In exercises 26 - 29, evaluate the integrals with irreducible quadratic factors in the denominators.

26) $\int \frac{2}{(x-4)(x^2+2x+6)}\,dx$

Answer

$$\int \frac{2}{(x-4)(x^2+2x+6)}\,dx \quad = \quad \tfrac{1}{30}(-2\sqrt{5}\arctan\left[\frac{1+x}{\sqrt{5}}\right] + 2\ln|-4+x| - \ln|6+2x+x^2|) + C$$

27) $\int \frac{x^2}{x^3-x^2+4x-4}\,dx$

28) $\int \frac{x^3+6x^2+3x+6}{x^3+2x^2}\,dx$

Answer

Note that we need to use long division first, since the degree of the numerator is greater than the degree of the denominator.

$$\int \frac{x^3+6x^2+3x+6}{x^3+2x^2}\,dx \quad = \quad -\frac{3}{x} + 4\ln|x+2| + x + C$$

29) $\int \frac{x}{(x-1)(x^2+2x+2)^2}\,dx$

In exercises 30 - 32, use the method of partial fractions to evaluate the integrals.

30) $\int \frac{3x+4}{(x^2+4)(3-x)}\,dx$

Answer

$$\int \frac{3x+4}{(x^2+4)(3-x)}\,dx \quad = \quad -\ln|3-x| + \tfrac{1}{2}\ln|x^2+4| + C$$

31) $\int \frac{2}{(x+2)^2(2-x)}\,dx$

32) $\int \frac{3x+4}{x^3-2x-4}\,dx$ (Hint: Use the rational root theorem.)

Answer

$$\int \frac{3x+4}{x^3-2x-4}\,dx \quad = \quad \ln|x-2| - \tfrac{1}{2}\ln|x^2+2x+2| + C$$

In exercises 33 - 46, use substitution to convert the integrals to integrals of rational functions. Then use partial fractions to evaluate the integrals.

33) $\int_0^1 \frac{e^x}{36-e^{2x}}\,dx$ (Give the exact answer and the decimal equivalent. Round to five decimal places.)

34) $\int \frac{e^x}{e^{2x}-e^x}\,dx$

Answer

$$\int \frac{e^x}{e^{2x}-e^x}\,dx \quad = \quad -x + \ln|1-e^x| + C$$

35) $\int \frac{\sin x\,dx}{1-\cos^2 x}$

36) $\int \frac{\sin x}{\cos^2 x + \cos x - 6}\,dx$

Answer

$$\int \frac{\sin x}{\cos^2 x + \cos x - 6}\,dx \quad = \quad \tfrac{1}{5}\ln\left|\frac{\cos x+3}{\cos x-2}\right| + C$$

37) $\int \frac{1-\sqrt{x}}{1+\sqrt{x}}\,dx$

38) $\int \frac{dt}{(e^t - e^{-t})^2}$

Answer

$$\int \frac{dt}{(e^t - e^{-t})^2} \quad = \quad \frac{1}{2 - 2e^{2t}} + C$$

39) $\int \frac{1 + e^x}{1 - e^x}\, dx$

40) $\int \frac{dx}{1 + \sqrt{x+1}}$

Answer

$$\int \frac{dx}{1 + \sqrt{x+1}} \quad = \quad 2\sqrt{1+x} - 2\ln|1 + \sqrt{1+x}| + C$$

41) $\int \frac{dx}{\sqrt{x} + \sqrt[4]{x}}$

42) $\int \frac{\cos x}{\sin x(1 - \sin x)}\, dx$

Answer

$$\int \frac{\cos x}{\sin x(1 - \sin x)}\, dx \quad = \quad \ln\left|\frac{\sin x}{1 - \sin x}\right| + C$$

43) $\int \frac{e^x}{(e^{2x} - 4)^2}\, dx$

44) $\int_1^2 \frac{1}{x^2\sqrt{4 - x^2}}\, dx$

Answer

$$\int_1^2 \frac{1}{x^2\sqrt{4 - x^2}}\, dx \quad = \quad \frac{\sqrt{3}}{4}$$

45) $\int \frac{1}{2 + e^{-x}}\, dx$

46) $\int \frac{1}{1 + e^x}\, dx$

Answer

$$\int \frac{1}{1 + e^x}\, dx \quad = \quad x - \ln(1 + e^x) + C$$

In exercises 47 - 48, use the given substitution to convert the integral to an integral of a rational function, then evaluate.

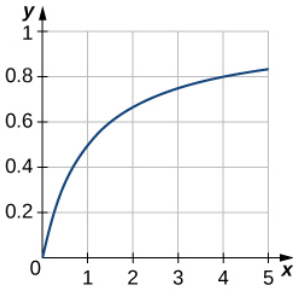
47) $\int \frac{1}{t - \sqrt[3]{t}}\, dt; \quad t = x^3$

48) $\int \frac{1}{\sqrt{x} + \sqrt[3]{x}}\, dx; \quad x = u^6$

Answer

$$\int \frac{1}{\sqrt{x} + \sqrt[3]{x}}\, dx \quad = \quad 6x^{1/6} - 3x^{1/3} + 2\sqrt{x} - 6\ln(1 + x^{1/6}) + C$$

49) Graph the curve $y = \frac{x}{1+x}$ over the interval $[0, 5]$. Then, find the area of the region bounded by the curve, the x-axis, and the line $x = 4$.



50) Find the volume of the solid generated when the region bounded by $y = \dfrac{1}{\sqrt{x(3-x)}}$, $y = 0$, $x = 1$, and $x = 2$ is revolved about the x-axis.

Answer

$V = \frac{4}{3}\pi \operatorname{arctanh}\left[\frac{1}{3}\right] = \frac{1}{3}\pi \ln 4 \text{ units}^3$

51) The velocity of a particle moving along a line is a function of time given by $v(t) = \dfrac{88t^2}{t^2+1}$. Find the distance that the particle has traveled after $t = 5$ sec.

In exercises 52 - 54, solve the initial-value problem for x as a function of t.

52) $(t^2 - 7t + 12)\dfrac{dx}{dt} = 1, \quad t > 4,\ x(5) = 0$

Answer

$x = -\ln|t-3| + \ln|t-4| + \ln 2 = \ln\left|\dfrac{2(t-4)}{t-3}\right|$

53) $(t+5)\dfrac{dx}{dt} = x^2 + 1, \quad t > -5,\ x(1) = \tan 1$

54) $(2t^3 - 2t^2 + t - 1)\dfrac{dx}{dt} = 3, \quad x(2) = 0$

Answer

$x = \ln|t-1| - \sqrt{2}\arctan(\sqrt{2}t) - \frac{1}{2}\ln(t^2 + \frac{1}{2}) + \sqrt{2}\arctan(2\sqrt{2}) + \frac{1}{2}\ln 4.5$

55) Find the x-coordinate of the centroid of the area bounded by $y(x^2 - 9) = 1$, $y = 0$, $x = 4$, and $x = 5$. (Round the answer to two decimal places.)

56) Find the volume generated by revolving the area bounded by $y = \dfrac{1}{x^3 + 7x^2 + 6x}$, $x = 1$, $x = 7$, and $y = 0$ about the y-axis.

Answer

$V = \frac{2}{5}\pi \ln \frac{28}{13} \text{ units}^3$

57) Find the area bounded by $y = \dfrac{x - 12}{x^2 - 8x - 20}$, $y = 0$, $x = 2$, and $x = 4$. (Round the answer to the nearest hundredth.)

58) Evaluate the integral $\displaystyle\int \frac{dx}{x^3 + 1}$.

Answer

$$\int \frac{dx}{x^3+1} = \frac{\arctan\left[\frac{-1+2x}{\sqrt{3}}\right]}{\sqrt{3}} + \frac{1}{3}\ln|1+x| - \frac{1}{6}\ln|1 - x + x^2| + C$$

For problems 59 - 62, use the substitutions $\tan(\frac{x}{2}) = t$, $dx = \dfrac{2}{1+t^2}\,dt$, $\sin x = \dfrac{2t}{1+t^2}$, and $\cos x = \dfrac{1-t^2}{1+t^2}$.

59) $\displaystyle\int \frac{dx}{3 - 5\sin x}$

60) Find the area under the curve $y = \dfrac{1}{1+\sin x}$ between $x = 0$ and $x = \pi$. (Assume the dimensions are in inches.)

Answer

2.0 in.^{2}

61) Given $\tan\left(\frac{x}{2}\right) = t$, derive the formulas $dx = \dfrac{2}{1+t^2}dt$, $\sin x = \dfrac{2t}{1+t^2}$, and $\cos x = \dfrac{1-t^2}{1+t^2}$.

62) Evaluate $\displaystyle\int \frac{\sqrt[3]{x-8}}{x}\,dx$.

Answer

$$\int \frac{\sqrt[3]{x-8}}{x}\,dx = 3(-8+x)^{1/3} - 2\sqrt{3}\arctan\left[\frac{-1+(-8+x)^{1/3}}{\sqrt{3}}\right] - 2\ln\left[2+(-8+x)^{1/3}\right] + \ln\left[4-2(-8+x)^{1/3}+(-8+x)^{2/3}\right] + C$$

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7.5: Other Strategies for Integration

Learning Objectives

- Use a table of integrals to solve integration problems.
- Use a computer algebra system (CAS) to solve integration problems.

In addition to the techniques of integration we have already seen, several other tools are widely available to assist with the process of integration. Among these tools are **integration tables**, which are readily available in many books, including the appendices to this one. Also widely available are **computer algebra systems (CAS)**, which are found on calculators and in many campus computer labs, and are free online.

Tables of Integrals

Integration tables, if used in the right manner, can be a handy way either to evaluate or check an integral quickly. Keep in mind that when using a table to check an answer, it is possible for two completely correct solutions to look very different. For example, in Trigonometric Substitution, we found that, by using the substitution $x = \tan\theta$, we can arrive at

$$\int \frac{dx}{\sqrt{1+x^2}} = \ln\left|x+\sqrt{x^2+1}\right| + C.$$

However, using $x = \sinh\theta$, we obtained a different solution—namely,

$$\int \frac{dx}{\sqrt{1+x^2}} = \sinh^{-1} x + C.$$

We later showed algebraically that the two solutions are equivalent. That is, we showed that $\sinh^{-1} x = \ln\left|x+\sqrt{x^2+1}\right|$. In this case, the two antiderivatives that we found were actually equal. This need not be the case. However, as long as the difference in the two antiderivatives is a constant, they are equivalent.

Example 7.5.1: Using a Formula from a Table to Evaluate an Integral

Use the table formula

$$\int \frac{\sqrt{a^2-u^2}}{u^2}du = -\frac{\sqrt{a^2-u^2}}{u} - \sin^{-1}\frac{u}{a} + C$$

to evaluate $\displaystyle\int \frac{\sqrt{16-e^{2x}}}{e^x}dx$.

Solution

If we look at integration tables, we see that several formulas contain expressions of the form $\sqrt{a^2-u^2}$. This expression is actually similar to $\sqrt{16-e^{2x}}$, where $a=4$ and $u=e^x$. Keep in mind that we must also have $du = e^x$. Multiplying the numerator and the denominator of the given integral by e^x should help to put this integral in a useful form. Thus, we now have

$$\int \frac{\sqrt{16-e^{2x}}}{e^x}dx = \int \frac{\sqrt{16-e^{2x}}}{e^{2x}}e^x dx.$$

Substituting $u = e^x$ and $du = e^x\,dx$ produces $\displaystyle\int \frac{\sqrt{a^2-u^2}}{u^2}du$. From the integration table (#88 in Appendix A),

$$\int \frac{\sqrt{a^2-u^2}}{u^2}du = -\frac{\sqrt{a^2-u^2}}{u} - \sin^{-1}\frac{u}{a} + C.$$

Thus,

$$\begin{aligned}\int \frac{\sqrt{16-e^{2x}}}{e^x}dx &= \int \frac{\sqrt{16-e^{2x}}}{e^{2x}}e^x dx \text{ Substitute } u=e^x \text{ and } du = e^x dx.\\ &= \int \frac{\sqrt{4^2-u^2}}{u^2}du \text{ Apply the formula using } a=4.\\ &= -\frac{\sqrt{4^2-u^2}}{u} - \sin^{-1}\frac{u}{4} + C \text{ Substitute } u=e^x.\end{aligned}$$

$$= -\frac{\sqrt{16 - e^{2x}}}{e^x} - \sin^{-1}\left(\frac{e^x}{4}\right) + C$$

Computer Algebra Systems

If available, a CAS is a faster alternative to a table for solving an integration problem. Many such systems are widely available and are, in general, quite easy to use.

Example 7.5.2: Using a Computer Algebra System to Evaluate an Integral

Use a computer algebra system to evaluate $\int \frac{dx}{\sqrt{x^2-4}}$. Compare this result with $\ln\left|\frac{\sqrt{x^2-4}}{2} + \frac{x}{2}\right| + C$, a result we might have obtained if we had used trigonometric substitution.

Solution

Using Wolfram Alpha, we obtain

$$\int \frac{dx}{\sqrt{x^2-4}} = \ln\left|\sqrt{x^2-4} + x\right| + C.$$

Notice that

$$\ln\left|\frac{\sqrt{x^2-4}}{2} + \frac{x}{2}\right| + C = \ln\left|\frac{\sqrt{x^2-4}+x}{2}\right| + C = \ln\left|\sqrt{x^2-4} + x\right| - \ln 2 + C.$$

Since these two antiderivatives differ by only a constant, the solutions are equivalent. We could have also demonstrated that each of these antiderivatives is correct by differentiating them.

You can access an integral calculator for more examples.

Example 7.5.3: Using a CAS to Evaluate an Integral

Evaluate $\int \sin^3 x\, dx$ using a CAS. Compare the result to $\frac{1}{3}\cos^3 x - \cos x + C$, the result we might have obtained using the technique for integrating odd powers of $\sin x$ discussed earlier in this chapter.

Solution

Using Wolfram Alpha, we obtain

$$\int \sin^3 x\, dx = \frac{1}{12}(\cos(3x) - 9\cos x) + C.$$

This looks quite different from $\frac{1}{3}\cos^3 x - \cos x + C$. To see that these antiderivatives are equivalent, we can make use of a few trigonometric identities:

$$\begin{aligned}
\frac{1}{12}(\cos(3x) - 9\cos x) &= \frac{1}{12}(\cos(x+2x) - 9\cos x) \\
&= \frac{1}{12}(\cos(x)\cos(2x) - \sin(x)\sin(2x) - 9\cos x) \\
&= \frac{1}{12}(\cos x(2\cos^2 x - 1) - \sin x(2\sin x\cos x) - 9\cos x) \\
&= \frac{1}{12}(2\cos^3 x - \cos x - 2\cos x(1 - \cos^2 x) - 9\cos x) \\
&= \frac{1}{12}(4\cos^3 x - 12\cos x) \\
&= \frac{1}{3}\cos^3 x - \cos x.
\end{aligned}$$

Thus, the two antiderivatives are identical.

We may also use a CAS to compare the graphs of the two functions, as shown in the following figure.

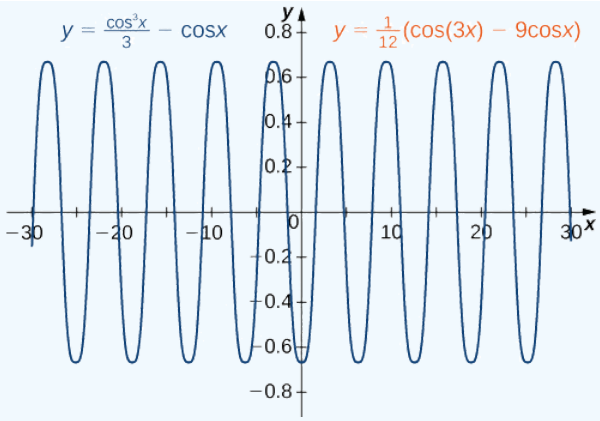


Figure 7.5.1: The graphs of $y = \frac{1}{3}\cos^3 x - \cos x$ and $y = \frac{1}{12}(\cos(3x) - 9\cos x)$ are identical.

? Exercise 7.5.1

Use a CAS to evaluate $\int \frac{dx}{\sqrt{x^2+4}}$.

Hint

Answers may vary.

Answer

Possible solutions include $\sinh^{-1}\left(\frac{x}{2}\right) + C$ and $\ln\left|\sqrt{x^2+4} + x\right| + C$.

Key Concepts

- An integration table may be used to evaluate indefinite integrals.
- A CAS (or computer algebra system) may be used to evaluate indefinite integrals.
- It may require some effort to reconcile equivalent solutions obtained using different methods.

Glossary

computer algebra system (CAS)

technology used to perform many mathematical tasks, including integration

integration table

a table that lists integration formulas

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7.5E: Exercises for Section 7.5

Use a table of integrals to evaluate the following integrals.

1) $\int_0^4 \frac{x}{\sqrt{1+2x}}\,dx$

2) $\int \frac{x+3}{x^2+2x+2}\,dx$

Answer

$$\int \frac{x+3}{x^2+2x+2}\,dx = \tfrac{1}{2}\ln|x^2+2x+2| + 2\arctan(x+1) + C$$

3) $\int x^3\sqrt{1+2x^2}\,dx$

4) $\int \frac{1}{\sqrt{x^2+6x}}\,dx$

Answer

$$\int \frac{1}{\sqrt{x^2+6x}}\,dx = \cosh^{-1}\left(\frac{x+3}{3}\right) + C$$

5) $\int \frac{x}{x+1}\,dx$

6) $\int x\cdot 2^{x^2}\,dx$

Answer

$$\int x\cdot 2^{x^2}\,dx = \frac{2^{x^2-1}}{\ln 2} + C$$

7) $\int \frac{1}{4x^2+25}\,dx$

8) $\int \frac{dy}{\sqrt{4-y^2}}$

Answer

$$\int \frac{dy}{\sqrt{4-y^2}} = \arcsin\left(\frac{y}{2}\right) + C$$

9) $\int \sin^3(2x)\cos(2x)\,dx$

10) $\int \csc(2w)\cot(2w)\,dw$

Answer

$$\int \csc(2w)\cot(2w)\,dw = -\tfrac{1}{2}\csc(2w) + C$$

11) $\int 2^y\,dy$

12) $\int_0^1 \frac{3x}{\sqrt{x^2+8}}\,dx$

Answer

$$\int_0^1 \frac{3x}{\sqrt{x^2+8}}\,dx = 9 - 6\sqrt{2}$$

13) $\int_{-1/4}^{1/4} \sec^2(\pi x)\tan(\pi x)\,dx$

14) $\int_0^{\pi/2} \tan^2\left(\frac{x}{2}\right)\,dx$

Answer

$$\int_0^{\pi/2} \tan^2\left(\frac{x}{2}\right)\,dx = 2 - \frac{\pi}{2}$$

15) $\int \cos^3 x\,dx$

16) $\int \tan^5(3x)\,dx$

Answer

$$\int \tan^5(3x)\,dx = \tfrac{1}{12}\tan^4(3x) - \tfrac{1}{6}\tan^2(3x) + \tfrac{1}{3}\ln|\sec 3x| + C$$

17) $\int \sin^2 y \cos^3 y\,dy$

Use a CAS to evaluate the following integrals. Tables can also be used to verify the answers.

18) [T] $\int \frac{dw}{1+\sec\left(\frac{w}{2}\right)}$

Answer

$$\int \frac{dw}{1+\sec\left(\frac{w}{2}\right)} = 2\cot\left(\tfrac{w}{2}\right) - 2\csc\left(\tfrac{w}{2}\right) + w + C$$

19) [T] $\int \frac{dw}{1-\cos(7w)}$

20) [T] $\int_0^t \frac{dt}{4\cos t + 3\sin t}$

Answer

$$\int_0^t \frac{dt}{4\cos t + 3\sin t} = \tfrac{1}{5}\ln\left|\frac{2(5+4\sin t - 3\cos t)}{4\cos t + 3\sin t}\right|$$

21) [T] $\int \frac{\sqrt{x^2-9}}{3x}\,dx$

22) [T] $\int \frac{dx}{x^{1/2}+x^{1/3}}$

Answer

$$\int \frac{dx}{x^{1/2}+x^{1/3}} = 6x^{1/6} - 3x^{1/3} + 2\sqrt{x} - 6\ln[1+x^{1/6}] + C$$

23) [T] $\int \frac{dx}{x\sqrt{x-1}}$

24) [T] $\int x^3 \sin x\,dx$

Answer

$$\int x^3 \sin x\,dx = -x^3\cos x + 3x^2\sin x + 6x\cos x - 6\sin x + C$$

25) [T] $\int x\sqrt{x^4-9}\,dx$

26) [T] $\int \frac{x}{1+e^{-x^2}}\,dx$

Answer

$$\int \frac{x}{1+e^{-x^2}}\,dx = \tfrac{1}{2}\left(x^2 + \ln|1+e^{-x^2}|\right) + C$$

27) [T] $\int \frac{\sqrt{3-5x}}{2x}\,dx$

28) [T] $\int \frac{dx}{x\sqrt{x-1}}$

Answer

$$\int \frac{dx}{x\sqrt{x-1}} = 2\arctan\left(\sqrt{x-1}\right) + C$$

29) [T] $\int e^x \cos^{-1}(e^x)\,dx$

Use a calculator or CAS to evaluate the following integrals.

30) [T] $\int_0^{\pi/4} \cos 2x\,dx$

Answer

$$\int_0^{\pi/4} \cos 2x\,dx = 0.5 = \frac{1}{2}$$

31) [T] $\int_0^1 x \cdot e^{-x^2}\,dx$

32) [T] $\int_0^8 \frac{2x}{\sqrt{x^2+36}}\,dx$

Answer

$$\int_0^8 \frac{2x}{\sqrt{x^2+36}}\,dx = 8.0$$

33) [T] $\int_0^{2/\sqrt{3}} \frac{1}{4+9x^2}\,dx$

34) [T] $\int \frac{dx}{x^2+4x+13}$

Answer

$$\int \frac{dx}{x^2+4x+13} = \tfrac{1}{3}\arctan\left(\tfrac{1}{3}(x+2)\right) + C$$

35) [T] $\int \frac{dx}{1+\sin x}$

Use tables to evaluate the integrals. You may need to complete the square or change variables to put the integral into a form given in the table.

36) $\int \frac{dx}{x^2+2x+10}$

Answer

$$\int \frac{dx}{x^2+2x+10} = \tfrac{1}{3}\arctan\left(\frac{x+1}{3}\right) + C$$

37) $\int \frac{dx}{\sqrt{x^2-6x}}$

38) $\int \frac{e^x}{\sqrt{e^{2x}-4}}\,dx$

Answer

$$\int \frac{e^x}{\sqrt{e^{2x}-4}}\,dx = \ln\left(e^x + \sqrt{4+e^{2x}}\right) + C$$

39) $\int \frac{\cos x}{\sin^2 x + 2\sin x}\,dx$

40) $\int \frac{\arctan(x^3)}{x^4}\,dx$

Answer

$$\int \frac{\arctan(x^3)}{x^4}\,dx = \ln x - \tfrac{1}{6}\ln(x^6+1) - \frac{\arctan(x^3)}{3x^3} + C$$

41) $\int \frac{\ln|x|\arcsin(\ln|x|)}{x}\,dx$

Use tables to perform the integration.

42) $\int \frac{dx}{\sqrt{x^2+16}}$

Answer

$$\int \frac{dx}{\sqrt{x^2+16}} = \ln|x| + \sqrt{16+x^2}\,| + C$$

43) $\int \frac{3x}{2x+7}\,dx$

44) $\int \frac{dx}{1-\cos 4x}$

Answer

$$\int \frac{dx}{1-\cos 4x} = -\frac{1}{4}\cot 2x + C$$

45) $\int \frac{dx}{\sqrt{4x+1}}$

46) Find the area bounded by $y(4+25x^2)=5,\ x=0,\ y=0$, and $x=4$. Use a table of integrals or a CAS.

Answer

$\frac{1}{2}\arctan 10$ units^2

47) The region bounded between the curve $y=\frac{1}{\sqrt{1+\cos x}},\ 0.3\le x\le 1.1$, and the x-axis is revolved about the x-axis to generate a solid. Use a table of integrals to find the volume of the solid generated. (Round the answer to two decimal places.)

48) Use substitution and a table of integrals to find the area of the surface generated by revolving the curve $y=e^x,\ 0\le x\le 3$, about the x-axis. (Round the answer to two decimal places.)

Answer

1276.14 units^2

49) [T] Use an integral table and a calculator to find the area of the surface generated by revolving the curve $y=\frac{x^2}{2},\ 0\le x\le 1$, about the x-axis. (Round the answer to two decimal places.)

50) [T] Use a CAS or tables to find the area of the surface generated by revolving the curve $y=\cos x,\ 0\le x\le \frac{\pi}{2}$, about the x-axis. (Round the answer to two decimal places.)

Answer

7.21 units^2

51) Find the length of the curve $y=\frac{x^2}{4}$ over $[0,8]$.

52) Find the length of the curve $y=e^x$ over $[0,\ \ln(2)]$.

Answer

$\left(\sqrt{5}-\sqrt{2}+\ln\left|\frac{2+2\sqrt{2}}{1+\sqrt{5}}\right|\right)$ units

53) Find the area of the surface formed by revolving the graph of $y = 2\sqrt{x}$ over the interval $[0, 9]$ about the x-axis.

54) Find the average value of the function $f(x) = \dfrac{1}{x^2 + 1}$ over the interval $[-3, 3]$.

Answer

$\frac{1}{3}\arctan(3) \approx 0.416$

55) Approximate the arc length of the curve $y = \tan \pi x$ over the interval $\left[0, \frac{1}{4}\right]$. (Round the answer to three decimal places.)

Contributors

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7.6: Numerical Integration

Learning Objectives

- Approximate the value of a definite integral by using the midpoint and trapezoidal rules.
- Determine the absolute and relative error in using a numerical integration technique.
- Estimate the absolute and relative error using an error-bound formula.
- Recognize when the midpoint and trapezoidal rules over- or underestimate the true value of an integral.
- Use Simpson's rule to approximate the value of a definite integral to a given accuracy.

The antiderivatives of many functions either cannot be expressed or cannot be expressed easily in closed form (that is, in terms of known functions). Consequently, rather than evaluate definite integrals of these functions directly, we resort to various techniques of **numerical integration** to approximate their values. In this section, we explore several of these techniques. In addition, we examine the process of estimating the error in using these techniques.

The Midpoint Rule

Earlier in this text we defined the definite integral of a function over an interval as the limit of **Riemann sums**. In general, any Riemann sum of a function $f(x)$ over an interval $[a, b]$ may be viewed as an estimate of $\int_a^b f(x)\,dx$. Recall that a Riemann sum of a function $f(x)$ over an interval $[a, b]$ is obtained by selecting a partition

$$P = \{x_0, x_1, x_2, \ldots, x_n\}$$

where $a = x_0 < x_1 < x_2 < \cdots < x_n = b$

and a set

$$S = \{x_1^*, x_2^*, \ldots, x_n^*\}$$

where $x_{i-1} \le x_i^* \le x_i \quad \text{for all } i.$

The Riemann sum corresponding to the partition P and the set S is given by $\sum_{i=1}^{n} f(x_i^*)\Delta x_i$, where $\Delta x_i = x_i - x_{i-1}$, the length of the ith subinterval.

The midpoint rule for estimating a definite integral uses a Riemann sum with subintervals of equal width and the midpoints, m_i, of each subinterval in place of x_i^*. Formally, we state a theorem regarding the convergence of the midpoint rule as follows.

The Midpoint Rule

Assume that $f(x)$ is continuous on $[a, b]$. Let n be a positive integer and $\Delta x = \dfrac{b-a}{n}$. If $[a, b]$ is divided into n subintervals, each of length Δx, and m_i is the midpoint of the ith subinterval, set

$$M_n = \sum_{i=1}^{n} f(m_i)\Delta x.$$

Then $\lim_{n\to\infty} M_n = \int_a^b f(x)\,dx.$

As we can see in Figure 7.6.1, if $f(x) \ge 0$ over $[a, b]$, then $\sum_{i=1}^{n} f(m_i)\Delta x$ corresponds to the sum of the areas of rectangles approximating the area between the graph of $f(x)$ and the x-axis over $[a, b]$. The graph shows the rectangles corresponding to M_4 for a nonnegative function over a closed interval $[a, b]$.

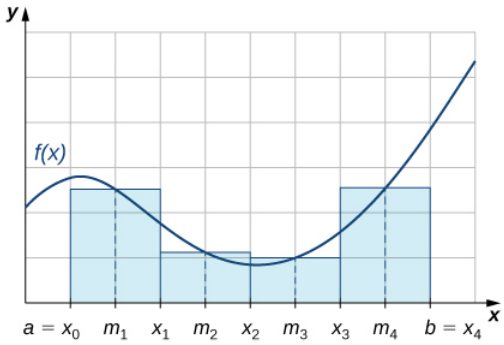


Figure 7.6.1: The midpoint rule approximates the area between the graph of $f(x)$ and the x-axis by summing the areas of rectangles with midpoints that are points on $f(x)$.

Example 7.6.1: Using the Midpoint Rule with M_4

Use the midpoint rule to estimate $\int_0^1 x^2\,dx$ using four subintervals. Compare the result with the actual value of this integral.

Solution: Each subinterval has length $\Delta x = \dfrac{1-0}{4} = \dfrac{1}{4}$. Therefore, the subintervals consist of

$$\left[0, \tfrac{1}{4}\right],\ \left[\tfrac{1}{4}, \tfrac{1}{2}\right],\ \left[\tfrac{1}{2}, \tfrac{3}{4}\right],\ \text{and}\ \left[\tfrac{3}{4}, 1\right].$$

The midpoints of these subintervals are $\left\{\frac{1}{8}, \frac{3}{8}, \frac{5}{8}, \frac{7}{8}\right\}$. Thus,

$$\begin{aligned} M_4 &= \frac{1}{4}\cdot f\left(\frac{1}{8}\right) + \frac{1}{4}\cdot f\left(\frac{3}{8}\right) + \frac{1}{4}\cdot f\left(\frac{5}{8}\right) + \frac{1}{4}\cdot f\left(\frac{7}{8}\right) \\ &= \frac{1}{4}\cdot\frac{1}{64} + \frac{1}{4}\cdot\frac{9}{64} + \frac{1}{4}\cdot\frac{25}{64} + \frac{1}{4}\cdot\frac{49}{64} \\ &= \frac{21}{64} = 0.328125. \end{aligned}$$

Since

$$\int_0^1 x^2\,dx = \frac{1}{3},$$

the absolute error in this approximation is:

$$\left|\frac{1}{3} - \frac{21}{64}\right| = \frac{1}{192} \approx 0.0052,$$

and we see that the midpoint rule produces an estimate that is somewhat close to the actual value of the definite integral.

Example 7.6.2: Using the Midpoint Rule with M_6

Use M_6 to estimate the length of the curve $y = \frac{1}{2}x^2$ on $[1, 4]$.

Solution: The length of $y = \frac{1}{2}x^2$ on $[1, 4]$ is

$$s = \int_1^4 \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx.$$

Since $\dfrac{dy}{dx} = x$, this integral becomes $\int_1^4 \sqrt{1+x^2}\,dx$.

If $[1, 4]$ is divided into six subintervals, then each subinterval has length $\Delta x = \dfrac{4-1}{6} = \dfrac{1}{2}$ and the midpoints of the subintervals are $\left\{\frac{5}{4}, \frac{7}{4}, \frac{9}{4}, \frac{11}{4}, \frac{13}{4}, \frac{15}{4}\right\}$. If we set $f(x) = \sqrt{1+x^2}$,

$$M_6 = \tfrac{1}{2}\cdot f\left(\frac{5}{4}\right) + \tfrac{1}{2}\cdot f\left(\frac{7}{4}\right) + \frac{1}{2}\cdot f\left(\frac{9}{4}\right) + \frac{1}{2}\cdot f\left(\frac{11}{4}\right) + \frac{1}{2}\cdot f\left(\frac{13}{4}\right) + \frac{1}{2}\cdot f\left(\frac{15}{4}\right)$$
$$\approx \frac{1}{2}(1.6008 + 2.0156 + 2.4622 + 2.9262 + 3.4004 + 3.8810) = 8.1431 \text{ units.}$$

? Exercise 7.6.1

Use the midpoint rule with $n = 2$ to estimate $\int_1^2 \frac{1}{x}\,dx$.

Hint

$\Delta x = \frac{1}{2}$, $m_1 = \frac{5}{4}$, and $m_2 = \frac{7}{4}$.

Answer

$\frac{24}{35} \approx 0.685714$

The Trapezoidal Rule

We can also approximate the value of a definite integral by using trapezoids rather than rectangles. In Figure 7.6.2, the area beneath the curve is approximated by trapezoids rather than by rectangles.

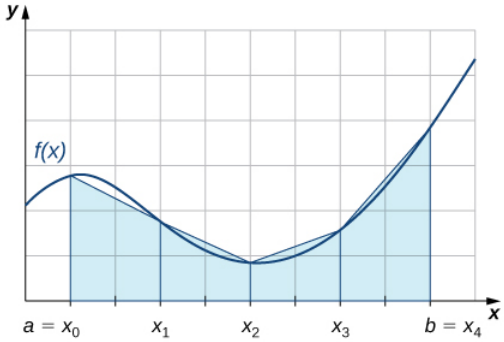


Figure 7.6.2: Trapezoids may be used to approximate the area under a curve, hence approximating the definite integral.

The **trapezoidal rule** for estimating definite integrals uses trapezoids rather than rectangles to approximate the area under a curve. To gain insight into the final form of the rule, consider the trapezoids shown in Figure 7.6.2. We assume that the length of each subinterval is given by Δx. First, recall that the area of a trapezoid with a height of h and bases of length b_1 and b_2 is given by $\text{Area} = \frac{1}{2}h(b_1 + b_2)$. We see that the first trapezoid has a height Δx and parallel bases of length $f(x_0)$ and $f(x_1)$. Thus, the area of the first trapezoid in Figure 7.6.2 is

$$\frac{1}{2}\Delta x\Big(f(x_0) + f(x_1)\Big).$$

The areas of the remaining three trapezoids are

$$\frac{1}{2}\Delta x\Big(f(x_1) + f(x_2)\Big),\ \frac{1}{2}\Delta x\Big(f(x_2) + f(x_3)\Big), \text{ and } \frac{1}{2}\Delta x\Big(f(x_3) + f(x_4)\Big).$$

Consequently,

$$\int_a^b f(x)\,dx$$
$$\approx \frac{1}{2}\Delta x\Big(f(x_0) + f(x_1)\Big) + \frac{1}{2}\Delta x\Big(f(x_1) + f(x_2)\Big) + \frac{1}{2}\Delta x\Big(f(x_2) + f(x_3)\Big) + \frac{1}{2}\Delta x\Big(f(x_3) + f(x_4)\Big).$$

After taking out a common factor of $\frac{1}{2}\Delta x$ and combining like terms, we have

$$\int_a^b f(x)\,dx \approx \frac{\Delta x}{2}\Big[f(x_0) + 2\,f(x_1) + 2\,f(x_2) + 2\,f(x_3) + f(x_4)\Big].$$

Generalizing, we formally state the following rule.

The Trapezoidal Rule

Assume that $f(x)$ is continuous over $[a,b]$. Let n be a positive integer and $\Delta x = \dfrac{b-a}{n}$. Let $[a,b]$ be divided into n subintervals, each of length Δx, with endpoints at $P = \{x_0, x_1, x_2 \ldots, x_n\}$.

Set

$$T_n = \frac{\Delta x}{2}\Big[f(x_0) + 2\,f(x_1) + 2\,f(x_2) + \cdots + 2\,f(x_{n-1}) + f(x_n)\Big].$$

Then, $\lim\limits_{n\to+\infty} T_n = \displaystyle\int_a^b f(x)\,dx.$

Before continuing, let's make a few observations about the trapezoidal rule. First of all, it is useful to note that

$$T_n = \frac{1}{2}(L_n + R_n) \text{ where } L_n = \sum_{i=1}^{n} f(x_{i-1})\Delta x \text{ and } R_n = \sum_{i=1}^{n} f(x_i)\Delta x.$$

That is, L_n and R_n approximate the integral using the left-hand and right-hand endpoints of each subinterval, respectively. In addition, a careful examination of Figure 7.6.3 leads us to make the following observations about using the trapezoidal rules and midpoint rules to estimate the definite integral of a nonnegative function. The trapezoidal rule tends to overestimate the value of a definite integral systematically over intervals where the function is concave up and to underestimate the value of a definite integral systematically over intervals where the function is concave down. On the other hand, the midpoint rule tends to average out these errors somewhat by partially overestimating and partially underestimating the value of the definite integral over these same types of intervals. This leads us to hypothesize that, in general, the midpoint rule tends to be more accurate than the trapezoidal rule.

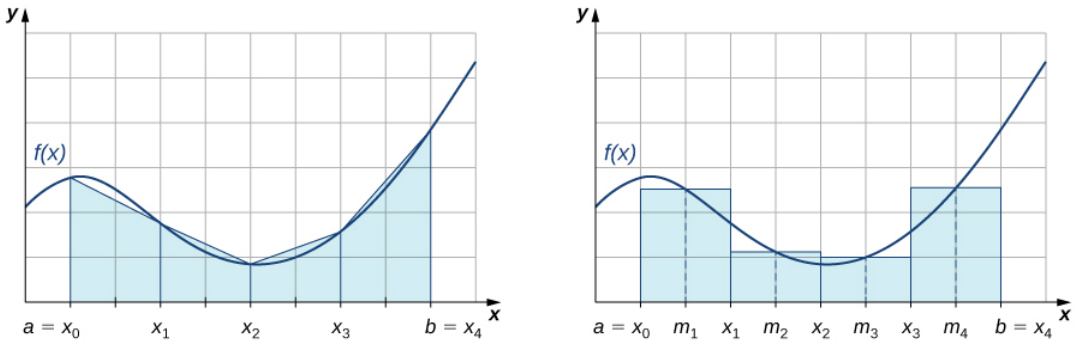


Figure 7.6.3:The trapezoidal rule tends to be less accurate than the midpoint rule.

Example 7.6.3: Using the Trapezoidal Rule

Use the trapezoidal rule to estimate $\displaystyle\int_0^1 x^2\,dx$ using four subintervals.

Solution

The endpoints of the subintervals consist of elements of the set $P = \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$ and $\Delta x = \frac{1-0}{4} = \frac{1}{4}$. Thus,

$$\begin{aligned}\int_0^1 x^2 dx &\approx \frac{1}{2}\cdot\frac{1}{4}\Big[f(0) + 2\,f\left(\tfrac{1}{4}\right) + 2\,f\left(\tfrac{1}{2}\right) + 2\,f\left(\tfrac{3}{4}\right) + f(1)\Big] \\ &= \tfrac{1}{8}\Big(0 + 2\cdot\tfrac{1}{16} + 2\cdot\tfrac{1}{4} + 2\cdot\tfrac{9}{16} + 1\Big) \\ &= \frac{11}{32} = 0.34375\end{aligned}$$

Exercise 7.6.2

Use the trapezoidal rule with $n = 2$ to estimate $\displaystyle\int_1^2 \frac{1}{x}\,dx.$

Hint

Set $\Delta x = \frac{1}{2}$. The endpoints of the subintervals are the elements of the set $P = \left\{1, \frac{3}{2}, 2\right\}$.

Answer

$\frac{17}{24} \approx 0.708333$

Absolute and Relative Error

An important aspect of using these numerical approximation rules consists of calculating the error in using them for estimating the value of a definite integral. We first need to define absolute error and relative error.

Definition: absolute and relative error

If B is our estimate of some quantity having an actual value of A, then the **absolute error** is given by $|A - B|$.

The **relative error** is the error as a percentage of the actual value and is given by

$$\left|\frac{A-B}{A}\right| \cdot 100\%.$$

Example 7.6.4: Calculating Error in the Midpoint Rule

Calculate the absolute and relative error in the estimate of $\int_0^1 x^2\,dx$ using the midpoint rule, found in Example 7.6.1.

Solution: The calculated value is $\int_0^1 x^2\,dx = \frac{1}{3}$ and our estimate from the example is $M_4 = \frac{21}{64}$. Thus, the absolute error is given by $\left|\frac{1}{3} - \frac{21}{64}\right| = \frac{1}{192} \approx 0.0052$.

The relative error is

$$\frac{1/192}{1/3} = \frac{1}{64} \approx 0.015625 \approx 1.6\%.$$

Example 7.6.5: Calculating Error in the Trapezoidal Rule

Calculate the absolute and relative error in the estimate of $\int_0^1 x^2\,dx$ using the trapezoidal rule, found in Example 7.6.3.

Solution: The calculated value is $\int_0^1 x^2\,dx = \frac{1}{3}$ and our estimate from the example is $T_4 = \frac{11}{32}$. Thus, the absolute error is given by $\left|\frac{1}{3} - \frac{11}{32}\right| = \frac{1}{96} \approx 0.0104$.

The relative error is given by

$$\frac{1/96}{1/3} = 0.03125 \approx 3.1\%.$$

Exercise 7.6.3

In an earlier checkpoint, we estimated $\int_1^2 \frac{1}{x}\,dx$ to be $\frac{24}{35}$ using M_2. The actual value of this integral is $\ln 2$. Using $\frac{24}{35} \approx 0.6857$ and $\ln 2 \approx 0.6931$, calculate the absolute error and the relative error.

Hint

Use the previous examples as a guide.

Answer

absolute error ≈ 0.0074, and relative error $\approx 1.1\%$

Error Bounds on the Midpoint and Trapezoidal Rules

In the two previous examples, we were able to compare our estimate of an integral with the actual value of the integral; however, we do not typically have this luxury. In general, if we are approximating an integral, we are doing so because we cannot compute the exact value of the integral itself easily. Therefore, it is often helpful to be able to determine an upper bound for the error in an approximation of an integral. The following theorem provides error bounds for the midpoint and trapezoidal rules. The theorem is stated without proof.

Error Bounds for the Midpoint and Trapezoidal Rules

Let $f(x)$ be a continuous function over $[a,b]$, having a second derivative $f''(x)$ over this interval. If M is the maximum value of $|f''(x)|$ over $[a,b]$, then the upper bounds for the error in using M_n and T_n to estimate $\int_a^b f(x)\,dx$ are

$$\text{Error in } M_n \le \frac{M(b-a)^3}{24n^2} \tag{7.6.1}$$

and

$$\text{Error in } T_n \le \frac{M(b-a)^3}{12n^2}$$

.

We can use these bounds to determine the value of n necessary to guarantee that the error in an estimate is less than a specified value.

Example 7.6.6: Determining the Number of Intervals to Use

What value of n should be used to guarantee that an estimate of $\int_0^1 e^{x^2}\,dx$ is accurate to within 0.01 if we use the midpoint rule?

Solution

We begin by determining the value of M, the maximum value of $|f''(x)|$ over $[0,1]$ for $f(x)=e^{x^2}$. Since $f'(x)=2xe^{x^2}$, we have

$$f''(x)=2e^{x^2}+4x^2e^{x^2}.$$

Thus,

$$|f''(x)|=2e^{x^2}(1+2x^2)\le 2\cdot e\cdot 3=6e.$$

From the error-bound Equation 7.6.1, we have

$$\text{Error in } M_n \le \frac{M(b-a)^3}{24n^2} \le \frac{6e(1-0)^3}{24n^2}=\frac{6e}{24n^2}.$$

Now we solve the following inequality for n:

$$\frac{6e}{24n^2}\le 0.01.$$

Thus, $n \ge \sqrt{\frac{600e}{24}} \approx 8.24$. Since n must be an integer satisfying this inequality, a choice of $n=9$ would guarantee that

$$\left|\int_0^1 e^{x^2}\,dx - M_n\right| < 0.01.$$

Analysis

We might have been tempted to round 8.24 down and choose $n=8$, but this would be incorrect because we must have an integer greater than or equal to 8.24. We need to keep in mind that the error estimates provide an upper bound only for the error. The actual estimate may, in fact, be a much better approximation than is indicated by the error bound.

Exercise 7.6.4

Use Equation 7.6.1 to find an upper bound for the error in using M_4 to estimate $\int_0^1 x^2\,dx$.

Hint

$f''(x) = 2$, so $M = 2$.

Answer

$\frac{1}{192}$

Simpson's Rule

With the midpoint rule, we estimated areas of regions under curves by using rectangles. In a sense, we approximated the curve with piecewise constant functions. With the trapezoidal rule, we approximated the curve by using piecewise linear functions. What if we were, instead, to approximate a curve using piecewise quadratic functions? With **Simpson's rule,** we do just this. We partition the interval into an even number of subintervals, each of equal width. Over the first pair of subintervals we approximate $\int_{x_0}^{x_2} f(x)\,dx$ with $\int_{x_0}^{x_2} p(x)\,dx$, where $p(x) = Ax^2 + Bx + C$ is the quadratic function passing through $(x_0, f(x_0))$, $(x_1, f(x_1))$, and $(x_2, f(x_2))$ (Figure 7.6.4). Over the next pair of subintervals we approximate $\int_{x_2}^{x_4} f(x)\,dx$ with the integral of another quadratic function passing through $(x_2, f(x_2))$, $(x_3, f(x_3))$, and $(x_4, f(x_4))$. This process is continued with each successive pair of subintervals.

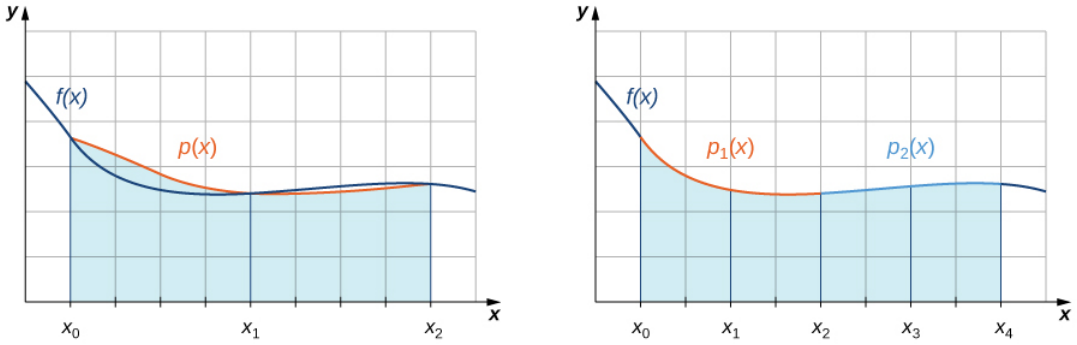


Figure 7.6.4: With Simpson's rule, we approximate a definite integral by integrating a piecewise quadratic function.

To understand the formula that we obtain for Simpson's rule, we begin by deriving a formula for this approximation over the first two subintervals. As we go through the derivation, we need to keep in mind the following relationships:

$$f(x_0) = p(x_0) = Ax_0^2 + Bx_0 + C$$

$$f(x_1) = p(x_1) = Ax_1^2 + Bx_1 + C$$

$$f(x_2) = p(x_2) = Ax_2^2 + Bx_2 + C$$

$x_2 - x_0 = 2\Delta x$, where Δx is the length of a subinterval.

$$x_2 + x_0 = 2x_1, \text{ since } x_1 = \frac{(x_2 + x_0)}{2}.$$

Thus,

$$
\begin{aligned}
\int_{x_0}^{x_2} f(x)\,dx &\approx \int_{x_0}^{x_2} p(x)\,dx \\
&= \int_{x_0}^{x_2} (Ax^2 + Bx + C)\,dx \\
&= \left(\frac{A}{3}x^3 + \frac{B}{2}x^2 + Cx\right)\Big|_{x_0}^{x_2} && \text{Find the antiderivative.} \\
&= \frac{A}{3}(x_2^3 - x_0^3) + \frac{B}{2}(x_2^2 - x_0^2) + C(x_2 - x_0) && \text{Evaluate the antiderivative.} \\
&= \frac{A}{3}(x_2 - x_0)(x_2^2 + x_2x_0 + x_0^2) \\
&\quad + \frac{B}{2}(x_2 - x_0)(x_2 + x_0) + C(x_2 - x_0) \\
&= \frac{x_2 - x_0}{6} && \text{Factor out } \frac{x_2 - x_0}{6}. \\
&\quad \Big(2A(x_2^2 + x_2x_0 + x_0^2) + 3B(x_2 + x_0) + 6C\Big) \\
&= \frac{\Delta x}{3}\Big((Ax_2^2 + Bx_2 + C) + (Ax_0^2 + Bx_0 + C) && \text{Rearrange the terms. Note:} \\
&\quad + A(x_2^2 + 2x_2x_0 + x_0^2) + 2B(x_2 + x_0) + 4C\Big) && \Delta x = \frac{x_2 - x_0}{2} \\
&= \frac{\Delta x}{3}\Big(f(x_2) + f(x_0) + A(x_2 + x_0)^2 && \text{Factor and substitute:} \\
&\quad + 2B(x_2 + x_0) + 4C\Big) \\
& && f(x_2) = Ax_2^2 + Bx_2 + C \text{ and} \\
& && f(x_0) = Ax_0^2 + Bx_0 + C. \\
&= \frac{\Delta x}{3}\Big(f(x_2) + f(x_0) + A(2x_1)^2 + 2B(2x_1) + 4C\Big) && \text{Substitute } x_2 + x_0 = 2x_1. \\
& && \text{Note: } x_1 = \frac{x_2 + x_0}{2} \text{ is} \\
& && \text{the midpoint.} \\
&= \frac{\Delta x}{3}\Big(f(x_2) + 4f(x_1) + f(x_0)\Big). && \text{Expand and substitute} \\
& && f(x_1) = Ax_1^2 + Bx_1 + C.
\end{aligned}
$$

If we approximate $\int_{x_2}^{x_4} f(x)\,dx$ using the same method, we see that we have

$$\int_{x_2}^{x_4} f(x)\,dx \approx \frac{\Delta x}{3}(f(x_4) + 4\,f(x_3) + f(x_2)).$$

Combining these two approximations, we get

$$\int_{x_0}^{x_4} f(x)\,dx \approx \frac{\Delta x}{3}(f(x_0) + 4\,f(x_1) + 2\,f(x_2) + 4\,f(x_3) + f(x_4)).$$

The pattern continues as we add pairs of subintervals to our approximation. The general rule may be stated as follows.

Simpson's Rule

Assume that $f(x)$ is continuous over $[a, b]$. Let n be a positive even integer and $\Delta x = \frac{b-a}{n}$. Let $[a, b]$ be divided into n subintervals, each of length Δx, with endpoints at $P = \{x_0, x_1, x_2, \ldots, x_n\}$. Set

$$S_n = \frac{\Delta x}{3}\Big[f(x_0) + 4\,f(x_1) + 2\,f(x_2) + 4\,f(x_3) + 2\,f(x_4) + \cdots + 2\,f(x_{n-2}) + 4\,f(x_{n-1}) + f(x_n)\Big].$$

Then,

$$\lim_{n\to+\infty} S_n = \int_a^b f(x)\,dx.$$

Just as the trapezoidal rule is the average of the left-hand and right-hand rules for estimating definite integrals, Simpson's rule may be obtained from the midpoint and trapezoidal rules by using a weighted average. It can be shown that $S_{2n} = \left(\frac{2}{3}\right) M_n + \left(\frac{1}{3}\right) T_n$.

It is also possible to put a bound on the error when using Simpson's rule to approximate a definite integral. The bound in the error is given by the following rule:

Rule: Error Bound for Simpson's Rule

Let $f(x)$ be a continuous function over $[a, b]$ having a fourth derivative, $f^{(4)}(x)$, over this interval. If M is the maximum value of $| f^{(4)}(x) |$ over $[a, b]$, then the upper bound for the error in using S_n to estimate $\int_a^b f(x)\, dx$ is given by

$$\text{Error in } S_n \leq \frac{M(b-a)^5}{180n^4}.$$

Example 7.6.7: Applying Simpson's Rule 1

Use S_2 to approximate $\int_0^1 x^3\, dx$. Estimate a bound for the error in S_2.

Solution

Since $[0, 1]$ is divided into two intervals, each subinterval has length $\Delta x = \frac{1-0}{2} = \frac{1}{2}$. The endpoints of these subintervals are $\left\{0, \frac{1}{2}, 1\right\}$. If we set $f(x) = x^3$, then

$$S_2 = \frac{1}{3} \cdot \frac{1}{2}(f(0) + 4\, f(\frac{1}{2}) + f(1)) = \frac{1}{6}(0 + 4 \cdot \frac{1}{8} + 1) = \frac{1}{4}.$$

Since $f^{(4)}(x) = 0$ and consequently $M = 0$, we see that

$$\text{Error in } S_2 \leq \tfrac{0(1)^5}{180 \cdot 2^4} = 0.$$

This bound indicates that the value obtained through Simpson's rule is exact. A quick check will verify that, in fact, $\int_0^1 x^3\, dx = \frac{1}{4}$.

Example 7.6.8: Applying Simpson's Rule 2

Use S_6 to estimate the length of the curve $y = \frac{1}{2}x^2$ over $[1, 4]$.

Solution

The length of $y = \frac{1}{2}x^2$ over $[1, 4]$ is $\int_1^4 \sqrt{1 + x^2}\, dx$. If we divide $[1, 4]$ into six subintervals, then each subinterval has length $\Delta x = \frac{4-1}{6} = \frac{1}{2}$, and the endpoints of the subintervals are $\left\{1, \frac{3}{2}, 2, \frac{5}{2}, 3, \frac{7}{2}, 4\right\}$. Setting $f(x) = \sqrt{1 + x^2}$,

$$S_6 = \frac{1}{3} \cdot \frac{1}{2}(f(1) + 4f(\frac{3}{2}) + 2f(2) + 4f(\frac{5}{2}) + 2f(3) + 4f(\frac{7}{2}) + f(4)).$$

After substituting, we have

$$S_6 = \frac{1}{6}(1.4142 + 4 \cdot 1.80278 + 2 \cdot 2.23607 + 4 \cdot 2.69258 + 2 \cdot 3.16228 + 4 \cdot 3.64005 + 4.12311)$$
$$\approx 8.14594 \text{ units}.$$

Exercise 7.6.5

Use S_2 to estimate $\int_1^2 \frac{1}{x}\, dx$.

Hint

$$S_2 = \frac{1}{3}\Delta x\, (f(x_0) + 4f(x_1) + f(x_2))$$

Answer

$\frac{25}{36} \approx 0.694444$

Key Concepts

- We can use numerical integration to estimate the values of definite integrals when a closed form of the integral is difficult to find or when an approximate value only of the definite integral is needed.
- The most commonly used techniques for numerical integration are the midpoint rule, trapezoidal rule, and Simpson's rule.
- The midpoint rule approximates the definite integral using rectangular regions whereas the trapezoidal rule approximates the definite integral using trapezoidal approximations.
- Simpson's rule approximates the definite integral by first approximating the original function using piecewise quadratic functions.

Key Equations

- **Midpoint rule**

  $M_n = \sum_{i=1}^{n} f(m_i)\Delta x$

- **Trapezoidal rule**

  $T_n = \frac{\Delta x}{2}\left[f(x_0) + 2\,f(x_1) + 2\,f(x_2) + \cdots + 2\,f(x_{n-1}) + f(x_n)\right]$

- **Simpson's rule**

  $S_n = \frac{\Delta x}{3}\left[f(x_0) + 4\,f(x_1) + 2\,f(x_2) + 4\,f(x_3) + 2\,f(x_4) + 4\,f(x_5) + \cdots + 2\,f(x_{n-2}) + 4\,f(x_{n-1}) + f(x_n)\right]$

- **Error bound for midpoint rule**

  Error in $M_n \leq \frac{M(b-a)^3}{24n^2}$, where M is the maximum value of $|f''(x)|$ over $[a, b]$.

- **Error bound for trapezoidal rule**

  Error in $T_n \leq \frac{M(b-a)^3}{12n^2}$, where M is the maximum value of $|f''(x)|$ over $[a, b]$.

- **Error bound for Simpson's rule**

  Error in $S_n \leq \frac{M(b-a)^5}{180n^4}$, where M is the maximum value of $|\,f^{(4)}(x)\,|$ over $[a, b]$.

Glossary

absolute error

if B is an estimate of some quantity having an actual value of A, then the absolute error is given by $|A - B|$

midpoint rule

a rule that uses a Riemann sum of the form $M_n = \sum_{i=1}^{n} f(m_i)\Delta x$, where m_i is the midpoint of the ith subinterval to approximate $\int_a^b f(x)\,dx$

numerical integration

the variety of numerical methods used to estimate the value of a definite integral, including the midpoint rule, trapezoidal rule, and Simpson's rule

relative error

error as a percentage of the actual value, given by

$$\text{relative error} = \left|\frac{A-B}{A}\right| \cdot 100\%$$

Simpson's rule

a rule that approximates $\int_a^b f(x)\,dx$ using the area under a piecewise quadratic function.

The approximation S_n to $\int_a^b f(x)\,dx$ is given by

$$S_n = \frac{\Delta x}{3}\Big(f(x_0) + 4\,f(x_1) + 2\,f(x_2) + 4\,f(x_3) + 2\,f(x_4) + \cdots + 2\,f(x_{n-2}) + 4\,f(x_{n-1}) + f(x_n)\Big).$$

trapezoidal rule

a rule that approximates $\int_a^b f(x)\,dx$ using the area of trapezoids.

The approximation T_n to $\int_a^b f(x)\,dx$ is given by

$$T_n = \frac{\Delta x}{2}\Big(f(x_0) + 2\,f(x_1) + 2\,f(x_2) + \cdots + 2\,f(x_{n-1}) + f(x_n)\Big).$$

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- Edited by Paul Seeburger (Monroe Community College). Notes added to development of area under a parabola and typos fixed in original text.

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7.6E: Exercises for Section 7.6

In exercises 1 - 5, approximate the following integrals using either the midpoint rule, trapezoidal rule, or Simpson's rule as indicated. (Round answers to three decimal places.)

1) $\int_1^2 \frac{dx}{x}$; trapezoidal rule; $n = 5$

Answer

0.696

2) $\int_0^3 \sqrt{4 + x^3}\ dx$; trapezoidal rule; $n = 6$

3) $\int_0^3 \sqrt{4 + x^3}\ dx$; Simpson's rule; $n = 6$

Answer

9.279

4) $\int_0^{12} x^2\ dx$; midpoint rule; $n = 6$

5) $\int_0^1 \sin^2(\pi x)\ dx$; midpoint rule; $n = 3$

Answer

0.500

6) Use the midpoint rule with eight subdivisions to estimate $\int_2^4 x^2\ dx$.

7) Use the trapezoidal rule with four subdivisions to estimate $\int_2^4 x^2\ dx$.

Answer

$T_4 = 18.75$

8) Find the exact value of $\int_2^4 x^2\ dx$. Find the error of approximation between the exact value and the value calculated using the trapezoidal rule with four subdivisions. Draw a graph to illustrate.

Approximate the integral to four decimal places using the indicated rule.

9) $\int_0^1 \sin^2(\pi x)\ dx$; trapezoidal rule; $n = 6$

Answer

0.5000

10) $\int_0^3 \frac{1}{1 + x^3}\ dx$; trapezoidal rule; $n = 6$

11) $\int_0^3 \frac{1}{1 + x^3}\ dx$; Simpson's rule; $n = 6$

Answer

1.1614

12) $\int_0^{0.8} e^{-x^2}\ dx$; trapezoidal rule; $n = 4$

13) $\int_0^{0.8} e^{-x^2}\ dx$; Simpson's rule; $n = 4$

Answer

0.6577

14) $\int_0^{0.4} \sin(x^2)\ dx$; trapezoidal rule; $n = 4$

15) $\int_0^{0.4} \sin(x^2)\ dx$; Simpson's rule; $n = 4$

Answer

0.0213

16) $\int_{0.1}^{0.5} \frac{\cos x}{x}\ dx$; trapezoidal rule; $n = 4$

17) $\int_{0.1}^{0.5} \frac{\cos x}{x}\ dx$; Simpson's rule; $n = 4$

Answer

1.5629

18) Evaluate $\int_0^1 \frac{dx}{1+x^2}$ exactly and show that the result is $\pi/4$. Then, find the approximate value of the integral using the trapezoidal rule with $n = 4$ subdivisions. Use the result to approximate the value of π.

19) Approximate $\int_2^4 \frac{1}{\ln x}\ dx$ using the midpoint rule with four subdivisions to four decimal places.

Answer

1.9133

20) Approximate $\int_2^4 \frac{1}{\ln x}\ dx$ using the trapezoidal rule with eight subdivisions to four decimal places.

21) Use the trapezoidal rule with four subdivisions to estimate $\int_0^{0.8} x^3\ dx$ to four decimal places.

Answer

$T(4) = 0.1088$

22) Use the trapezoidal rule with four subdivisions to estimate $\int_0^{0.8} x^3\ dx$. Compare this value with the exact value and find the error estimate.

23) Using Simpson's rule with four subdivisions, find $\int_0^{\pi/2} \cos(x)\ dx$.

Answer

$\int_0^{\pi/2} \cos(x)\ dx \approx \quad 1.0$

24) Show that the exact value of $\int_0^1 xe^{-x}\ dx = 1 - \frac{2}{e}$. Find the absolute error if you approximate the integral using the midpoint rule with 16 subdivisions.

25) Given $\int_0^1 xe^{-x}\ dx = 1 - \frac{2}{e}$, use the trapezoidal rule with 16 subdivisions to approximate the integral and find the absolute error.

Answer

Approximate error is 0.000325.

26) Find an upper bound for the error in estimating $\int_0^3 (5x + 4)\ dx$ using the trapezoidal rule with six steps.

27) Find an upper bound for the error in estimating $\int_4^5 \frac{1}{(x-1)^2}\,dx$ using the trapezoidal rule with seven subdivisions.

Answer

$\frac{1}{7938}$

28) Find an upper bound for the error in estimating $\int_0^3 (6x^2-1)\,dx$ using Simpson's rule with $n=10$ steps.

29) Find an upper bound for the error in estimating $\int_2^5 \frac{1}{x-1}\,dx$ using Simpson's rule with $n=10$ steps.

Answer

$\frac{81}{25{,}000}$

30) Find an upper bound for the error in estimating $\int_0^\pi 2x\cos(x)\,dx$ using Simpson's rule with four steps.

31) Estimate the minimum number of subintervals needed to approximate the integral $\int_1^4 (5x^2+8)\,dx$ with an error magnitude of less than 0.0001 using the trapezoidal rule.

Answer

475

32) Determine a value of n such that the trapezoidal rule will approximate $\int_0^1 \sqrt{1+x^2}\,dx$ with an error of no more than 0.01.

33) Estimate the minimum number of subintervals needed to approximate the integral $\int_2^3 (2x^3+4x)\,dx$ with an error of magnitude less than 0.0001 using the trapezoidal rule.

Answer

174

34) Estimate the minimum number of subintervals needed to approximate the integral $\int_3^4 \frac{1}{(x-1)^2}\,dx$ with an error magnitude of less than 0.0001 using the trapezoidal rule.

35) Use Simpson's rule with four subdivisions to approximate the area under the probability density function $y=\frac{1}{\sqrt{2\pi}}e^{-x^2/2}$ from $x=0$ to $x=0.4$.

Answer

0.1554

36) Use Simpson's rule with $n=14$ to approximate (to three decimal places) the area of the region bounded by the graphs of $y=0, x=0,$ and $x=\pi/2$.

37) The length of one arch of the curve $y=3\sin(2x)$ is given by $L=\int_0^{\pi/2}\sqrt{1+36\cos^2(2x)}\,dx$. Estimate L using the trapezoidal rule with $n=6$.

Answer

6.2807

38) The length of the ellipse $x=a\cos(t), y=b\sin(t), 0\le t\le 2\pi$ is given by $L=4a\int_0^{\pi/2}\sqrt{1-e^2\cos^2(t)}dt$, where e is the eccentricity of the ellipse. Use Simpson's rule with $n=6$ subdivisions to estimate the length of the ellipse when $a=2$ and $e=1/3$.

39) Estimate the area of the surface generated by revolving the curve $y=\cos(2x), 0\le x\le \frac{\pi}{4}$ about the x-axis. Use the trapezoidal rule with six subdivisions.

Answer

4.606

40) Estimate the area of the surface generated by revolving the curve $y = 2x^2, 0 \leq x \leq 3$ about the x-axis. Use Simpson's rule with $n = 6$.

41) The growth rate of a certain tree (in feet) is given by $y = \frac{2}{t+1} + e^{-t^2/2}$, where t is time in years. Estimate the growth of the tree through the end of the second year by using Simpson's rule, using two subintervals. (Round the answer to the nearest hundredth.)

Answer

3.41 ft

42) [T] Use a calculator to approximate $\int_0^1 \sin(\pi x)\ dx$ using the midpoint rule with 25 subdivisions. Compute the relative error of approximation.

43) [T] Given $\int_1^5 (3x^2 - 2x)\ dx = 100$, approximate the value of this integral using the midpoint rule with 16 subdivisions and determine the absolute error.

Answer

$T_{16} = 100.125$; absolute error = 0.125

44) Given that we know the Fundamental Theorem of Calculus, why would we want to develop numerical methods for definite integrals?

45) The table represents the coordinates (x, y) that give the boundary of a lot. The units of measurement are meters. Use the trapezoidal rule to estimate the number of square meters of land that is in this lot.

x	y	x	y
0	125	600	95
100	125	700	88
200	120	800	75
300	112	900	35
400	90	1000	0
500	90		

Answer

about 89,250 m^2

46) Choose the correct answer. When Simpson's rule is used to approximate the definite integral, it is necessary that the number of partitions be____

a. an even number

b. odd number

c. either an even or an odd number

d. a multiple of 4

47) The "Simpson" sum is based on the area under a ____.

Answer

parabola

48) The error formula for Simpson's rule depends on___.

a. $f(x)$

b. $f'(x)$

c. $f^{(4)}(x)$

d. the number of steps

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7.7: Improper Integrals

Learning Objectives

- Evaluate an integral over an infinite interval.
- Evaluate an integral over a closed interval with an infinite discontinuity within the interval.
- Use the comparison theorem to determine whether a definite integral is convergent.

Is the area between the graph of $f(x) = \dfrac{1}{x}$ and the x-axis over the interval $[1, +\infty)$ finite or infinite? If this same region is revolved about the x-axis, is the volume finite or infinite? Surprisingly, the area of the region described is infinite, but the volume of the solid obtained by revolving this region about the x-axis is finite.

In this section, we define integrals over an infinite interval as well as integrals of functions containing a discontinuity on the interval. Integrals of these types are called improper integrals. We examine several techniques for evaluating improper integrals, all of which involve taking limits.

Integrating over an Infinite Interval

How should we go about defining an integral of the type $\displaystyle\int_a^{+\infty} f(x)\,dx$? We can integrate $\displaystyle\int_a^t f(x)\,dx$ for any value of t, so it is reasonable to look at the behavior of this integral as we substitute larger values of t. Figure 7.7.1 shows that $\displaystyle\int_a^t f(x)\,dx$ may be interpreted as area for various values of t. In other words, we may define an improper integral as a limit, taken as one of the limits of integration increases or decreases without bound.

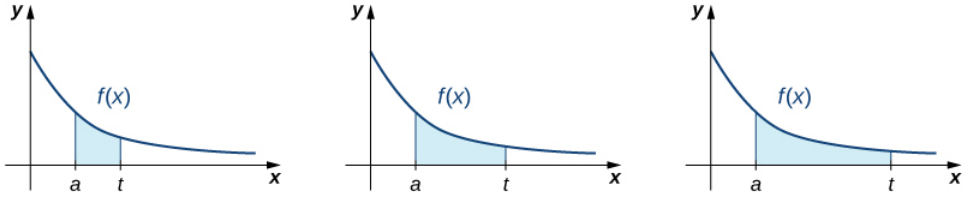


Figure 7.7.1: To integrate a function over an infinite interval, we consider the limit of the integral as the upper limit increases without bound.

Definition: Improper Integral

1. Let $f(x)$ be continuous over an interval of the form $[a, +\infty)$. Then

$$\int_a^{+\infty} f(x)\,dx = \lim_{t\to+\infty} \int_a^t f(x)\,dx, \tag{7.7.1}$$

   provided this limit exists.

2. Let $f(x)$ be continuous over an interval of the form $(-\infty, b]$. Then

$$\int_{-\infty}^b f(x)\,dx = \lim_{t\to-\infty} \int_t^b f(x)\,dx, \tag{7.7.2}$$

   provided this limit exists.

In each case, if the limit exists, then the *improper integral* is said to converge. If the limit does not exist, then the improper integral is said to diverge.

3. Let $f(x)$ be continuous over $(-\infty, +\infty)$. Then

$$\int_{-\infty}^{+\infty} f(x)\,dx = \int_{-\infty}^0 f(x)\,dx + \int_0^{+\infty} f(x)\,dx, \tag{7.7.3}$$

provided that $\int_{-\infty}^{0} f(x)\,dx$ and $\int_{0}^{+\infty} f(x)\,dx$ both converge.

If either of these two integrals diverge, then $\int_{-\infty}^{+\infty} f(x)\,dx$ diverges.

(It can be shown that, in fact, $\int_{-\infty}^{+\infty} f(x)\,dx = \int_{-\infty}^{a} f(x)\,dx + \int_{a}^{+\infty} f(x)\,dx$ for any value of a.).

In our first example, we return to the question we posed at the start of this section: Is the area between the graph of $f(x) = \frac{1}{x}$ and the x-axis over the interval $[1, +\infty)$ finite or infinite?

Example 7.7.1: Finding an Area

Determine whether the area between the graph of $f(x) = \frac{1}{x}$ and the x-axis over the interval $[1, +\infty)$ is finite or infinite.

Solution

We first do a quick sketch of the region in question, as shown in Figure 7.7.2.

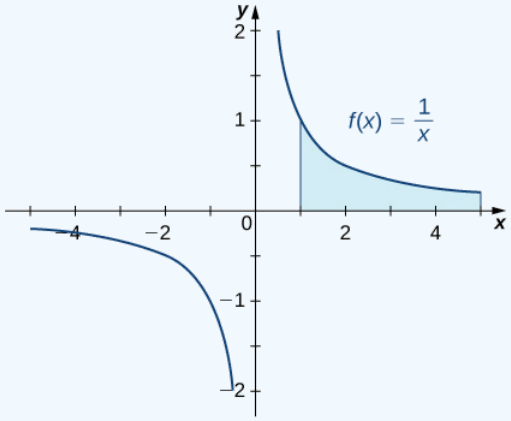


Figure 7.7.2: We can find the area between the curve $f(x) = 1/x$ and the x-axis on an infinite interval.

We can see that the area of this region is given by

$$A = \int_{1}^{\infty} \frac{1}{x}\,dx.$$

which can be evaluated using Equation 7.7.1:

$$\begin{aligned} A &= \int_{1}^{\infty} \frac{1}{x}\,dx \\ &= \lim_{t\to+\infty} \int_{1}^{t} \frac{1}{x}\,dx && \text{Rewrite the improper integral as a limit} \\ &= \lim_{t\to+\infty} \ln|x|\,|_{1}^{t} && \text{Find the antiderivative} \\ &= \lim_{t\to+\infty} (\ln|t| - \ln 1) && \text{Evaluate the antiderivative} \\ &= +\infty. && \text{Evaluate the limit.} \end{aligned}$$

Since the improper integral diverges to $+\infty$, the area of the region is infinite.

Example 7.7.2: Finding a Volume

Find the volume of the solid obtained by revolving the region bounded by the graph of $f(x) = \frac{1}{x}$ and the x-axis over the interval $[1, +\infty)$ about the x-axis.

Solution

The solid is shown in Figure 7.7.3. Using the disk method, we see that the volume V is

$$V = \pi \int_1^{+\infty} \frac{1}{x^2}\, dx.$$

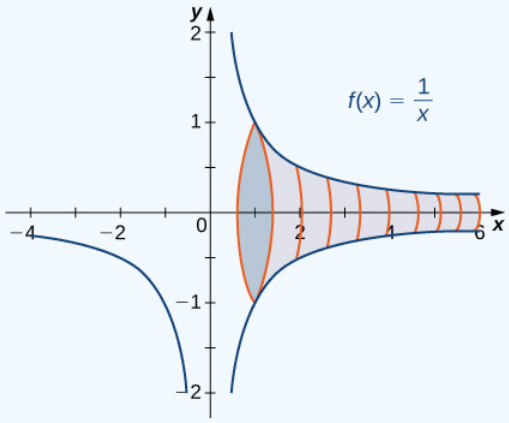


Figure 7.7.3: The solid of revolution can be generated by rotating an infinite area about the x-axis.

Then we have

$$\begin{aligned}
V &= \pi \int_1^{+\infty} \frac{1}{x^2}\, dx \\
&= \pi \lim_{t\to+\infty} \int_1^t \frac{1}{x^2}\, dx && \text{Rewrite as a limit.} \\
&= \pi \lim_{t\to+\infty} -\frac{1}{x}\,\Big|_1^t && \text{Find the antiderivative.} \\
&= \pi \lim_{t\to+\infty} \left(-\frac{1}{t} + 1\right) && \text{Evaluate the antiderivative.} \\
&= \pi \lim_{t\to+\infty} \left(-\frac{1}{\not{t}^{\,0}} + 1\right) && \text{Evaluate the antiderivative.} \\
&= \pi
\end{aligned}$$

The improper integral converges to π. Therefore, the volume of the solid of revolution is π.

In conclusion, although the area of the region between the x-axis and the graph of $f(x) = 1/x$ over the interval $[1, +\infty)$ is infinite, the volume of the solid generated by revolving this region about the x-axis is finite. The solid generated is known as Gabriel's Horn.

Note: Gabriel's horn (also called Torricelli's trumpet) is a geometric figure which has *infinite* surface area, but *finite* volume. The name refers to the tradition identifying the Archangel Gabriel as the angel who blows the horn to announce Judgment Day, associating the divine, or infinite, with the finite. The properties of this figure were first studied by Italian physicist and mathematician Evangelista Torricelli in the 17th century.

Example 7.7.3: Traffic Accidents in a City

Suppose that at a busy intersection, traffic accidents occur at an average rate of one every three months. After residents complained, changes were made to the traffic lights at the intersection. It has now been eight months since the changes were made and there have been no accidents. Were the changes effective or is the 8-month interval without an accident a result of chance?



Figure 7.7.4: Modification of work by David McKelvey, Flickr.

Probability theory tells us that if the average time between events is k, the *probability* that X, the time between events, is between a and b is given by

$$(P(a \le x \le b) = \int_a^b f(x)\,dx$$

where

$$f(x) = \begin{cases} 0, & \text{if } x < 0 \\ ke^{-kx}, & \text{if } x \ge 0 \end{cases}.$$

Thus, if accidents are occurring at a rate of one every 3 months, then the probability that X, the time between accidents, is between a and b is given by

$$P(a \le x \le b) = \int_a^b f(x)\,dx$$

where

$$f(x) = \begin{cases} 0, & \text{if } x < 0 \\ 3e^{-3x}, & \text{if } x \ge 0 \end{cases}.$$

To answer the question, we must compute $P(X \ge 8) = \int_8^{+\infty} 3e^{-3x}\,dx$ and decide whether it is likely that 8 months could have passed without an accident if there had been no improvement in the traffic situation.

Solution

We need to calculate the probability as an improper integral:

$$\begin{aligned} P(X \ge 8) &= \int_8^{+\infty} 3e^{-3x}\,dx \\ &= \lim_{t \to +\infty} \int_8^t 3e^{-3x}\,dx \\ &= \lim_{t \to +\infty} -e^{-3x}\,|_8^t \\ &= \lim_{t \to +\infty} (-e^{-3t} + e^{-24}) \\ &\approx 3.8 \times 10^{-11}. \end{aligned}$$

The value 3.8×10^{-11} represents the probability of no accidents in 8 months under the initial conditions. Since this value is very, very small, it is reasonable to conclude the changes were effective.

Example 7.7.4: Evaluating an Improper Integral over an Infinite Interval

Evaluate $\int_{-\infty}^0 \frac{1}{x^2+4}\,dx$. State whether the improper integral converges or diverges.

Solution

Begin by rewriting $\int_{-\infty}^0 \frac{1}{x^2+4}\,dx$ as a limit using Equation 7.7.2 from the definition. Thus,

$$\int_{-\infty}^{0} \frac{1}{x^2+4}\,dx = \lim_{t\to-\infty}\int_{t}^{0} \frac{1}{x^2+4}\,dx \qquad \text{Rewrite as a limit.}$$

$$= \lim_{t\to-\infty} \frac{1}{2}\tan^{-1}\frac{x}{2}\,|_{t}^{0} \qquad \text{Find the antiderivative.}$$

$$= \lim_{t\to-\infty} \left(\frac{1}{2}\tan^{-1}0 - \frac{1}{2}\tan^{-1}\frac{t}{2}\right) \qquad \text{Evaluate the antiderivative.}$$

$$= \frac{\pi}{4}. \qquad \text{Evaluate the limit and simplify.}$$

The improper integral converges to $\frac{\pi}{4}$.

Example 7.7.5: Evaluating an Improper Integral on $(-\infty, +\infty)$

Evaluate $\int_{-\infty}^{+\infty} xe^x\,dx$. State whether the improper integral converges or diverges.

Solution

Start by splitting up the integral:

$$\int_{-\infty}^{+\infty} xe^x\,dx = \int_{-\infty}^{0} xe^x\,dx + \int_{0}^{+\infty} xe^x\,dx.$$

If either $\int_{-\infty}^{0} xe^x\,dx$ or $\int_{0}^{+\infty} xe^x\,dx$ diverges, then $\int_{-\infty}^{+\infty} xe^x\,dx$ diverges.

Compute each integral separately.

For the first integral,

$$\int_{-\infty}^{0} xe^x\,dx = \lim_{t\to-\infty}\int_{t}^{0} xe^x\,dx \qquad \text{Rewrite as a limit.}$$

$$= \lim_{t\to-\infty} (xe^x - e^x)\,|_{t}^{0} \qquad \text{Use integration by parts to find the antiderivative. (Here } u = x \text{ and } dv = e^x\,dx.\text{)}$$

$$= \lim_{t\to-\infty} (-1 - te^t + e^t) \qquad \text{Evaluate the antiderivative.}$$

$$= -1.$$

Evaluate the limit. Note: $\lim_{t\to-\infty} te^t$ is indeterminate of the form $0\cdot\infty$.Thus, $\lim_{t\to-\infty} te^t = \lim_{t\to-\infty} \frac{t}{e^{-t}} = \lim_{t\to-\infty} \frac{-1}{e^{-t}} = \lim_{t\to-\infty} -e^t = 0$ by L'Hôpital's Rule.

The first improper integral converges. For the second integral,

$$\int_{0}^{+\infty} xe^x\,dx = \lim_{t\to+\infty}\int_{0}^{t} xe^x\,dx \qquad \text{Rewrite as a limit.}$$

$$= \lim_{t\to+\infty} (xe^x - e^x)\,|_{0}^{t} \qquad \text{Find the antiderivative.}$$

$$= \lim_{t\to+\infty} (te^t - e^t + 1) \qquad \text{Evaluate the antiderivative.}$$

$$= \lim_{t\to+\infty} ((t-1)e^t + 1) \qquad \text{Rewrite. } (te^t - e^t \text{ is indeterminate.})$$

$$= +\infty. \qquad \text{Evaluate the limit.}$$

Thus, $\int_{0}^{+\infty} xe^x\,dx$ diverges. Since this integral diverges, $\int_{-\infty}^{+\infty} xe^x\,dx$ diverges as well.

Exercise 7.7.1

Evaluate $\int_{-3}^{+\infty} e^{-x}\,dx$. State whether the improper integral converges or diverges.

Hint

$$\int_{-3}^{+\infty} e^{-x}\,dx = \lim_{t\to+\infty}\int_{-3}^{t} e^{-x}\,dx$$

Answer

It converges to e^3.

Integrating a Discontinuous Integrand

Now let's examine integrals of functions containing an infinite discontinuity in the interval over which the integration occurs. Consider an integral of the form $\int_a^b f(x)\,dx$, where $f(x)$ is continuous over $[a,b)$ and discontinuous at b. Since the function $f(x)$ is continuous over $[a,t]$ for all values of t satisfying $a \le t < b$, the integral $\int_a^t f(x)\,dx$ is defined for all such values of t. Thus, it makes sense to consider the values of $\int_a^t f(x)\,dx$ as t approaches b for $a \le t < b$. That is, we define $\int_a^b f(x)\,dx = \lim_{t\to b^-} \int_a^t f(x)\,dx$, provided this limit exists. Figure 7.7.5 illustrates $\int_a^t f(x)\,dx$ as areas of regions for values of t approaching b.

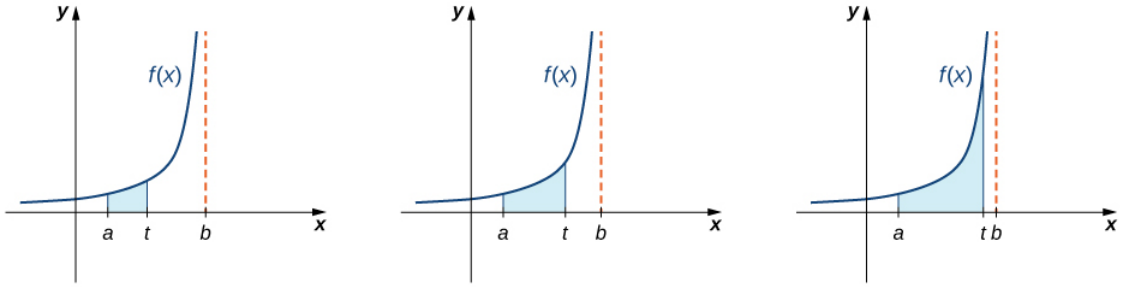


Figure 7.7.5: As t approaches b from the left, the value of the area from a to t approaches the area from a to b.

We use a similar approach to define $\int_a^b f(x)\,dx$, where $f(x)$ is continuous over $(a,b]$ and discontinuous at a. We now proceed with a formal definition.

Definition: Converging and Diverging Improper Integral

1. Let $f(x)$ be continuous over $[a,b)$. Then,

$$\int_a^b f(x)\,dx = \lim_{t\to b^-} \int_a^t f(x)\,dx. \tag{7.7.4}$$

2. Let $f(x)$ be continuous over $(a,b]$. Then,

$$\int_a^b f(x)\,dx = \lim_{t\to a^+} \int_t^b f(x)\,dx. \tag{7.7.5}$$

In each case, if the limit exists, then the improper integral is said to converge. If the limit does not exist, then the improper integral is said to diverge.

3. If $f(x)$ is continuous over $[a,b]$ except at a point c in (a,b), then

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx, \tag{7.7.6}$$

provided both $\int_a^c f(x)\,dx$ and $\int_c^b f(x)\,dx$ converge. If either of these integrals diverges, then $\int_a^b f(x)\,dx$ diverges.

The following examples demonstrate the application of this definition.

Example 7.7.6: Integrating a Discontinuous Integrand

Evaluate $\int_0^4 \frac{1}{\sqrt{4-x}}\,dx$, if possible. State whether the integral converges or diverges.

Solution

The function $f(x) = \dfrac{1}{\sqrt{4-x}}$ is continuous over $[0, 4)$ and discontinuous at 4. Using Equation 7.7.4 from the definition, rewrite $\displaystyle\int_0^4 \frac{1}{\sqrt{4-x}}\,dx$ as a limit:

$$\begin{aligned}
\int_0^4 \frac{1}{\sqrt{4-x}}\,dx &= \lim_{t\to 4^-}\int_0^t \frac{1}{\sqrt{4-x}}\,dx && \text{Rewrite as a limit.}\\
&= \lim_{t\to 4^-}\left(-2\sqrt{4-x}\right)\Big|_0^t && \text{Find the antiderivative.}\\
&= \lim_{t\to 4^-}\left(-2\sqrt{4-t}+4\right) && \text{Evaluate the antiderivative.}\\
&= 4. && \text{Evaluate the limit.}
\end{aligned}$$

The improper integral converges.

Example 7.7.7: Integrating a Discontinuous Integrand

Evaluate $\displaystyle\int_0^2 x\ln x\,dx$. State whether the integral converges or diverges.

Solution

Since $f(x) = x\ln x$ is continuous over $(0, 2]$ and is discontinuous at zero, we can rewrite the integral in limit form using Equation 7.7.5:

$$\begin{aligned}
\int_0^2 x\ln x\,dx &= \lim_{t\to 0^+}\int_t^2 x\ln x\,dx && \text{Rewrite as a limit.}\\
&= \lim_{t\to 0^+}\left(\frac{1}{2}x^2\ln x - \frac{1}{4}x^2\right)\Big|_t^2 && \text{Evaluate } \int x\ln x\,dx \text{ using integration by parts with } u=\ln x \text{ and } dv = x.\\
&= \lim_{t\to 0^+}\left(2\ln 2 - 1 - \frac{1}{2}t^2\ln t + \frac{1}{4}t^2\right) && \text{Evaluate the antiderivative.}\\
&= 2\ln 2 - 1. && \text{Evaluate the limit. Note that } \lim_{t\to 0^+} t^2\ln t \text{ is indeterminate.}\\
& && \text{To evaluate it, rewrite it as a quotient and apply L'Hôpital's rule.}
\end{aligned}$$

The improper integral converges.

Example 7.7.8: Integrating a Discontinuous Integrand

Evaluate $\displaystyle\int_{-1}^1 \frac{1}{x^3}\,dx$. State whether the improper integral converges or diverges.

Solution

Since $f(x) = 1/x^3$ is discontinuous at zero, using Equation 7.7.6, we can write

$$\int_{-1}^1 \frac{1}{x^3}\,dx = \int_{-1}^0 \frac{1}{x^3}\,dx \quad + \quad \int_0^1 \frac{1}{x^3}\,dx.$$

If either of the two integrals diverges, then the original integral diverges. Begin with $\displaystyle\int_{-1}^0 \frac{1}{x^3}\,dx$:

$$\begin{aligned}
\int_{-1}^0 \frac{1}{x^3}\,dx &= \lim_{t\to 0^-}\int_{-1}^t \frac{1}{x^3}\,dx && \text{Rewrite as a limit.}\\
&= \lim_{t\to 0^-}\left(-\frac{1}{2x^2}\right)\Big|_{-1}^t && \text{Find the antiderivative.}\\
&= \lim_{t\to 0^-}\left(-\frac{1}{2t^2}+\frac{1}{2}\right) && \text{Evaluate the antiderivative.}\\
&= +\infty. && \text{Evaluate the limit.}
\end{aligned}$$

Therefore, $\displaystyle\int_{-1}^0 \frac{1}{x^3}\,dx$ diverges.

Since $\int_{-1}^{0} \frac{1}{x^3}\,dx$ diverges, $\int_{-1}^{1} \frac{1}{x^3}\,dx$ also diverges.

Exercise 7.7.2

Evaluate $\int_0^2 \frac{1}{x}\,dx$. State whether the integral converges or diverges.

Hint

Write $\int_0^2 \frac{1}{x}\,dx$ in limit form using Equation 7.7.5.

Answer

$+\infty$, It diverges.

A Comparison Theorem

It is not always easy or even possible to evaluate an improper integral directly; however, by comparing it with another carefully chosen integral, it may be possible to determine its convergence or divergence. To see this, consider two continuous functions $f(x)$ and $g(x)$ satisfying $0 \le f(x) \le g(x)$ for $x \ge a$ (Figure 7.7.6). In this case, we may view integrals of these functions over intervals of the form $[a, t]$ as areas, so we have the relationship

$$0 \le \int_a^t f(x)\,dx \le \int_a^t g(x)\,dx$$

for $t \ge a$.

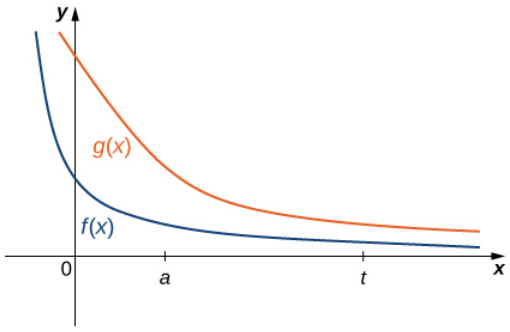


Figure 7.7.6: If $0 \le f(x) \le g(x)$ for $x \ge a$, then for $t \ge a$, $\int_a^t f(x)\,dx \le \int_a^t g(x)\,dx$.

Thus, if

$$\int_a^{+\infty} f(x)\,dx = \lim_{t\to+\infty} \int_a^t f(x)\,dx = +\infty,$$

then

$$\int_a^{+\infty} g(x)\,dx = \lim_{t\to+\infty} \int_a^t g(x)\,dx = +\infty$$

as well. That is, if the area of the region between the graph of $f(x)$ and the x-axis over $[a, +\infty)$ is infinite, then the area of the region between the graph of $g(x)$ and the x-axis over $[a, +\infty)$ is infinite too.

On the other hand, if

$$\int_a^{+\infty} g(x)\,dx = \lim_{t\to+\infty} \int_a^t g(x)\,dx = L$$

for some real number L, then

$$\int_a^{+\infty} f(x)\,dx = \lim_{t\to+\infty}\int_a^t f(x)\,dx$$

must converge to some value less than or equal to L, since $\int_a^t f(x)\,dx$ increases as t increases and $\int_a^t f(x)\,dx \le L$ for all $t \ge a$.

If the area of the region between the graph of $g(x)$ and the x-axis over $[a,+\infty)$ is finite, then the area of the region between the graph of $f(x)$ and the x-axis over $[a,+\infty)$ is also finite.

These conclusions are summarized in the following theorem.

A Comparison Theorem

Let $f(x)$ and $g(x)$ be continuous over $[a,+\infty)$. Assume that $0 \le f(x) \le g(x)$ for $x \ge a$.

i. If

$$\int_a^{+\infty} f(x)\,dx = \lim_{t\to+\infty}\int_a^t f(x)\,dx = +\infty,$$

then

$$\int_a^{+\infty} g(x)\,dx = \lim_{t\to+\infty}\int_a^t g(x)\,dx = +\infty.$$

ii. If

$$\int_a^{+\infty} g(x)\,dx = \lim_{t\to+\infty}\int_a^t g(x)\,dx = L,$$

where L is a real number, then

$$\int_a^{+\infty} f(x)\,dx = \lim_{t\to+\infty}\int_a^t f(x)\,dx = M$$

for some real number $M \le L$.

Example 7.7.9: Applying the Comparison Theorem

Use a comparison to show that $\int_1^{+\infty} \frac{1}{xe^x}\,dx$.

converges.

Solution

We can see that $0 \le \frac{1}{xe^x} \le \frac{1}{e^x} = e^{-x}$,

so if $\int_1^{+\infty} e^{-x}\,dx$ converges, then so does $\int_1^{+\infty} \frac{1}{xe^x}\,dx$.

To evaluate $\int_1^{+\infty} e^{-x}\,dx$, first rewrite it as a limit:

$$\begin{aligned}\int_1^{+\infty} e^{-x}\,dx &= \lim_{t\to+\infty}\int_1^t e^{-x}\,dx \\ &= \lim_{t\to+\infty}\left(-e^{-x}\right)\Big|_1^t \\ &= \lim_{t\to+\infty}\left(-e^{-t}+e^{-1}\right) \\ &= e^{-1}.\end{aligned}$$

Since $\int_1^{+\infty} e^{-x}\,dx$ converges, so does $\int_1^{+\infty} \frac{1}{xe^x}\,dx$.

Example 7.7.10: Applying the Comparison Theorem

Use the comparison theorem to show that $\int_1^{+\infty} \frac{1}{x^p}\,dx$ diverges for all $p < 1$.

Solution

For $p < 1$, $\frac{1}{x} \leq \frac{1}{x^p}$ over $[1, +\infty)$.

In Example 7.7.1, we showed that $\int_1^{+\infty} \frac{1}{x}\,dx = +\infty$.

Therefore, $\int_1^{+\infty} \frac{1}{x^p}\,dx$ diverges for all $p < 1$.

Exercise 7.7.3

Use a comparison to show that $\int_e^{+\infty} \frac{\ln x}{x}\,dx$ diverges.

Hint

$\frac{1}{x} \leq \frac{\ln x}{x}$ on $[e, +\infty)$

Answer

Since $\int_e^{+\infty} \frac{1}{x}\,dx = +\infty$, $\int_e^{+\infty} \frac{\ln x}{x}\,dx$ diverges.

Laplace Transforms

Laplace Transforms

In the last few chapters, we have looked at several ways to use integration for solving real-world problems. For this next project, we are going to explore a more advanced application of integration: integral transforms. Specifically, we describe the Laplace transform and some of its properties. The **Laplace transform** is used in engineering and physics to simplify the computations needed to solve some problems. It takes functions expressed in terms of time and *transforms* them to functions expressed in terms of frequency. It turns out that, in many cases, the computations needed to solve problems in the frequency domain are much simpler than those required in the time domain.

The Laplace transform is defined in terms of an integral as

$$\mathscr{L}\big(f(t)\big) = F(s) = \int_0^{\infty} e^{-st} f(t)\,dt.$$

Note that the input to a Laplace transform is a function of time, $f(t)$, and the output is a function of frequency, $F(s)$. Although many real-world examples require the use of complex numbers (involving the imaginary number $i = \sqrt{-1}$), in this project we limit ourselves to functions of real numbers.

Let's start with a simple example. Here we calculate the Laplace transform of $f(t) = t$. We have

$$\mathscr{L}(t) = \int_0^{\infty} te^{-st} dt.$$

This is an improper integral, so we express it in terms of a limit, which gives

$$\mathscr{L}(t) = \int_0^{\infty} te^{-st} dt = \lim_{z \to \infty} \int_0^{z} te^{-st} dt.$$

Now we use integration by parts to evaluate the integral. Note that we are integrating with respect to t, so we treat the variable s as a constant. We have

$$u = t \qquad dv = e^{-st}dt$$
$$du = dt \quad v = -\tfrac{1}{s}e^{-st}$$

Then we obtain

$$\begin{aligned}
\lim_{z\to\infty}\int_0^z te^{-st}dt &= \lim_{z\to\infty}\left[-\frac{t}{s}e^{-st}\Big|_0^z + \frac{1}{s}\int_0^z e^{-st}dt\right] \\
&= \lim_{z\to\infty}\left[\left[-\frac{z}{s}e^{-sz} + \frac{0}{s}e^{-0s}\right] + \frac{1}{s}\int_0^z e^{-st}dt\right] \\
&= \lim_{z\to\infty}\left[\left[-\frac{z}{s}e^{-sz} + 0\right] - \frac{1}{s}\left[\frac{e^{-st}}{s}\right]\Big|_0^z\right] \\
&= \lim_{z\to\infty}\left[-\frac{z}{s}e^{-sz} - \frac{1}{s^2}\left[e^{-sz} - 1\right]\right] \\
&= \lim_{z\to\infty}\left[-\frac{z}{se^{sz}}\right] - \lim_{z\to\infty}\frac{1}{s^2e^{sz}} + \lim_{z\to\infty}\frac{1}{s^2} \\
&= 0 - 0 + \frac{1}{s^2} \qquad \text{Note that the first limit requires L'Hôpital's rule. See work below.} \\
&= \frac{1}{s^2}.
\end{aligned}$$

As noted above, $\lim_{z\to\infty}\left[-\frac{z}{se^{sz}}\right]$ is indeterminate of the form $-\frac{\infty}{\infty}$ and requires L'Hôpital's rule to evaluate. Using L'Hôpital's rule,

$$\lim_{z\to\infty}\left[-\frac{z}{se^{sz}}\right] = \lim_{z\to\infty}\left[-\frac{1}{s^2e^{sz}}\right] = 0.$$

1. Calculate the Laplace transform of $f(t) = 1$.
2. Calculate the Laplace transform of $f(t) = e^{-3t}$.
3. Calculate the Laplace transform of $f(t) = t^2$. (Note, you will have to integrate by parts twice.)

Laplace transforms are often used to solve differential equations. Differential equations are not covered in detail until later in this book; but, for now, let's look at the relationship between the Laplace transform of a function and the Laplace transform of its derivative.

Let's start with the definition of the Laplace transform. We have

$$\mathscr{L}\big(f(t)\big) = \int_0^\infty e^{-st}f(t)\,dt = \lim_{z\to\infty}\int_0^z e^{-st}f(t)\,dt.$$

Use integration by parts to evaluate $\lim_{z\to\infty}\int_0^z e^{-st}f(t)\,dt$. (Let $u = f(t)$ and $dv = e^{-st}dt$.)

After integrating by parts and evaluating the limit, you should see that

$$\mathscr{L}\big(f(t)\big) = \frac{f(0)}{s} + \frac{1}{s}\big[\mathscr{L}\big(f'(t)\big)\big].$$

Then,

$$\mathscr{L}\big(f'(t)\big) = s\mathscr{L}\big(f(t)\big) - f(0).$$

Thus, differentiation in the time domain simplifies to multiplication by s in the frequency domain.

The final thing we look at in this project is how the Laplace transforms of $f(t)$ and its antiderivative are related. Let $g(t) = \int_0^t f(u)\,du$.

Then,

$$\mathscr{L}\big(g(t)\big) = \int_0^\infty e^{-st}g(t)\,dt = \lim_{z\to\infty}\int_0^z e^{-st}g(t)\,dt.$$

Use integration by parts to evaluate $\lim_{z\to\infty}\int_0^z e^{-st}g(t)\,dt$. (Let $u=g(t)$ and $dv=e^{-st}dt$. Note, by the way that we have defined $g(t)$, $du=f(t)\,dt$.)

As you might expect, you should see that

$$\mathscr{L}\big(g(t)\big)=\frac{1}{s}\cdot\mathscr{L}\big(f(t)\big).$$

Integration in the time domain simplifies to division by s in the frequency domain.

Key Concepts

- Integrals of functions over infinite intervals are defined in terms of limits.
- Integrals of functions over an interval for which the function has a discontinuity at an endpoint may be defined in terms of limits.
- The convergence or divergence of an improper integral may be determined by comparing it with the value of an improper integral for which the convergence or divergence is known.

Key Equations

- **Improper integrals**

$$\int_a^{+\infty} f(x)\,dx=\lim_{t\to+\infty}\int_a^t f(x)\,dx$$
$$\int_{-\infty}^b f(x)\,dx=\lim_{t\to-\infty}\int_t^b f(x)\,dx$$
$$\int_{-\infty}^{+\infty} f(x)\,dx=\int_{-\infty}^0 f(x)\,dx+\int_0^{+\infty} f(x)\,dx$$

Glossary

improper integral

an integral over an infinite interval or an integral of a function containing an infinite discontinuity on the interval; an improper integral is defined in terms of a limit. The improper integral converges if this limit is a finite real number; otherwise, the improper integral diverges

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7.7E: Exercises for Section 7.7

In exercises 1 - 8, evaluate the following integrals. If the integral is not convergent, answer "It diverges."

1) $\int_2^4 \frac{dx}{(x-3)^2}$

Answer

It diverges.

2) $\int_0^\infty \frac{1}{4+x^2}\,dx$

3) $\int_0^2 \frac{1}{\sqrt{4-x^2}}\,dx$

Answer

Converges to $\frac{\pi}{2}$

4) $\int_1^\infty \frac{1}{x\ln x}\,dx$

5) $\int_1^\infty xe^{-x}\,dx$

Answer

Converges to $\frac{2}{e}$

6) $\int_{-\infty}^\infty \frac{x}{x^2+1}\,dx$

7) Without integrating, determine whether the integral $\int_1^\infty \frac{1}{\sqrt{x^3+1}}\,dx$ converges or diverges by comparing the function $f(x) = \frac{1}{\sqrt{x^3+1}}$ with $g(x) = \frac{1}{\sqrt{x^3}}$.

Answer

It converges.

8) Without integrating, determine whether the integral $\int_1^\infty \frac{1}{\sqrt{x+1}}\,dx$ converges or diverges.

In exercises 9 - 25, determine whether the improper integrals converge or diverge. If possible, determine the value of the integrals that converge.

9) $\int_0^\infty e^{-x}\cos x\,dx$

Answer

Converges to $\frac{1}{2}$.

10) $\int_1^\infty \frac{\ln x}{x}\,dx$

11) $\int_0^1 \frac{\ln x}{\sqrt{x}}\,dx$

Answer

Converges to -4.

12) $\int_0^1 \ln x\,dx$

13) $\int_{-\infty}^\infty \frac{1}{x^2+1}\,dx$

Answer

Converges to π.

14) $\int_1^5 \frac{dx}{\sqrt{x-1}}$

15) $\int_{-2}^2 \frac{dx}{(1+x)^2}$

Answer

It diverges.

16) $\int_0^\infty e^{-x}\,dx$

17) $\int_0^\infty \sin x\,dx$

Answer

It diverges.

18) $\int_{-\infty}^\infty \frac{e^x}{1+e^{2x}}\,dx$

19) $\int_0^1 \frac{dx}{\sqrt[3]{x}}$

Answer

Converges to 1.5.

20) $\int_0^2 \frac{dx}{x^3}$

21) $\int_{-1}^2 \frac{dx}{x^3}$

Answer

It diverges.

22) $\int_0^1 \frac{dx}{\sqrt{1-x^2}}$

23) $\int_0^3 \frac{1}{x-1}\,dx$

Answer

It diverges.

24) $\int_1^\infty \frac{5}{x^3}\,dx$

25) $\int_3^5 \frac{5}{(x-4)^2}\,dx$

Answer

It diverges.

In exercises 26 and 27, determine the convergence of each of the following integrals by comparison with the given integral. If the integral converges, find the number to which it converges.

26) $\int_1^\infty \frac{dx}{x^2+4x}$; compare with $\int_1^\infty \frac{dx}{x^2}$.

27) $\int_1^\infty \frac{dx}{\sqrt{x}+1}$; compare with $\int_1^\infty \frac{dx}{2\sqrt{x}}$.

Answer

Both integrals diverge.

In exercises 28 - 38, evaluate the integrals. If the integral diverges, answer "It diverges."

28) $\int_1^{\infty} \frac{dx}{x^e}$

29) $\int_0^1 \frac{dx}{x^{\pi}}$

Answer

It diverges.

30) $\int_0^1 \frac{dx}{\sqrt{1-x}}$

31) $\int_0^1 \frac{dx}{1-x}$

Answer

It diverges.

32) $\int_{-\infty}^0 \frac{dx}{x^2+1}$

33) $\int_{-1}^1 \frac{dx}{\sqrt{1-x^2}}$

Answer

Converges to π.

34) $\int_0^1 \frac{\ln x}{x}\, dx$

35) $\int_0^e \ln(x)\, dx$

Answer

Converges to 0.

36) $\int_0^{\infty} xe^{-x}\, dx$

37) $\int_{-\infty}^{\infty} \frac{x}{(x^2+1)^2}\, dx$

Answer

Converges to 0.

38) $\int_0^{\infty} e^{-x}\, dx$

In exercises 39 - 44, evaluate the improper integrals. Each of these integrals has an infinite discontinuity either at an endpoint or at an interior point of the interval.

39) $\int_0^9 \frac{dx}{\sqrt{9-x}}$

Answer

Converges to 6.

40) $\int_{-27}^1 \frac{dx}{x^{2/3}}$

41) $\int_0^3 \frac{dx}{\sqrt{9-x^2}}$

Answer

Converges to $\frac{\pi}{2}$.

42) $\int_6^{24} \frac{dt}{t\sqrt{t^2-36}}$

43) $\int_0^4 x\ln(4x)\,dx$

Answer

Converges to $8\ln(16)-4$.

44) $\int_0^3 \frac{x}{\sqrt{9-x^2}}\,dx$

45) Evaluate $\int_{.5}^t \frac{dx}{\sqrt{1-x^2}}$. (Be careful!) (Express your answer using three decimal places.)

Answer

Converges to about 1.047.

46) Evaluate $\int_1^4 \frac{dx}{\sqrt{x^2-1}}$. (Express the answer in exact form.)

47) Evaluate $\int_2^\infty \frac{dx}{(x^2-1)^{3/2}}$.

Answer

Converges to $-1+\frac{2}{\sqrt{3}}$.

48) Find the area of the region in the first quadrant between the curve $y=e^{-6x}$ and the x-axis.

49) Find the area of the region bounded by the curve $y=\frac{7}{x^2}$, the x-axis, and on the left by $x=1$.

Answer

$A=7.0$ units.^{2}

50) Find the area under the curve $y=\frac{1}{(x+1)^{3/2}}$, bounded on the left by $x=3$.

51) Find the area under $y=\frac{5}{1+x^2}$ in the first quadrant.

Answer

$A=\frac{5\pi}{2}$ units.^{2}

52) Find the volume of the solid generated by revolving about the x-axis the region under the curve $y=\frac{3}{x}$ from $x=1$ to $x=\infty$.

53) Find the volume of the solid generated by revolving about the y-axis the region under the curve $y=6e^{-2x}$ in the first quadrant.

Answer

$V=3\pi$ units^3

54) Find the volume of the solid generated by revolving about the x-axis the area under the curve $y=3e^{-x}$ in the first quadrant.

The **Laplace transform** of a continuous function over the interval $[0,\infty)$ is defined by $F(s)=\int_0^\infty e^{-sx}f(x)\,dx$ (see the Student Project). This definition is used to solve some important initial-value problems in differential equations, as discussed later. The

domain of F is the set of all real numbers s such that the improper integral converges. Find the Laplace transform F of each of the following functions and give the domain of F.

55) $f(x) = 1$

Answer

$\frac{1}{s}, \quad s > 0$

56) $f(x) = x$

57) $f(x) = \cos(2x)$

Answer

$\frac{s}{s^2 + 4}, \quad s > 0$

58) $f(x) = e^{ax}$

59) Use the formula for arc length to show that the circumference of the circle $x^2 + y^2 = 1$ is 2π.

Answer

Answers will vary.

A function is a **probability density function** if it satisfies the following definition: $\int_{-\infty}^{\infty} f(t)\, dt = 1$. The probability that a random variable x lies between a and b is given by $P(a \leq x \leq b) = \int_a^b f(t)\, dt$.

60) Show that $f(x) = \begin{cases} 0, & \text{if } x < 0 \\ 7e^{-7x}, & \text{if } x \geq 0 \end{cases}$ is a probability density function.

61) Find the probability that x is between 0 and 0.3. (Use the function defined in the preceding problem.) Use four-place decimal accuracy.

Answer

0.8775

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7.R: Chapter 7 Review Exercises

In exercises 1 - 4, determine whether the statement is true or false. Justify your answer with a proof or a counterexample.

1) $\int e^x \sin(x)\, dx$ cannot be integrated by parts.

2) $\int \frac{1}{x^4+1}\, dx$ cannot be integrated using partial fractions.

Answer

False

3) In numerical integration, increasing the number of points decreases the error.

4) Integration by parts can always yield the integral.

Answer

False

In exercises 5 - 10, evaluate the integral using the specified method.

5) $\int x^2 \sin(4x)\, dx$, using integration by parts

6) $\int \frac{1}{x^2\sqrt{x^2+16}}\, dx$, using trigonometric substitution

Answer

$$\int \frac{1}{x^2\sqrt{x^2+16}}\, dx = -\frac{\sqrt{x^2+16}}{16x} + C$$

7) $\int \sqrt{x} \ln x\, dx$, using integration by parts

8) $\int \frac{3x}{x^3+2x^2-5x-6}\, dx$, using partial fractions

Answer

$$\int \frac{3x}{x^3+2x^2-5x-6}\, dx = \frac{1}{10}\left(4\ln|2-x| + 5\ln|x+1| - 9\ln|x+3|\right) + C$$

9) $\int \frac{x^5}{(4x^2+4)^{5/2}}\, dx$, using trigonometric substitution

10) $\int \frac{\sqrt{4-\sin^2(x)}}{\sin^2(x)}\cos(x)\, dx$, using a table of integrals or a CAS

Answer

$$\int \frac{\sqrt{4-\sin^2(x)}}{\sin^2(x)}\cos(x)\, dx = -\frac{\sqrt{4-\sin^2(x)}}{\sin(x)} - \frac{x}{2} + C$$

In exercises 11 - 15, integrate using whatever method you choose.

11) $\int \sin^2 x \cos^2 x\, dx$

12) $\int x^3\sqrt{x^2+2}\, dx$

Answer

$$\int x^3\sqrt{x^2+2}\, dx = \frac{1}{15}(x^2+2)^{3/2}(3x^2-4) + C$$

13) $\int \frac{3x^2+1}{x^4-2x^3-x^2+2x}\,dx$

14) $\int \frac{1}{x^4+4}\,dx$

Answer

$$\int \frac{1}{x^4+4}\,dx = \frac{1}{16}\ln\left(\frac{x^2+2x+2}{x^2-2x+2}\right) - \frac{1}{8}\tan^{-1}(1-x) + \frac{1}{8}\tan^{-1}(x+1) + C$$

15) $\int \frac{\sqrt{3+16x^4}}{x^4}\,dx$

In exercises 16 - 18, approximate the integrals using the midpoint rule, trapezoidal rule, and Simpson's rule using four subintervals, rounding to three decimals.

16) [T] $\int_1^2 \sqrt{x^5+2}\,dx$

Answer

$M_4 = 3.312,$
$T_4 = 3.354,$
$S_4 = 3.326$

17) [T] $\int_0^{\sqrt{\pi}} e^{-\sin(x^2)}\,dx$

18) [T] $\int_1^4 \frac{\ln(1/x)}{x}\,dx$

Answer

$M_4 = -0.982,$
$T_4 = -0.917,$
$S_4 = -0.952$

In exercises 19 - 20, evaluate the integrals, if possible.

19) $\int_1^\infty \frac{1}{x^n}\,dx$, for what values of n does this integral converge or diverge?

20) $\int_1^\infty \frac{e^{-x}}{x}\,dx$

Answer

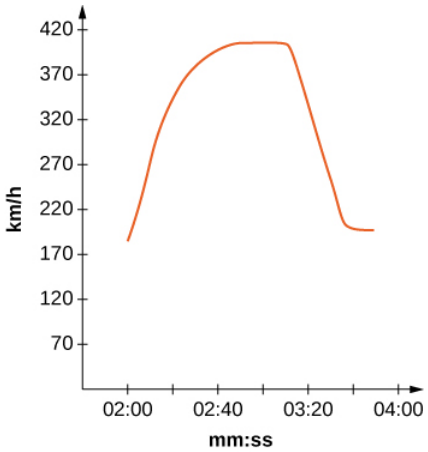
approximately 0.2194

In exercises 21 - 22, consider the gamma function given by $\Gamma(a) = \int_0^\infty e^{-y}y^{a-1}\,dy.$

21) Show that $\Gamma(a) = (a-1)\Gamma(a-1)$.

22) Extend to show that $\Gamma(a) = (a-1)!$, assuming a is a positive integer.

The fastest car in the world, the Bugati Veyron, can reach a top speed of 408 km/h. The graph represents its velocity.



23) [T] Use the graph to estimate the velocity every 20 sec and fit to a graph of the form $v(t) = ae^{bx}\sin(cx) + d$. (Hint: Consider the time units.)

24) [T] Using your function from the previous problem, find exactly how far the Bugati Veyron traveled in the 1 min 40 sec included in the graph.

Answer

Answers may vary. Ex: 9.405 km

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CHAPTER OVERVIEW

8: Introduction to Differential Equations

Many real-world phenomena can be modeled mathematically by using differential equations. Population growth, radioactive decay, predator-prey models, and spring-mass systems are four examples of such phenomena. In this chapter we study some of these applications. A goal of this chapter is to develop solution techniques for different types of differential equations. As the equations become more complicated, the solution techniques also become more complicated, and in fact an entire course could be dedicated to the study of these equations. In this chapter we study several types of differential equations and their corresponding methods of solution.

8.0: Prelude to Differential Equations
8.1: Basics of Differential Equations
8.1E: Exercises for Section 8.1
8.2: Direction Fields and Numerical Methods
8.2E: Exercises for Section 8.2
8.3: Separable Equations
8.3E: Exercises for Section 8.3
8.4: The Logistic Equation
8.4E: Exercises for Section 8.4
8.5: First-order Linear Equations
8.5E: Exercises for Section 8.5
8.R: Chapter 8 Review Exercises

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8.0: Prelude to Differential Equations

Many real-world phenomena can be modeled mathematically by using differential equations. Population growth, radioactive decay, predator-prey models, and spring-mass systems are four examples of such phenomena. In this chapter we study some of these applications. Suppose we wish to study a population of deer over time and determine the total number of animals in a given area. We can first observe the population over a period of time, estimate the total number of deer, and then use various assumptions to derive a mathematical model for different scenarios. Some factors that are often considered are environmental impact, threshold population values, and predators. In this chapter we see how differential equations can be used to predict populations over time.



Figure 8.0.1: The white-tailed deer (Odocoileus virginianus) of the eastern United States. Differential equations can be used to study animal populations. (credit: modification of work by Rachel Kramer, Flickr)

Another goal of this chapter is to develop solution techniques for different types of differential equations. As the equations become more complicated, the solution techniques also become more complicated, and in fact an entire course could be dedicated to the study of these equations. In this chapter we study several types of differential equations and their corresponding methods of solution.

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8.1: Basics of Differential Equations

Learning Objectives

- Identify the order of a differential equation.
- Explain what is meant by a solution to a differential equation.
- Distinguish between the general solution and a particular solution of a differential equation.
- Identify an initial-value problem.
- Identify whether a given function is a solution to a differential equation or an initial-value problem.

Calculus is the mathematics of change, and rates of change are expressed by derivatives. Thus, one of the most common ways to use calculus is to set up an equation containing an unknown function $y = f(x)$ and its derivative, known as a differential equation. Solving such equations often provides information about how quantities change and frequently provides insight into how and why the changes occur.

Techniques for solving differential equations can take many different forms, including direct solution, use of graphs, or computer calculations. We introduce the main ideas in this chapter and describe them in a little more detail later in the course. In this section we study what differential equations are, how to verify their solutions, some methods that are used for solving them, and some examples of common and useful equations.

General Differential Equations

Consider the equation $y' = 3x^2$, which is an example of a differential equation because it includes a derivative. There is a relationship between the variables x and y : y is an unknown function of x. Furthermore, the left-hand side of the equation is the derivative of y. Therefore we can interpret this equation as follows: Start with some function $y = f(x)$ and take its derivative. The answer must be equal to $3x^2$. What function has a derivative that is equal to $3x^2$? One such function is $y = x^3$, so this function is considered a *solution* to a differential equation.

Definition: differential equation

A *differential equation* is an equation involving an unknown function $y = f(x)$ and one or more of its derivatives. A solution to a differential equation is a function $y = f(x)$ that satisfies the differential equation when f and its derivatives are substituted into the equation.

Go to this website to explore more on this topic.

Some examples of differential equations and their solutions appear in Table 8.1.1.

Table 8.1.1: Examples of Differential Equations and Their Solutions

Equation	Solution
$y' = 2x$	$y = x^2$
$y' + 3y = 6x + 11$	$y = e^{-3x} + 2x + 3$
$y'' - 3y' + 2y = 24e^{-2x}$	$y = 3e^x - 4e^{2x} + 2e^{-2x}$

Note that a solution to a differential equation is not necessarily unique, primarily because the derivative of a constant is zero. For example, $y = x^2 + 4$ is also a solution to the first differential equation in Table 8.1.1. We will return to this idea a little bit later in this section. For now, let's focus on what it means for a function to be a solution to a differential equation.

Example 8.1.1: Verifying Solutions of Differential Equations

Verify that the function $y = e^{-3x} + 2x + 3$ is a solution to the differential equation $y' + 3y = 6x + 11$.

Solution

To verify the solution, we first calculate y' using the chain rule for derivatives. This gives $y' = -3e^{-3x} + 2$. Next we substitute y and y' into the left-hand side of the differential equation:

$$(-3e^{-2x} + 2) + 3(e^{-2x} + 2x + 3).$$

The resulting expression can be simplified by first distributing to eliminate the parentheses, giving

$$-3e^{-2x} + 2 + 3e^{-2x} + 6x + 9.$$

Combining like terms leads to the expression $6x + 11$, which is equal to the right-hand side of the differential equation. This result verifies that $y = e^{-3x} + 2x + 3$ is a solution of the differential equation.

Exercise 8.1.1

Verify that $y = 2e^{3x} - 2x - 2$ is a solution to the differential equation $y' - 3y = 6x + 4$.

Hint

First calculate y' then substitute both y' and y into the left-hand side.

It is convenient to define characteristics of differential equations that make it easier to talk about them and categorize them. The most basic characteristic of a differential equation is its order.

Definition: order of a differential equation

The *order of a differential equation* is the highest order of any derivative of the unknown function that appears in the equation.

Example 8.1.2: Identifying the Order of a Differential Equation

The highest derivative in the equation is y',

What is the order of each of the following differential equations?

a. $y' - 4y = x^2 - 3x + 4$
b. $x^2y''' - 3xy'' + xy' - 3y = \sin x$
c. $\frac{4}{x}y^{(4)} - \frac{6}{x^2}y'' + \frac{12}{x^4}y = x^3 - 3x^2 + 4x - 12$

Solution

a. The highest derivative in the equation is y',so the order is 1.
b. The highest derivative in the equation is y''', so the order is 3.
c. The highest derivative in the equation is $y^{(4)}$, so the order is 4.

Exercise 8.1.2

What is the order of the following differential equation?

$$(x^4 - 3x)y^{(5)} - (3x^2 + 1)y' + 3y = \sin x \cos x$$

Hint

What is the highest derivative in the equation?

Answer

5

General and Particular Solutions

We already noted that the differential equation $y' = 2x$ has at least two solutions: $y = x^2$ and $y = x^2 + 4$. The only difference between these two solutions is the last term, which is a constant. What if the last term is a different constant? Will this expression still be a solution to the differential equation? In fact, any function of the form $y = x^2 + C$, where C represents any constant, is a solution as well. The reason is that the derivative of $x^2 + C$ is $2x$, regardless of the value of C. It can be shown that any solution of this differential equation must be of the form $y = x^2 + C$. This is an example of a **general solution** to a differential equation. A graph of some of these solutions is given in Figure 8.1.1. (Note: in this graph we used even integer values for C ranging between -4 and 4. In fact, there is no restriction on the value of C; it can be an integer or not.)

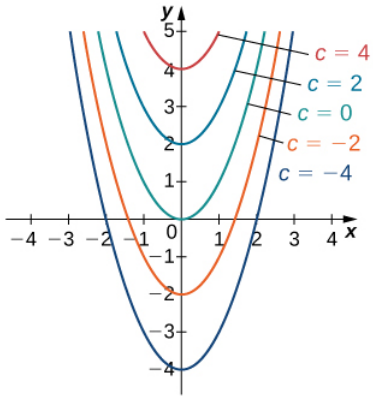


Figure 8.1.1: Family of solutions to the differential equation $y' = 2x$.

In this example, we are free to choose any solution we wish; for example, $y = x^2 - 3$ is a member of the family of solutions to this differential equation. This is called a particular solution to the differential equation. A particular solution can often be uniquely identified if we are given additional information about the problem.

Example 8.1.3: Finding a Particular Solution

Find the particular solution to the differential equation $y' = 2x$ passing through the point $(2, 7)$.

Solution

Any function of the form $y = x^2 + C$ is a solution to this differential equation. To determine the value of C, we substitute the values $x = 2$ and $y = 7$ into this equation and solve for C:

$$\begin{aligned} y &= x^2 + C \\ 7 &= 2^2 + C \\ &= 4 + C \\ C &= 3. \end{aligned}$$

Therefore the particular solution passing through the point $(2, 7)$ is $y = x^2 + 3$.

Exercise 8.1.3

Find the particular solution to the differential equation

$$y' = 4x + 3$$

passing through the point $(1, 7)$, given that $y = 2x^2 + 3x + C$ is a general solution to the differential equation.

Hint

First substitute $x = 1$ and $y = 7$ into the equation, then solve for C.

Answer

$$y = 2x^2 + 3x + 2$$

Initial-Value Problems

Usually a given differential equation has an infinite number of solutions, so it is natural to ask which one we want to use. To choose one solution, more information is needed. Some specific information that can be useful is an *initial value*, which is an ordered pair that is used to find a particular solution.

A differential equation together with one or more initial values is called an *initial-value problem*. The general rule is that the number of initial values needed for an initial-value problem is equal to the order of the differential equation. For example, if we have the differential equation $y' = 2x$, then $y(3) = 7$ is an initial value, and when taken together, these equations form an initial-value problem. The differential equation $y'' - 3y' + 2y = 4e^x$ is second order, so we need two initial values. With initial-value problems of order greater than one, the same value should be used for the independent variable. An example of initial values for this second-order equation would be $y(0) = 2$ and $y'(0) = -1$. These two initial values together with the differential equation form an initial-value problem. These problems are so named because often the independent variable in the unknown function is t, which represents time. Thus, a value of $t = 0$ represents the beginning of the problem.

Example 8.1.4: Verifying a Solution to an Initial-Value Problem

Verify that the function $y = 2e^{-2t} + e^t$ is a solution to the initial-value problem

$$y' - y = -6e^{-2t}, \quad y(0) = 3.$$

Solution

For a function to satisfy an initial-value problem, it must satisfy both the differential equation and the initial condition. To show that y satisfies the differential equation, we start by calculating y'. This gives $y' = -4e^{-2t} + e^t$. Next we substitute both y and y' into the left-hand side of the differential equation and simplify:

$$\begin{aligned} y' - y &= (-4e^{-2t} + e^t) - (2e^{-2t} + e^t) \\ &= -4e^{-2t} + e^t - 2e^{-2t} - e^t = -6e^{-2t}. \end{aligned}$$

This is equal to the right-hand side of the differential equation, so $y = 2e^{-2t} + e^t$ solves the differential equation. Next we calculate $y(0)$:

$$y(0) = 2e^{-2(0)} + e^0 = 2 + 1 = 3.$$

This result verifies the initial value. Therefore the given function satisfies the initial-value problem.

Exercise 8.1.4

Verify that $y = 3e^{2t} + 4\sin t$ is a solution to the initial-value problem

$$y' - 2y = 4\cos t - 8\sin t, y(0) = 3.$$

Hint

First verify that y solves the differential equation. Then check the initial value.

In Example 8.1.4, the initial-value problem consisted of two parts. The first part was the differential equation $y' - y = -6e^{-2t}$, and the second part was the initial value $y(0) = 3$. These two equations together formed the initial-value problem.

The same is true in general. An initial-value problem will consists of two parts: the differential equation and the initial condition. The differential equation has a family of solutions, and the initial condition determines the value of C. The family of solutions to the differential equation in Example 8.1.4 is given by $y = 2e^{-2t} + Ce^t$. This family of solutions is shown in Figure 8.1.2, with the particular solution $y = 2e^{-2t} + e^t$ labeled.

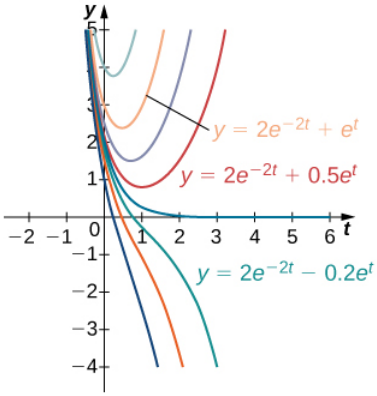


Figure 8.1.2: A family of solutions to the differential equation $y' - y = -6e^{-2t}$. The particular solution $y = 2e^{-2t} + e^t$ is labeled.

Example 8.1.5: Solving an Initial-value Problem

Solve the following initial-value problem:

$$y' = 3e^x + x^2 - 4, y(0) = 5.$$

Solution

The first step in solving this initial-value problem is to find a general family of solutions. To do this, we find an antiderivative of both sides of the differential equation

$$\int y'\,dx = \int (3e^x + x^2 - 4)\,dx,$$

namely,

$$y + C_1 = 3e^x + \tfrac{1}{3}x^3 - 4x + C_2.$$

We are able to integrate both sides because the y term appears by itself. Notice that there are two integration constants: C_1 and C_2. Solving this equation for y gives

$$y = 3e^x + \tfrac{1}{3}x^3 - 4x + C_2 - C_1.$$

Because C_1 and C_2 are both constants, $C_2 - C_1$ is also a constant. We can therefore define $C = C_2 - C_1$, which leads to the equation

$$y = 3e^x + \tfrac{1}{3}x^3 - 4x + C.$$

Next we determine the value of C. To do this, we substitute $x = 0$ and $y = 5$ into this equation and solve for C:

$$\begin{aligned} 5 &= 3e^0 + \frac{1}{3}0^3 - 4(0) + C \\ 5 &= 3 + C \\ C &= 2 \end{aligned}$$

Now we substitute the value $C = 2$ into the general equation. The solution to the initial-value problem is $y = 3e^x + \frac{1}{3}x^3 - 4x + 2$.

Analysis

The difference between a general solution and a particular solution is that a general solution involves a family of functions, either explicitly or implicitly defined, of the independent variable. The initial value or values determine which particular solution in the family of solutions satisfies the desired conditions.

Exercise 8.1.5

Solve the initial-value problem

$$y' = x^2 - 4x + 3 - 6e^x, y(0) = 8.$$

Hint

First take the antiderivative of both sides of the differential equation. Then substitute $x = 0$ and $y = 8$ into the resulting equation and solve for C.

Answer

$y = \frac{1}{3}x^3 - 2x^2 + 3x - 6e^x + 14$

Applications

In physics and engineering applications, we often consider the forces acting upon an object, and use this information to understand the resulting motion that may occur. For example, if we start with an object at Earth's surface, the primary force acting upon that object is gravity. Physicists and engineers can use this information, along with Newton's **second law of motion** (in equation form $F = ma$, where F represents force, m represents mass, and a represents acceleration), to derive an equation that can be solved.

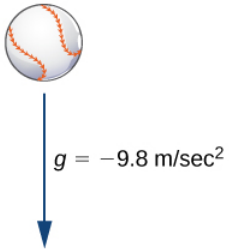


Figure 8.1.3: For a baseball falling in air, the only force acting on it is gravity (neglecting air resistance).

In Figure 8.1.3 we assume that the only force acting on a baseball is the force of gravity. This assumption ignores air resistance. (The force due to air resistance is considered in a later discussion.) The acceleration due to gravity at Earth's surface, g, is approximately $9.8\ \text{m/s}^2$. We introduce a frame of reference, where Earth's surface is at a height of 0 meters. Let $v(t)$ represent the velocity of the object in meters per second. If $v(t) > 0$, the ball is rising, and if $v(t) < 0$, the ball is falling (Figure).



Figure 8.1.4: Possible velocities for the rising/falling baseball.

Our goal is to solve for the velocity $v(t)$ at any time t. To do this, we set up an initial-value problem. Suppose the mass of the ball is m, where m is measured in kilograms. We use Newton's second law, which states that the force acting on an object is equal to its mass times its acceleration $(F = ma)$. Acceleration is the derivative of velocity, so $a(t) = v'(t)$. Therefore the force acting on the baseball is given by $F = mv'(t)$. However, this force must be equal to the force of gravity acting on the object, which (again using Newton's second law) is given by $F_g = -mg$, since this force acts in a downward direction. Therefore we obtain the equation $F = F_g$, which becomes $mv'(t) = -mg$. Dividing both sides of the equation by m gives the equation

$$v'(t) = -g.$$

Notice that this differential equation remains the same regardless of the mass of the object.

We now need an initial value. Because we are solving for velocity, it makes sense in the context of the problem to assume that we know the **initial velocity**, or the velocity at time $t = 0$. This is denoted by $v(0) = v_0$.

Example 8.1.6: Velocity of a Moving Baseball

A baseball is thrown upward from a height of 3 meters above Earth's surface with an initial velocity of 10 m/s, and the only force acting on it is gravity. The ball has a mass of 0.15 kg at Earth's surface.

a. Find the velocity $v(t)$ of the basevall at time t.
b. What is its velocity after 2 seconds?

Solution

a. From the preceding discussion, the differential equation that applies in this situation is

$$v'(t) = -g,$$

where $g = 9.8\ \text{m/s}^2$. The initial condition is $v(0) = v_0$, where $v_0 = 10$ m/s. Therefore the initial-value problem is $v'(t) = -9.8\ \text{m/s}^2$, $v(0) = 10$ m/s.

The first step in solving this initial-value problem is to take the antiderivative of both sides of the differential equation. This gives

$$\int v'(t)\,dt = \int -9.8\,dt$$

$$v(t) = -9.8t + C.$$

The next step is to solve for C. To do this, substitute $t = 0$ and $v(0) = 10$:

$$\begin{aligned} v(t) &= -9.8t + C \\ v(0) &= -9.8(0) + C \\ 10 &= C. \end{aligned}$$

Therefore $C = 10$ and the velocity function is given by $v(t) = -9.8t + 10$.

b. To find the velocity after 2 seconds, substitute $t = 2$ into $v(t)$.

$$\begin{aligned} v(t) &= -9.8t + 10 \\ v(2) &= -9.8(2) + 10 \\ v(2) &= -9.6 \end{aligned}$$

The units of velocity are meters per second. Since the answer is negative, the object is falling at a speed of 9.6 m/s.

Exercise 8.1.6

Suppose a rock falls from rest from a height of 100 meters and the only force acting on it is gravity. Find an equation for the velocity $v(t)$ as a function of time, measured in meters per second.

Hint

What is the initial velocity of the rock? Use this with the differential equation in Example 8.1.6 to form an initial-value problem, then solve for $v(t)$.

Answer

$v(t) = -9.8t$

A natural question to ask after solving this type of problem is how high the object will be above Earth's surface at a given point in time. Let $s(t)$ denote the height above Earth's surface of the object, measured in meters. Because velocity is the derivative of position (in this case height), this assumption gives the equation $s'(t) = v(t)$. An initial value is necessary; in this case the initial height of the object works well. Let the initial height be given by the equation $s(0) = s_0$. Together these assumptions give the initial-value problem

$$s'(t) = v(t), s(0) = s_0.$$

If the velocity function is known, then it is possible to solve for the position function as well.

Example 8.1.7: Height of a Moving Baseball

A baseball is thrown upward from a height of 3 meters above Earth's surface with an initial velocity of $10m/s$, and the only force acting on it is gravity. The ball has a mass of 0.15 kilogram at Earth's surface.

a. Find the position $s(t)$ of the baseball at time t.
b. What is its height after 2 seconds?

Solution

We already know the velocity function for this problem is $v(t) = -9.8t + 10$. The initial height of the baseball is 3 meters, so $s_0 = 3$. Therefore the initial-value problem for this example is

To solve the initial-value problem, we first find the antiderivatives:

$$\int s'(t)\,dt = \int (-9.8t + 10)\,dt$$

$$s(t) = -4.9t^2 + 10t + C.$$

Next we substitute $t = 0$ and solve for C:

$$s(t) = -4.9t^2 + 10t + C$$

$$s(0) = -4.9(0)^2 + 10(0) + C$$

$$3 = C.$$

Therefore the position function is $s(t) = -4.9t^2 + 10t + 3$.

b. The height of the baseball after 2 sec is given by $s(2)$:

$$s(2) = -4.9(2)^2 + 10(2) + 3 = -4.9(4) + 23 = 3.4.$$

Therefore the baseball is 3.4 meters above Earth's surface after 2 seconds. It is worth noting that the mass of the ball cancelled out completely in the process of solving the problem.

Key Concepts

- A differential equation is an equation involving a function $y = f(x)$ and one or more of its derivatives. A solution is a function $y = f(x)$ that satisfies the differential equation when f and its derivatives are substituted into the equation.
- The order of a differential equation is the highest order of any derivative of the unknown function that appears in the equation.
- A differential equation coupled with an initial value is called an initial-value problem. To solve an initial-value problem, first find the general solution to the differential equation, then determine the value of the constant. Initial-value problems have many applications in science and engineering.

Glossary

differential equation

an equation involving a function $y = y(x)$ and one or more of its derivatives

general solution (or family of solutions)

the entire set of solutions to a given differential equation

initial value(s)

a value or set of values that a solution of a differential equation satisfies for a fixed value of the independent variable

initial velocity

the velocity at time $t = 0$

initial-value problem

a differential equation together with an initial value or values

order of a differential equation

the highest order of any derivative of the unknown function that appears in the equation

particular solution

member of a family of solutions to a differential equation that satisfies a particular initial condition

solution to a differential equation

a function $y = f(x)$ that satisfies a given differential equation

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8.1E: Exercises for Section 8.1

In exercises 1 - 7, determine the order of each differential equation.

1) $y'+y=3y^2$

Answer

1st-order

2) $(y')^2=y'+2y$

3) $y'''+y''y'=3x^2$

Answer

3rd-order

4) $y'=y''+3t^2$

5) $\frac{dy}{dt}=t$

Answer

1st-order

6) $\frac{dy}{dx}+\frac{d^2y}{dx^2}=3x^4$

7) $\left(\frac{dy}{dt}\right)^2+8\frac{dy}{dt}+3y=4t$

Answer

1st-order

In exercises 8 - 17, verify that the given function is a solution to the given differential equation.

8) $y=\frac{x^3}{3}$ solves $y'=x^2$

9) $y=2e^{-x}+x-1$ solves $y'=x-y$

10) $y=e^{3x}-\frac{e^x}{2}$ solves $y'=3y+e^x$

11) $y=\frac{1}{1-x}$ solves $y'=y^2$

12) $y=\frac{e^{x^2}}{2}$ solves $y'=2xy$

13) $y=4+\ln x$ solves $xy'=1$

14) $y=3-x+x\ln x$ solves $y'=\ln x$

15) $y=2e^x-x-1$ solves $y'=y+x$

16) $y=e^x+\frac{\sin x}{2}-\frac{\cos x}{2}$ solves $y'=\cos x+y$

17) $y=\pi e^{-\cos x}$ solves $y'=y\sin x$

In exercises 18 - 27, verify the given general solution and find the particular solution.

18) Find the particular solution to the differential equation $y'=4x^2$ that passes through $(-3,-30)$, given that $y=C+\frac{4x^3}{3}$ is a general solution.

19) Find the particular solution to the differential equation $y'=3x^3$ that passes through $(1,4.75)$, given that $y=C+\frac{3x^4}{4}$ is a general solution.

Answer

$y = 4 + \frac{3x^4}{4}$

20) Find the particular solution to the differential equation $y' = 3x^2y$ that passes through $(0, 12)$, given that $y = Ce^{x^3}$ is a general solution.

21) Find the particular solution to the differential equation $y' = 2xy$ that passes through $(0, \frac{1}{2})$, given that $y = Ce^{x^2}$ is a general solution.

Answer

$y = \frac{1}{2}e^{x^2}$

22) Find the particular solution to the differential equation $y' = \left(2xy\right)^2$ that passes through $(1, -\frac{1}{2})$, given that $y = -\frac{3}{C + 4x^3}$ is a general solution.

23) Find the particular solution to the differential equation $y'x^2 = y$ that passes through $(1, \frac{2}{e})$, given that $y = Ce^{-1/x}$ is a general solution.

Answer

$y = 2e^{-1/x}$

24) Find the particular solution to the differential equation $8\frac{dx}{dt} = -2\cos(2t) - \cos(4t)$ that passes through (π, π), given that $x = C - \frac{1}{8}\sin(2t) - \frac{1}{32}\sin(4t)$ is a general solution.

25) Find the particular solution to the differential equation $\frac{du}{dt} = \tan u$ that passes through $(1, \frac{\pi}{2})$, given that $u = \sin^{-1}\left(e^{C+t}\right)$ is a general solution.

Answer

$u = \sin^{-1}\left(e^{-1+t}\right)$

26) Find the particular solution to the differential equation $\frac{dy}{dt} = e^{t+y}$ that passes through $(1, 0)$, given that $y = -\ln(C - e^t)$ is a general solution.

27) Find the particular solution to the differential equation $y'(1 - x^2) = 1 + y$ that passes through $(0, -2)$, given that $y = C\frac{\sqrt{x+1}}{\sqrt{1-x}} - 1$ is a general solution.

Answer

$y = -\frac{\sqrt{x+1}}{\sqrt{1-x}} - 1$

In exercises 28 - 37, find the general solution to the differential equation.

28) $y' = 3x + e^x$

29) $y' = \ln x + \tan x$

Answer

$y = C - x + x\ln x - \ln(\cos x)$

30) $y' = \sin x e^{\cos x}$

31) $y' = 4^x$

Answer

$y = C + \frac{4^x}{\ln 4}$

32) $y' = \sin^{-1}(2x)$

33) $y' = 2t\sqrt{t^2+16}$

Answer

$y = \frac{2}{3}\sqrt{t^2+16}\left(t^2+16\right) + C$

34) $x' = \coth t + \ln t + 3t^2$

35) $x' = t\sqrt{4+t}$

Answer

$x = \frac{2}{15}\sqrt{4+t}\left(3t^2+4t-32\right) + C$

36) $y' = y$

37) $y' = \frac{y}{x}$

Answer

$y = Cx$

In exercises 38 - 42, solve the initial-value problems starting from $y(t=0) = 1$ and $y(t=0) = -1$. Draw both solutions on the same graph.

38) $\frac{dy}{dt} = 2t$

39) $\frac{dy}{dt} = -t$

Answer

$y = 1 - \frac{t^2}{2}$, and $y = -\frac{t^2}{2} - 1$

40) $\frac{dy}{dt} = 2y$

41) $\frac{dy}{dt} = -y$

Answer

$y = e^{-t}$ and $y = -e^{-t}$

42) $\frac{dy}{dt} = 2$

In exercises 43 - 47, solve the initial-value problems starting from $y_0 = 10$. At what time does y increase to 100 or drop to 1?

43) $\frac{dy}{dt} = 4t$

Answer

$y = 2(t^2+5)$, When $t = 3\sqrt{5}$, y will increase to 100.

44) $\frac{dy}{dt} = 4y$

45) $\frac{dy}{dt} = -2y$

Answer

$y = 10e^{-2t}$, When $t = -\frac{1}{2}\ln\left(\frac{1}{10}\right)$, y will decrease to 1.

46) $\frac{dy}{dt} = e^{4t}$

47) $\frac{dy}{dt} = e^{-4t}$

Answer

$y = \frac{1}{4}(41 - e^{-4t})$, Neither condition will ever happen.

Recall that a family of solutions includes solutions to a differential equation that differ by a constant. For exercises 48 - 52, use your calculator to graph a family of solutions to the given differential equation. Use initial conditions from $y(t = 0) = -10$ to $y(t = 0) = 10$ increasing by 2. Is there some critical point where the behavior of the solution begins to change?

48) [T] $y' = y(x)$

49) [T] $xy' = y$

Answer

Solution changes from increasing to decreasing at $y(0) = 0$.

50) [T] $y' = t^3$

51) [T] $y' = x + y$ (Hint: $y = Ce^x - x - 1$ is the general solution)

Answer

Solution changes from increasing to decreasing at $y(0) = 0$.

52) [T] $y' = x \ln x + \sin x$

53) Find the general solution to describe the velocity of a ball of mass 1 lb that is thrown upward at a rate of a ft/sec.

Answer

$v(t) = -32t + a$

54) In the preceding problem, if the initial velocity of the ball thrown into the air is $a = 25$ ft/s, write the particular solution to the velocity of the ball. Solve to find the time when the ball hits the ground.

55) You throw two objects with differing masses m_1 and m_2 upward into the air with the same initial velocity of a ft/s. What is the difference in their velocity after 1 second?

Answer

0 ft/s

56) [T] You throw a ball of mass 1 kilogram upward with a velocity of $a = 25$ m/s on Mars, where the force of gravity is $g = -3.711$ m/s^2. Use your calculator to approximate how much longer the ball is in the air on Mars.

57) [T] For the previous problem, use your calculator to approximate how much higher the ball went on Mars.

Answer

4.86 meters

58) [T] A car on the freeway accelerates according to $a = 15\cos(\pi t)$, where t is measured in hours. Set up and solve the differential equation to determine the velocity of the car if it has an initial speed of 51 mph. After 40 minutes of driving, what is the driver's velocity?

59) [T] For the car in the preceding problem, find the expression for the distance the car has traveled in time t, assuming an initial distance of 0. How long does it take the car to travel 100 miles? Round your answer to hours and minutes.

Answer

$x = 50t - \frac{15}{\pi^2}\cos(\pi t) + \frac{3}{\pi^2}$, 2 hours 1 minute

60) [T] For the previous problem, find the total distance traveled in the first hour.

61) Substitute $y = Be^{3t}$ into $y' - y = 8e^{3t}$ to find a particular solution.

Answer

$y = 4e^{3t}$

62) Substitute $y = a\cos(2t) + b\sin(2t)$ into $y' + y = 4\sin(2t)$ to find a particular solution.

63) Substitute $y = a + bt + ct^2$ into $y' + y = 1 + t^2$ to find a particular solution.

Answer

$y = 1 - 2t + t^2$

64) Substitute $y = ae^t \cos t + be^t \sin t$ into $y' = 2e^t \cos t$ to find a particular solution.

65) Solve $y' = e^{kt}$ with the initial condition $y(0) = 0$ and solve $y' = 1$ with the same initial condition. As k approaches 0, what do you notice?

Answer

$y = \frac{1}{k}(e^{kt} - 1)$ and $y = t$

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8.2: Direction Fields and Numerical Methods

Learning Objectives

- Draw the direction field for a given first-order differential equation.
- Use a direction field to draw a solution curve of a first-order differential equation.
- Use Euler's Method to approximate the solution to a first-order differential equation.

For the rest of this chapter we will focus on various methods for solving differential equations and analyzing the behavior of the solutions. In some cases it is possible to predict properties of a solution to a differential equation without knowing the actual solution. We will also study numerical methods for solving differential equations, which can be programmed by using various computer languages or even by using a spreadsheet program, such as Microsoft Excel.

Creating Direction Fields

Direction fields (also called slope fields) are useful for investigating first-order differential equations. In particular, we consider a first-order differential equation of the form

$$y' = f(x, y).$$

An applied example of this type of differential equation appears in Newton's law of cooling, which we will solve explicitly later in this chapter. First, though, let us create a direction field for the differential equation

$$T'(t) = -0.4(T - 72).$$

Here $T(t)$ represents the temperature (in degrees Fahrenheit) of an object at time t, and the ambient temperature is $72°F$. Figure 8.2.1 shows the direction field for this equation.

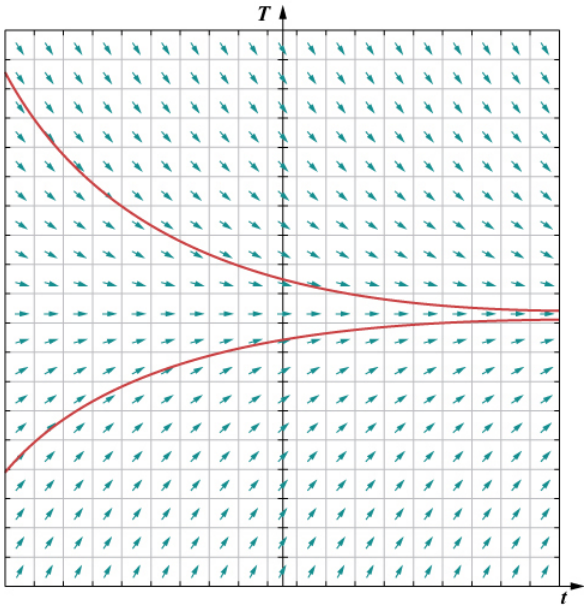


Figure 8.2.1: Direction field for the differential equation $T'(t) = -0.4(T - 72)$. Two solutions are plotted: one with initial temperature less than $72°F$ and the other with initial temperature greater than $72°F$.

The idea behind a direction field is the fact that the derivative of a function evaluated at a given point is the slope of the tangent line to the graph of that function at the same point. Other examples of differential equations for which we can create a direction field include

$$y' = 3x + 2y - 4$$

$$y' = x^2 - y^2$$

$$y' = \frac{2x+4}{y-2}.$$

To create a direction field, we start with the first equation: $y' = 3x + 2y - 4$. We let (x_0, y_0) be any ordered pair, and we substitute these numbers into the right-hand side of the differential equation. For example, if we choose $x = 1$ and $y = 2$, substituting into the right-hand side of the differential equation yields

$$y' = 3x + 2y - 4 = 3(1) + 2(2) - 4 = 3.$$

This tells us that if a solution to the differential equation $y' = 3x + 2y - 4$ passes through the point $(1, 2)$, then the slope of the solution at that point must equal 3. To start creating the direction field, we put a short line segment at the point $(1, 2)$ having slope 3. We can do this for any point in the domain of the function $f(x, y) = 3x + 2y - 4$, which consists of all ordered pairs (x, y) in R^2. Therefore any point in the Cartesian plane has a slope associated with it, assuming that a solution to the differential equation passes through that point. The direction field for the differential equation $y' = 3x + 2y - 4$ is shown in Figure $8.2.2$.

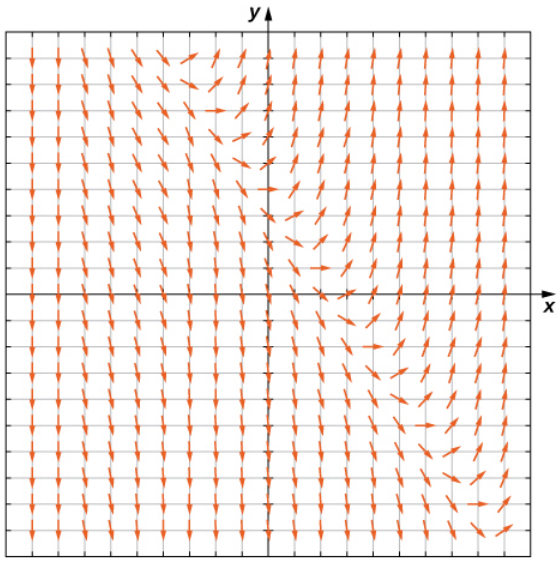


Figure $8.2.2$: Direction field for the differential equation $y' = 3x + 2y - 4$.

We can generate a direction field of this type for any differential equation of the form $y' = f(x, y)$.

Definition: Direction Field (Slope Field)

A direction field (slope field) is a mathematical object used to graphically represent solutions to a first-order differential equation. At each point in a direction field, a line segment appears whose slope is equal to the slope of a solution to the differential equation passing through that point.

Using Direction Fields

We can use a direction field to predict the behavior of solutions to a differential equation without knowing the actual solution. For example, the direction field in Figure $8.2.3$ serves as a guide to the behavior of solutions to the differential equation $y' = 3x + 2y - 4$.

To use a direction field, we start by choosing any point in the field. The line segment at that point serves as a signpost telling us what direction to go from there. For example, if a solution to the differential equation passes through the point $(0, 1)$, then the slope of the solution passing through that point is given by $y' = 3(0) + 2(1) - 4 = -2$. Now let x increase slightly, say to $x = 0.1$. Using the method of linear approximations gives a formula for the approximate value of y for $x = 0.1$. In particular,

$$L(x) = y_0 + f'(x_0)(x - x_0) = 1 - 2(x_0 - 0) = 1 - 2x_0.$$

Substituting $x_0 = 0.1$ into $L(x)$ gives an approximate y value of 0.8.

At this point the slope of the solution changes (again according to the differential equation). We can keep progressing, recalculating the slope of the solution as we take small steps to the right, and watching the behavior of the solution. Figure 8.2.3 shows a graph of the solution passing through the point $(0, 1)$.

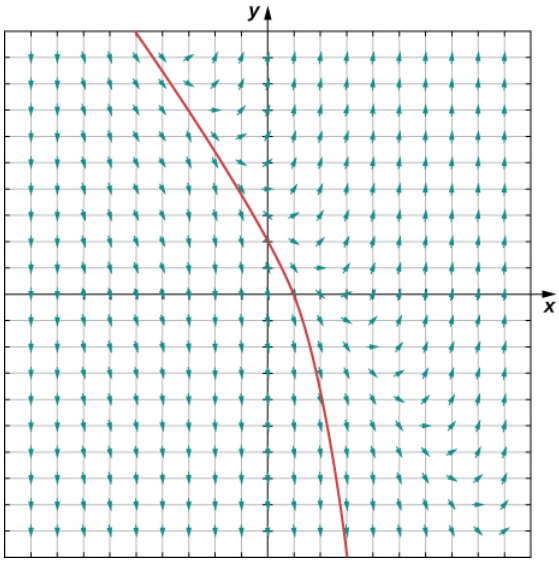


Figure 8.2.3: Direction field for the differential equation $y' = 3x + 2y - 4$ with the solution passing through the point $(0, 1)$.

The curve is the graph of the solution to the initial-value problem

$$y' = 3x + 2y - 4,\ y(0) = 1.$$

This curve is called a solution curve passing through the point $(0, 1)$. The exact solution to this initial-value problem is

$$y = -\frac{3}{2}x + \frac{5}{4} - \frac{1}{4}e^{2x},$$

and the graph of this solution is identical to the curve in Figure 8.2.3.

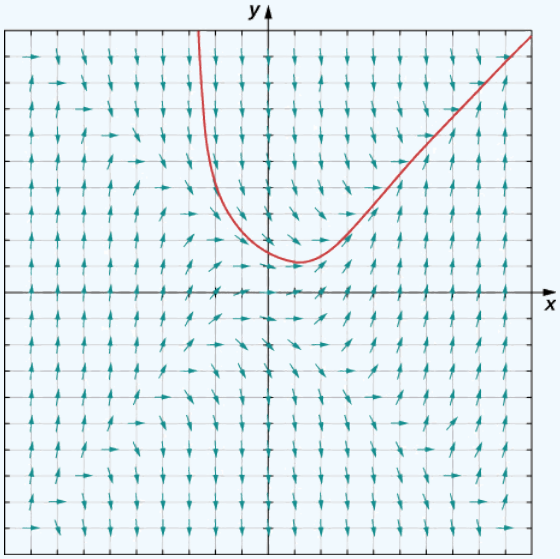
? Exercise 8.2.1

Create a direction field for the differential equation $y' = x^2 - y^2$ and sketch a solution curve passing through the point $(-1, 2)$.

Hint

Use x and y values ranging from -5 to 5. For each coordinate pair, calculate y' using the right-hand side of the differential equation.

Answer



Now consider the direction field for the differential equation $y' = (x-3)(y^2-4)$, shown in Figure 8.2.4. This direction field has several interesting properties. First of all, at $y = -2$ and $y = 2$, horizontal dashes appear all the way across the graph. This means that if $y = -2$, then $y' = 0$. Substituting this expression into the right-hand side of the differential equation gives

$$\begin{aligned}(x-3)(y^2-4) &= (x-3)((-2)^2-4)\\ &= (x-3)(0)\\ &= 0\\ &= y'.\end{aligned}$$

Therefore $y = -2$ is a solution to the differential equation. Similarly, $y = 2$ is a solution to the differential equation. These are the only constant-valued solutions to the differential equation, as we can see from the following argument. Suppose $y = k$ is a constant solution to the differential equation. Then $y' = 0$. Substituting this expression into the differential equation yields $0 = (x-3)(k^2-4)$. This equation must be true for all values of x, so the second factor must equal zero. This result yields the equation $k^2 - 4 = 0$. The solutions to this equation are $k = -2$ and $k = 2$, which are the constant solutions already mentioned. These are called the equilibrium solutions to the differential equation.

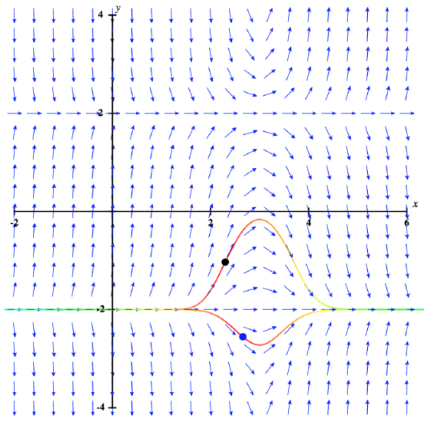


Figure 8.2.4: Direction field for the differential equation $y' = (x-3)(y^2-4)$ showing two solutions. These solutions are very close together, but one is barely above the equilibrium solution $y = -2$ and the other is barely below the same equilibrium solution.

Definition: Equilibrium Solutions

Consider the differential equation $y' = f(x, y)$. An *equilibrium solution* is any solution to the differential equation of the form $y = c$, where c is a constant.

To determine the equilibrium solutions to the differential equation $y' = f(x, y)$, set the right-hand side equal to zero. An equilibrium solution of the differential equation is any function of the form $y = k$ such that $f(x, k) = 0$ for all values of x in the domain of f.

An important characteristic of equilibrium solutions concerns whether or not they approach the line $y = k$ as an asymptote for large values of x.

Definition: asymptotically Stable, Unstable and Semi-Stable Solutions

Consider the differential equation $y' = f(x, y)$, and assume that all solutions to this differential equation are defined for $x \geq x_0$. Let $y = k$ be an equilibrium solution to the differential equation.

1. $y = k$ is an *asymptotically stable solution* to the differential equation if there exists $\varepsilon > 0$ such that for any value $c \in (k - \varepsilon,\ k + \varepsilon)$ the solution to the initial-value problem $y' = f(x, y),\ y(x_0) = c$ approaches k as x approaches infinity.
2. $y = k$ is an *asymptotically unstable solution* to the differential equation if there exists $\varepsilon > 0$ such that for any value $c \in (k - \varepsilon,\ k + \varepsilon)$ the solution to the initial-value problem $y' = f(x, y),\ y(x_0) = c$ never approaches k as x approaches infinity.
3. $y = k$ is an *asymptotically semi-stable solution* to the differential equation if it is neither asymptotically stable nor asymptotically unstable.

Now we return to the differential equation $y' = (x - 3)(y^2 - 4)$, with the initial condition $y(0) = 0.5$. The direction field for this initial-value problem, along with the corresponding solution, is shown in Figure 8.2.5.

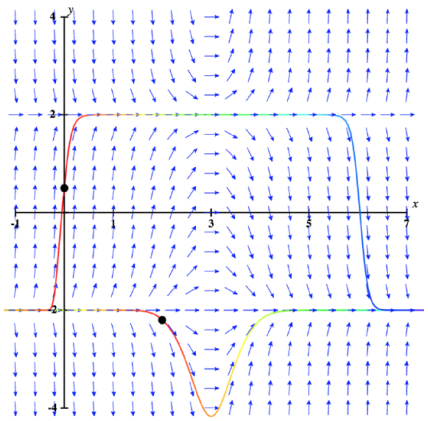


Figure 8.2.5: Direction field for the initial-value problem $y' = (x - 3)(y^2 - 4),\ y(0) = 0.5$. A second solution curve is graphed with initial condition $(2, -2.2)$.

The values of the solution to this initial-value problem stay between $y = -2$ and $y = 2$, which are the equilibrium solutions to the differential equation. Furthermore, as x approaches infinity, although they initially appear to approach the line y = 2, the y-coordinates clearly approach -2. The behavior of solutions is similar if the initial value is below -2, for example, $y(2) = -2.2$. In this case, the solutions increase and approach $y = -2$ as x approaches infinity. Therefore $y = -2$ is an asymptotically stable solution to this differential equation.

What happens when the initial value is above $y = 2$? This scenario is illustrated in Figure 8.2.6, with the initial value $y(0) = 3$.

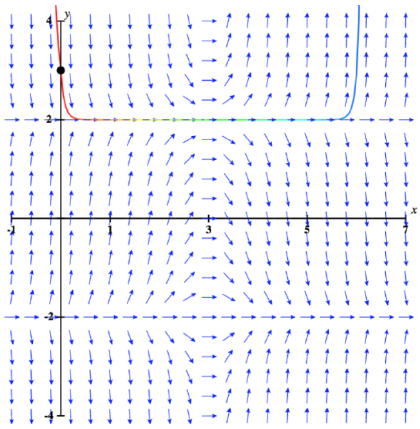


Figure 8.2.6: Direction field for the initial-value problem $y' = (x-3)(y^2-4),\ y(0) = 3$.

The solution increases rapidly toward positive infinity as x approaches infinity. Furthermore, if the initial value is slightly below 2, then the solution approaches -2, which is the other equilibrium solution. Therefore in neither case does the solution approach $y = 2$, so $y = 2$ is called an asymptotically unstable, or unstable, equilibrium solution.

Example 8.2.1: Stability of an Equilibrium Solution

Create a direction field for the differential equation $y' = (y-3)^2(y^2+y-2)$ and identify any equilibrium solutions. Classify each of the equilibrium solutions as stable, unstable, or semi-stable.

Solution

The direction field is shown in Figure 8.2.7.

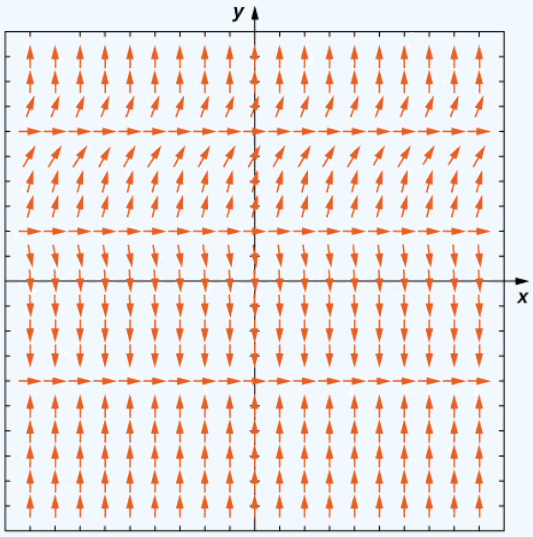


Figure 8.2.7: Direction field for the differential equation $y' = (y-3)^2(y^2+y-2)$.

The equilibrium solutions are $y = -2,\ y = 1$, and $y = 3$. To classify each of the solutions, look at an arrow directly above or below each of these values. For example, at $y = -2$ the arrows directly below this solution point up, and the arrows directly above the solution point down. Therefore all initial conditions close to $y = -2$ approach $y = -2$, and the solution is stable. For the solution $y = 1$, all initial conditions above and below $y = 1$ are repelled (pushed away) from $y = 1$, so this solution is unstable. The solution $y = 3$ is semi-stable, because for initial conditions slightly greater than 3, the solution approaches infinity, and for initial conditions slightly less than 3, the solution approaches $y = 1$.

Analysis

It is possible to find the equilibrium solutions to the differential equation by setting the right-hand side equal to zero and solving for y. This approach gives the same equilibrium solutions as those we saw in the direction field.

Exercise 8.2.2

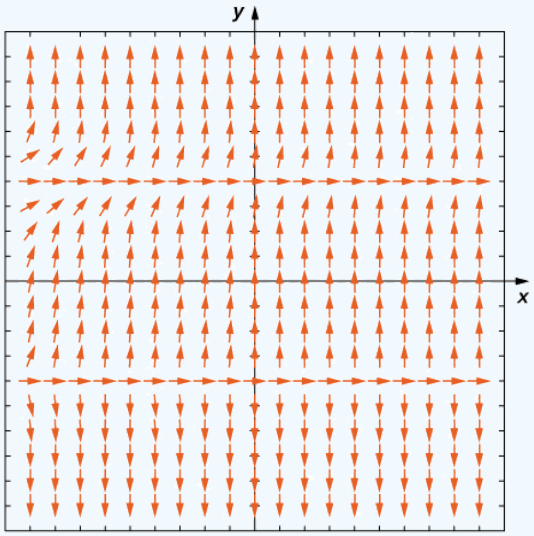
Create a direction field for the differential equation $y' = (x + 5)(y + 2)(y^2 - 4y + 4)$ and identify any equilibrium solutions. Classify each of the equilibrium solutions as stable, unstable, or semi-stable.

Hint

First create the direction field and look for horizontal dashes that go all the way across. Then examine the slope lines directly above and below the equilibrium solutions.

Answer

The equilibrium solutions are $y = -2$ and $y = 2$. For this equation, $y = -2$ is an unstable equilibrium solution, and $y = 2$ is a semi-stable equilibrium solution.



Euler's Method

Consider the initial-value problem

$$y' = 2x - 3,\ y(0) = 3.$$

Integrating both sides of the differential equation gives $y = x^2 - 3x + C$, and solving for C yields the particular solution $y = x^2 - 3x + 3$. The solution for this initial-value problem appears as the parabola in Figure 8.2.8.

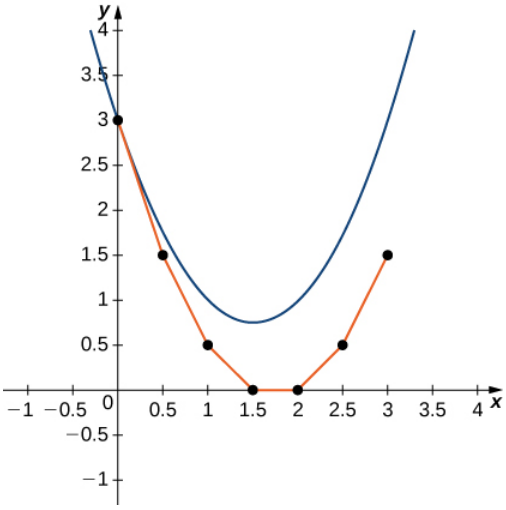


Figure 8.2.8: Euler's Method for the initial-value problem $y' = 2x - 3, y(0) = 3$.

The red graph consists of line segments that approximate the solution to the initial-value problem. The graph starts at the same initial value of $(0, 3)$. Then the slope of the solution at any point is determined by the right-hand side of the differential equation, and the length of the line segment is determined by increasing the x value by 0.5 each time (the step size). This approach is the basis of Euler's Method.

Before we state Euler's Method as a theorem, let's consider another initial-value problem:

$$y' = x^2 - y^2, \; y(-1) = 2.$$

The idea behind direction fields can also be applied to this problem to study the behavior of its solution. For example, at the point $(-1, 2)$, the slope of the solution is given by $y' = (-1)^2 - 2^2 = -3$, so the slope of the tangent line to the solution at that point is also equal to -3. Now we define $x_0 = -1$ and $y_0 = 2$. Since the slope of the solution at this point is equal to -3, we can use the method of linear approximation to approximate y near $(-1, 2)$.

$$L(x) = y_0 + f'(x_0)(x - x_0).$$

Here $x_0 = -1, y_0 = 2$, and $f'(x_0) = -3$, so the linear approximation becomes

$$L(x) = 2 - 3(x - (-1)) = 2 - 3x - 3 = -3x - 1.$$

Now we choose a **step size**. The step size is a small value, typically 0.1 or less, that serves as an increment for x; it is represented by the variable h. In our example, let $h = 0.1$. Incrementing x_0 by h gives our next x value:

$$x_1 = x_0 + h = -1 + 0.1 = -0.9.$$

We can substitute $x_1 = -0.9$ into the linear approximation to calculate y_1.

$$y_1 = L(x_1) = -3(-0.9) - 1 = 1.7.$$

Therefore the approximate y value for the solution when $x = -0.9$ is $y = 1.7$. We can then repeat the process, using $x_1 = -0.9$ and $y_1 = 1.7$ to calculate x_2 and y_2. The new slope is given by $y' = (-0.9)^2 - (1.7)^2 = -2.08$. First, $x_2 = x_1 + h = -0.9 + 0.1 = -0.8$. Using linear approximation gives

$$\begin{aligned} L(x) &= y_1 + f'(x_1)(x - x_1) \\ &= 1.7 - 2.08(x - (-0.9)) \\ &= 1.7 - 2.08x - 1.872 \\ &= -2.08x - 0.172. \end{aligned}$$

Finally, we substitute $x_2 = -0.8$ into the linear approximation to calculate y_2.

$$\begin{aligned} y_2 &= L(x_2) \\ &= -2.08x_2 - 0.172 \\ &= -2.08(-0.8) - 0.172 \\ &= 1.492. \end{aligned}$$

Therefore the approximate value of the solution to the differential equation is $y = 1.492$ when $x = -0.8$.

What we have just shown is the idea behind **Euler's Method**. Repeating these steps gives a list of values for the solution. These values are shown in the table, rounded off to four decimal places.

Table 8.2.1: Using Euler's Method to Approximate Solutions to a Differential Equation

n	0	1	2	3	4	5
x_n	−1	−0.9	−0.8	−0.7	−0.6	−0.5
y_n	2	1.7	1.492	1.3334	1.2046	1.0955
n	6	7	8	9	10	
x_n	−0.4	−0.3	−0.2	−0.1	0	
y_n	1.0004	1.9164	1.8414	1.7746	1.7156	

Euler's method

Consider the initial-value problem

$$y' = f(x, y),\ y(x_0) = y_0.$$

To approximate a solution to this problem using Euler's method, define

$$x_n = x_0 + nh$$

$$y_n = y_{n-1} + hf(x_{n-1}, y_{n-1}).$$

Here $h > 0$ represents the step size and n is an integer, starting with 1. The number of steps taken is counted by the variable n.

Typically h is a small value, say 0.1 or 0.05. The smaller the value of h, the more calculations are needed. The higher the value of h, the fewer calculations are needed. However, the tradeoff results in a lower degree of accuracy for larger step size, as illustrated in Figure 8.2.9.

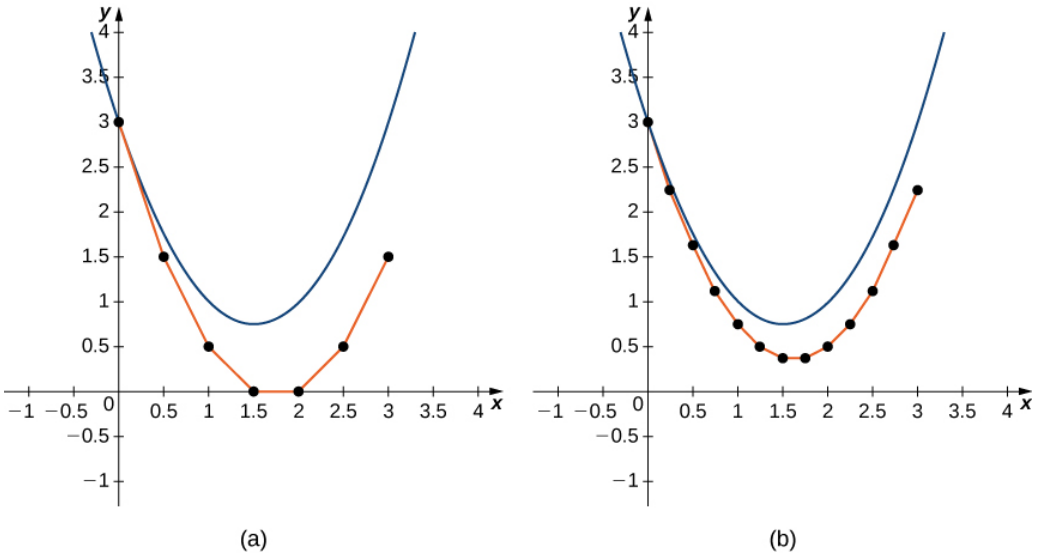


Figure 8.2.9: Euler's method for the initial-value problem $y' = 2x - 3,\ y(0) = 3$ with (a) a step size of $h = 0.5$; and (b) a step size of $h = 0.25$.

Example 8.2.2: Using Euler's Method

Consider the initial-value problem

$$y' = 3x^2 - y^2 + 1,\ y(0) = 2.$$

Use Euler's method with a step size of 0.1 to generate a table of values for the solution for values of x between 0 and 1.

Solution

We are given $h = 0.1$ and $f(x, y) = 3x^2 - y^2 + 1$. Furthermore, the initial condition $y(0) = 2$ gives $x_0 = 0$ and $y_0 = 2$. Using Equation with $n = 0$, we can generate this table.

Table 8.2.2: Using Euler's Method to Approximate Solutions to a Differential Equation

n	x_n	$y_n = y_{n-1} + hf(x_{n-1}, y_{n-1})$
0	0	2
1	0.1	$y_1 = y_0 + hf(x_0, y_0) = 1.7$
2	0.2	$y_2 = y_1 + hf(x_1, y_1) = 1.514$
3	0.3	$y_3 = y_2 + hf(x_2, y_2) = 1.3968$
4	0.4	$y_4 = y_3 + hf(x_3, y_3) = 1.3287$
5	0.5	$y_5 = y_4 + hf(x_4, y_4) = 1.3001$
6	0.6	$y_6 = y_5 + hf(x_5, y_5) = 1.3061$
7	0.7	$y_7 = y_6 + hf(x_6, y_6) = 1.3435$
8	0.8	$y_8 = y_7 + hf(x_7, y_7) = 1.4100$
9	0.9	$y_9 = y_8 + hf(x_8, y_8) = 1.5032$
10	1.0	$y_{10} = y_9 + hf(x_9, y_9) = 1.6202$

With ten calculations, we are able to approximate the values of the solution to the initial-value problem for values of x between 0 and 1.

Go to this website for a chance to visually explore Euler's method.

Exercise 8.2.3

Consider the initial-value problem

$$y' = x^3 + y^2,\ y(1) = -2.$$

Using a step size of 0.1, generate a table with approximate values for the solution to the initial-value problem for values of x between 1 and 2.

Hint

Start by identifying the value of h, then figure out what $f(x, y)$ is. Then use the formula for Euler's Method to calculate $y_1, y_2,$ and so on.

Answer

Table 8.2.3: Using Euler's Method to approximate solutions to the differential equation in Exercise 8.2.3.

n	\(x_n)	$y_n = y_{n-1} + hf(x_{n-1}, y_{n-1})$
0	1	-2
1	1.1	$y_1 = y_0 + hf(x_0, y_0) = -1.5$
2	1.2	$y_2 = y_1 + hf(x_1, y_1) = -1.1419$
3	1.3	$y_3 = y_2 + hf(x_2, y_2) = -0.8387$
4	1.4	$y_4 = y_3 + hf(x_3, y_3) = -0.5487$
5	1.5	$y_5 = y_4 + hf(x_4, y_4) = -0.2442$
6	1.6	$y_6 = y_5 + hf(x_5, y_5) = 0.0993$
7	1.7	$y_7 = y_6 + hf(x_6, y_6) = 0.5099$
8	1.8	$y_8 = y_7 + hf(x_7, y_7) = 1.0272$
9	1.9	$y_9 = y_8 + hf(x_8, y_8) = 1.7159$
10	2	$y_{10} = y_9 + hf(x_9, y_9) = 2.6962$

Key Concepts

- A direction field is a mathematical object used to graphically represent solutions to a first-order differential equation.
- Euler's Method is a numerical technique that can be used to approximate solutions to a differential equation.

Key Equations

- **Euler's Method**

  $x_n = x_0 + nh$

  $y_n = y_{n-1} + hf(x_{n-1}, y_{n-1})$, where h is the step size

Glossary

asymptotically semi-stable solution

$y = k$ if it is neither asymptotically stable nor asymptotically unstable

asymptotically stable solution

$y = k$ if there exists $\varepsilon > 0$ such that for any value $c \in (k - \varepsilon,\ k + \varepsilon)$ the solution to the initial-value problem $y' = f(x, y),\ y(x_0) = c$ approaches k as x approaches infinity

asymptotically unstable solution

$y = k$ if there exists $\varepsilon > 0$ such that for any value $c \in (k - \varepsilon,\ k + \varepsilon)$ the solution to the initial-value problem $y' = f(x, y),\ y(x_0) = c$ never approaches k as $x$approaches infinity

direction field (slope field)

a mathematical object used to graphically represent solutions to a first-order differential equation; at each point in a direction field, a line segment appears whose slope is equal to the slope of a solution to the differential equation passing through that point

equilibrium solution

any solution to the differential equation of the form $y = c$, where c is a constant

Euler's Method

a numerical technique used to approximate solutions to an initial-value problem

solution curve

a curve graphed in a direction field that corresponds to the solution to the initial-value problem passing through a given point in the direction field

step size

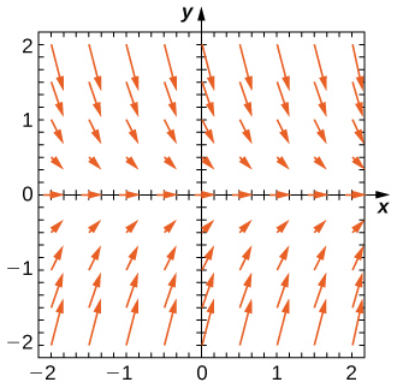
the increment hh that is added to the xx value at each step in Euler's Method

- Gilbert Strang (MIT) and Edwin "Jed" Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.
- Paul Seeburger (Monroe Community College) edited this section to adjust the explanation of equilibrium points in the example shown in Figures 8.2.4 - 8.2.6. He also created these figures.

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8.2E: Exercises for Section 8.2

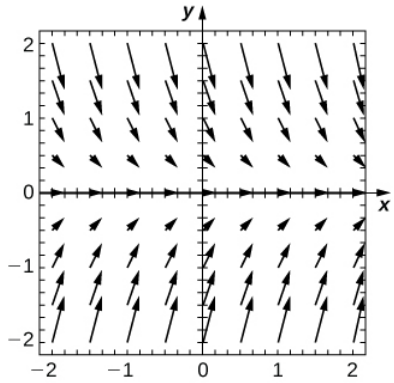
For exercises 1 - 3, use the direction field below from the differential equation $y' = -2y$. Sketch the graph of the solution for the given initial conditions.



1) $y(0) = 1$

2) $y(0) = 0$

Answer



3) $y(0) = -1$

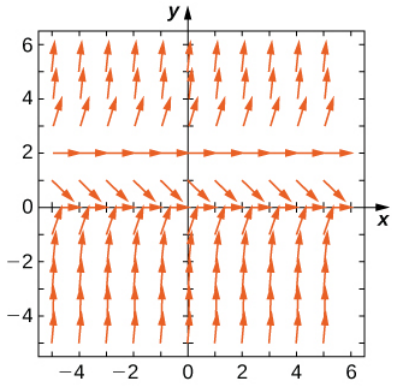
4) Are there any equilibria among the solutions of the differential equation from exercises 1 - 3? List any equilibria along with their stabilities.

Answer

$y = 0$ is a stable equilibrium

For exercises 5 - 7, use the direction field below from the differential equation $y' = y^2 - 2y$. Sketch the graph of the solution for the given initial conditions.

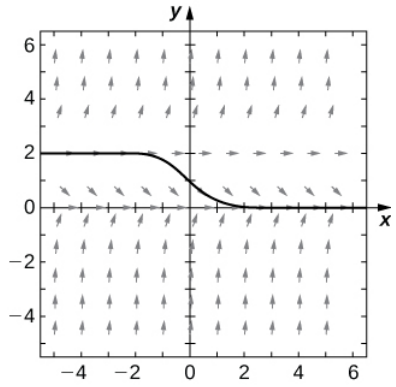
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5) $y(0) = 3$

6) $y(0) = 1$

Answer



7) $y(0) = -1$

8) Are there any equilibria among the solutions of the differential equation from exercises 5 - 7? List any equilibria along with their stabilities.

Answer

$y = 0$ is a stable equilibrium and $y = 2$ is unstable

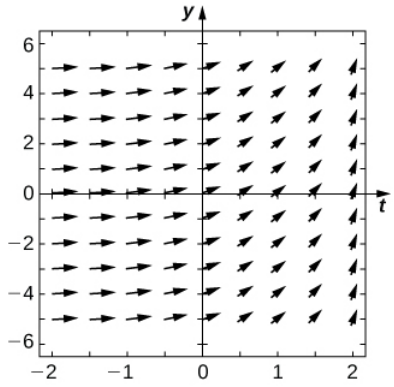
In exercises 9 - 13, draw the direction field for the following differential equations, then solve the differential equation. Draw your solution on top of the direction field. Does your solution follow along the arrows on your direction field?

9) $y' = t^3$

10) $y' = e^t$

Answer

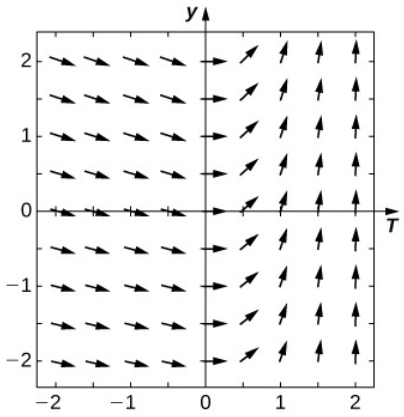
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11) $\frac{dy}{dx} = x^2 \cos x$

12) $\frac{dy}{dt} = te^t$

Answer



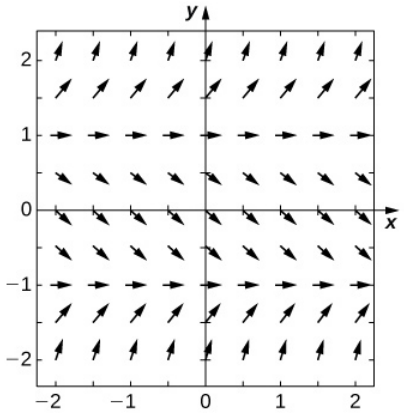
13) $\frac{dx}{dt} = \cosh(t)$

In exercises 14 - 18, draw the directional field for the following differential equations. What can you say about the behavior of the solution? Are there equilibria? What stability do these equilibria have?

14) $y' = y^2 - 1$

Answer

There appear to be equlibria at $y = -1$ (stable) and $y = 1$ (unstable).

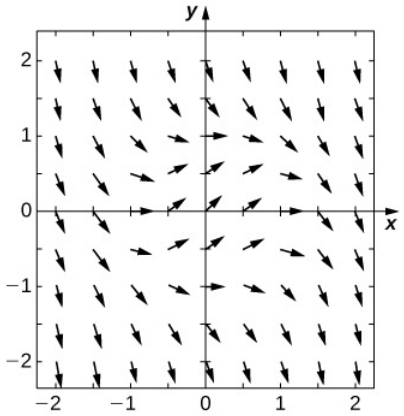


15) $y' = y - x$

16) $y' = 1 - y^2 - x^2$

Answer

There do not appear to be any equilibria.

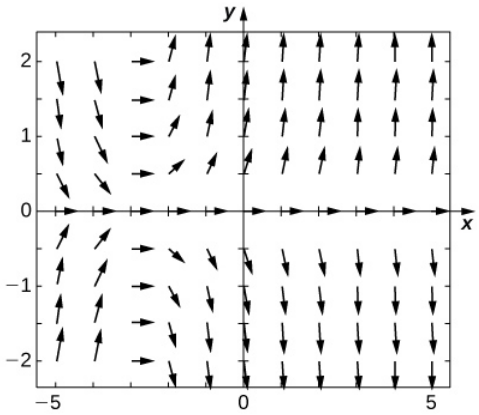


17) $y' = t^2 \sin y$

18) $y' = 3y + xy$

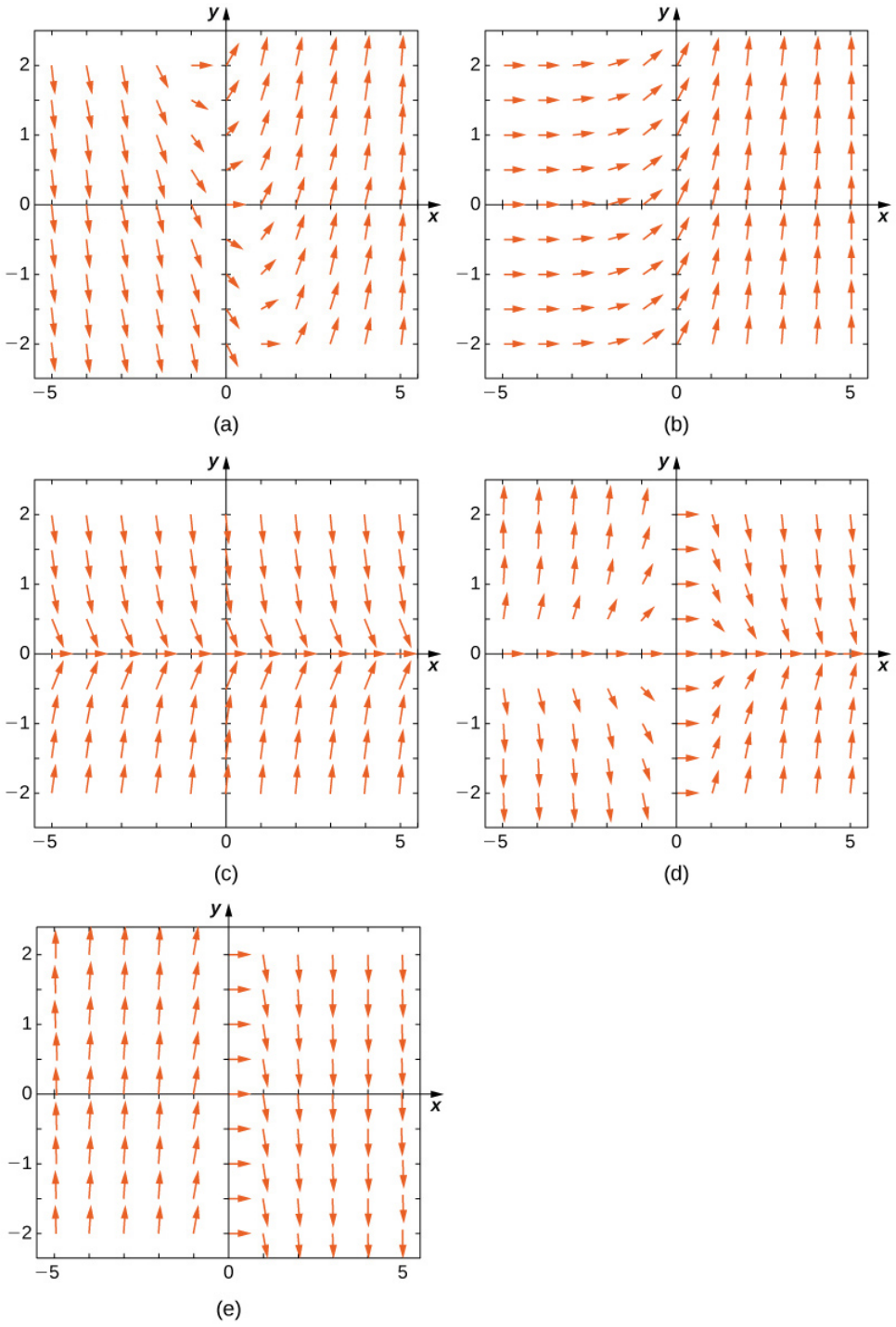
Answer

There appears to be an unstable equilibrium at $y = 0$.



Match the direction field with the given differential equations. Explain your selections.

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19) $y' = -3y$

20) $y' = -3t$

Answer

E

21) $y' = e^t$

22) $y' = \frac{1}{2}y + t$

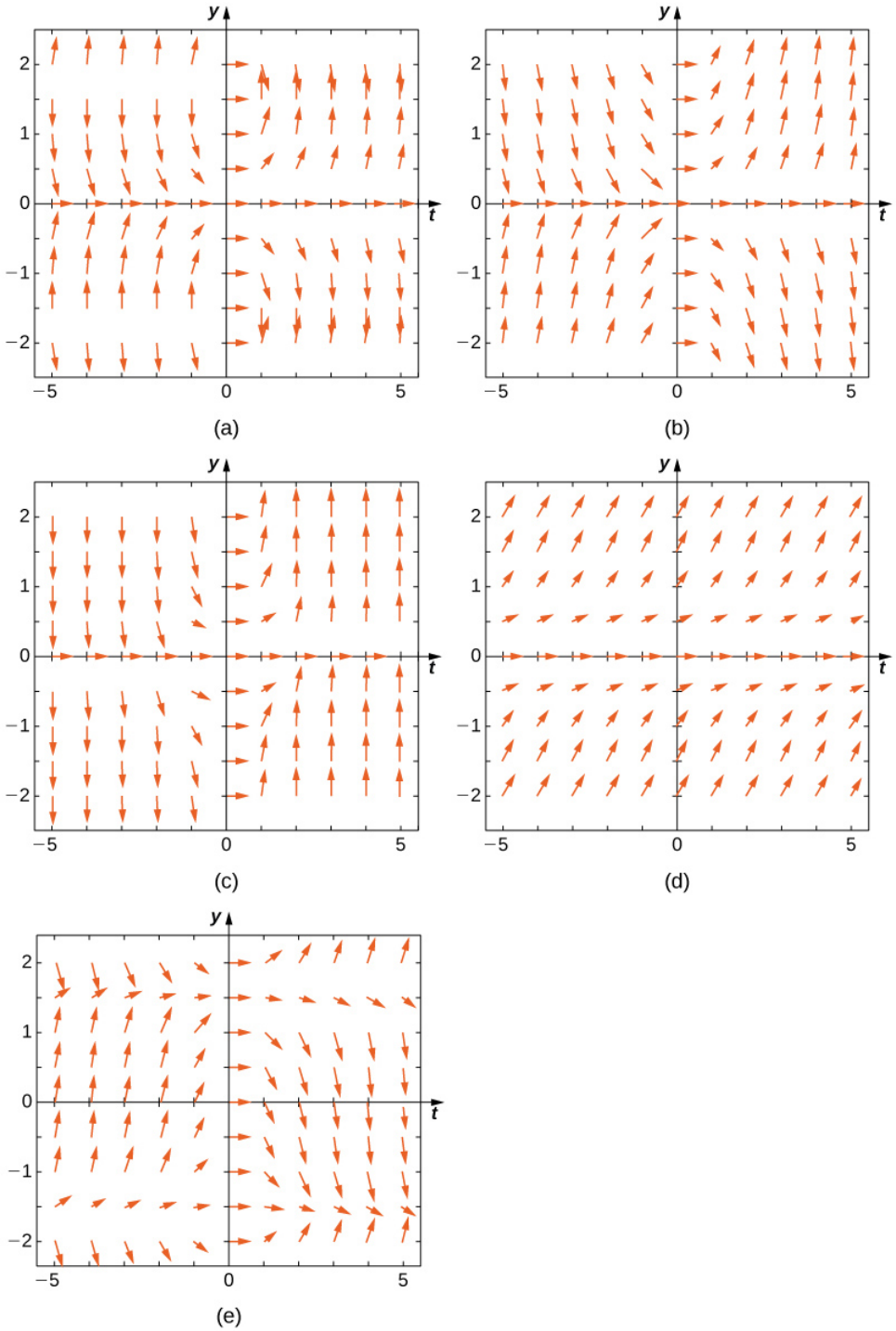
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Answer

A

23) $y' = -ty$

Match the direction field with the given differential equations. Explain your selections.



24) $y' = t \sin y$

Answer

B

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25) $y' = -t\cos y$

26) $y' = t\tan y$

Answer

A

27) $y' = \sin^2 y$

28) $y' = y^2 t^3$

Answer

C

Estimate the following solutions using Euler's method with $n = 5$ steps over the interval $t = [0, 1]$. If you are able to solve the initial-value problem exactly, compare your solution with the exact solution. If you are unable to solve the initial-value problem, the exact solution will be provided for you to compare with Euler's method. How accurate is Euler's method?

29) $y' = -3y, \quad y(0) = 1$

30) $y' = t^2, \quad y(0) = 1$

Answer

1.24, exact: $\frac{4}{3} \approx 1.33$

Solution:

31) $y' = 3t - y, \quad y(0) = 1$. Exact solution is $y = 3t + 4e^{-t} - 3$

32) $y' = y + t^2, \quad y(0) = 3$. Exact solution is $y = 5e^t - 2 - t^2 - 2t$

Answer

7.739364, exact: $5(e - 1)$

33) $y' = 2t, \quad y(0) = 0$

34) [T] $y' = e^{x+y}, y(0) = -1$. Exact solution is $y = -\ln(e + 1 - e^x)$

Answer

-0.2535, exact: 0

35) $y' = y^2 \ln(x + 1), \quad y(0) = 1$. Exact solution is $y = -\dfrac{1}{(x + 1)(\ln(x + 1) - 1)}$

36) $y' = 2^x, \quad y(0) = 0$. Exact solution is $y = \dfrac{2^x - 1}{\ln 2}$

Answer

1.345, exact: $\frac{1}{\ln(2)}$

37) $y' = y, \quad y(0) = -1$. Exact solution is $y = -e^x$.

38) $y' = -5t, \quad y(0) = -2$. Exact solution is $y = -\frac{5}{2}t^2 - 2$

Answer

-4, exact: $-9/2 = -4.5$

Differential equations can be used to model **disease epidemics.** In the next set of problems, we examine the change of size of two sub-populations of people living in a city: individuals who are infected and individuals who are susceptible to infection. S represents the size of the susceptible population, and I represents the size of the infected population. We assume that if a

susceptible person interacts with an infected person, there is a probability c that the susceptible person will become infected. Each infected person recovers from the infection at a rate r and becomes susceptible again. We consider the case of influenza, where we assume that no one dies from the disease, so we assume that the total population size of the two sub-populations is a constant number, N. The differential equations that model these population sizes are

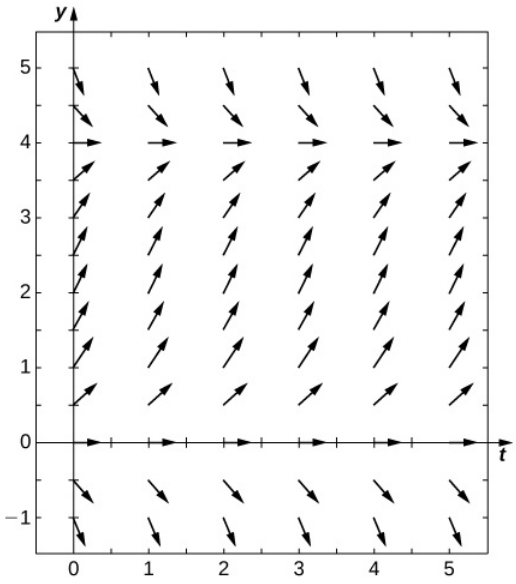
$$S' = rI - cSI \text{ and } I' = cSI - rI.$$

Here c represents the contact rate and r is the recovery rate.

39) Show that, by our assumption that the total population size is constant $(S + I = N)$, you can reduce the system to a single differential equation in $I : I' = c(N - I)I - rI$.

40) Assuming the parameters are $c = 0.5, N = 5$, and $r = 0.5$, draw the resulting directional field.

Answer



41) [T] Use computational software or a calculator to compute the solution to the initial-value problem $y' = ty, y(0) = 2$ using Euler's Method with the given step size h. Find the solution at $t = 1$. For a hint, here is "pseudo-code" for how to write a computer program to perform Euler's Method for $y' = f(t, y), y(0) = 2$:

Create function $f(t, y)$

Define parameters $y(1) = y_0, t(0) = 0$, step size h, and total number of steps, N

Write a for-loop:

for $k = 1$ to N

$fn = f(t(k), y(k))$

$y(k + 1) = y(k) + h * fn$

$t(k + 1) = t(k) + h$

42) Solve the initial-value problem for the exact solution.

Answer

$y' = 2e^{t^2/2}$

43) Draw the directional field

44) $h = 1$

Answer

2

45) [T] $h = 10$

46) [T] $h = 100$

Answer

3.2756

47) [T] $h = 1000$

48) [T] Evaluate the exact solution at $t = 1$. Make a table of errors for the relative error between the Euler's method solution and the exact solution. How much does the error change? Can you explain?

Answer

Exact solution: y $= 2\sqrt{e}$.

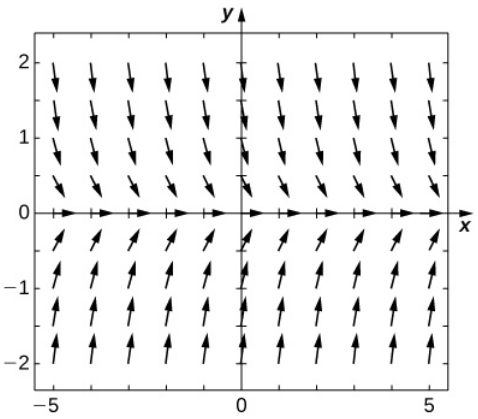
Step Size	Error
$h = 1$	0.3935
$h = 10$	0.06163
$h = 100$	0.006612
$h = 10000$	0.0006661

For exercises 49 - 53, consider the initial-value problem $y' = -2y$, with $y(0) = 2$.

49) Show that $y = 2e^{-2x}$ solves this initial-value problem.

50) Draw the directional field of this differential equation.

Answer



51) [T] By hand or by calculator or computer, approximate the solution using Euler's Method at $t = 10$ using $h = 5$.

52) [T] By calculator or computer, approximate the solution using Euler's Method at $t = 10$ using $h = 100$.

Answer

$4.0741e^{-10}$

53) [T] Plot exact answer and each Euler approximation (for $h = 5$ and $h = 100$) at each h on the directional field. What do you notice?

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8.3: Separable Equations

Learning Objectives

- Use separation of variables to solve a differential equation.
- Solve applications using separation of variables.

We now examine a solution technique for finding exact solutions to a class of differential equations known as separable differential equations. These equations are common in a wide variety of disciplines, including physics, chemistry, and engineering. We illustrate a few applications at the end of the section.

Separation of Variables

We start with a definition and some examples.

Definition: Separable Differential Equations

A *separable differential equation* is any equation that can be written in the form

$$y' = f(x)g(y). \tag{8.3.1}$$

The term 'separable' refers to the fact that the right-hand side of Equation 8.3.1 can be separated into a function of x times a function of y. Examples of separable differential equations include

$$y' = (x^2 - 4)(3y + 2) \tag{8.3.2}$$
$$y' = 6x^2 + 4x \tag{8.3.3}$$
$$y' = \sec y + \tan y \tag{8.3.4}$$
$$y' = xy + 3x - 2y - 6. \tag{8.3.5}$$

Equation 8.3.3 is separable with $f(x) = 6x^2 + 4x$ and $g(y) = 1$, Equation 8.3.4 is separable with $f(x) = 1$ and $g(y) = \sec y + \tan y$, and the right-hand side of Equation 8.3.5 can be factored as $(x + 3)(y - 2)$, so it is separable as well. Equation 8.3.4 is also called an *autonomous differential equation* because the right-hand side of the equation is a function of y alone. If a differential equation is separable, then it is possible to solve the equation using the method of *separation of variables*.

Problem-Solving Strategy: Separation of Variables

1. Check for any values of y that make $g(y) = 0$. These correspond to constant solutions.
2. Rewrite the differential equation in the form
$$\frac{dy}{g(y)} = f(x)dx.$$
3. Integrate both sides of the equation.
4. Solve the resulting equation for y if possible.
5. If an initial condition exists, substitute the appropriate values for x and y into the equation and solve for the constant.

Note that Step 4 states "Solve the resulting equation for y if possible." It is not always possible to obtain y as an explicit function of x. Quite often we have to be satisfied with finding y as an implicit function of x.

Example 8.3.1: Using Separation of Variables

Find a general solution to the differential equation $y' = (x^2 - 4)(3y + 2)$ using the method of separation of variables.

Solution

Follow the five-step method of separation of variables.

1. In this example, $f(x) = x^2 - 4$ and $g(y) = 3y + 2$. Setting $g(y) = 0$ gives $y = -\frac{2}{3}$ as a constant solution.

2. Rewrite the differential equation in the form

$$\frac{dy}{3y+2} = (x^2 - 4)\,dx.$$

3. Integrate both sides of the equation:

$$\int \frac{dy}{3y+2} = \int (x^2 - 4)\,dx.$$

Let $u = 3y + 2$. Then $du = 3\frac{dy}{dx}\,dx$, so the equation becomes

$$\frac{1}{3}\int \frac{1}{u}\,du = \frac{1}{3}x^3 - 4x + C$$

$$\frac{1}{3}\ln|u| = \frac{1}{3}x^3 - 4x + C$$

$$\frac{1}{3}\ln|3y+2| = \frac{1}{3}x^3 - 4x + C.$$

4. To solve this equation for y, first multiply both sides of the equation by 3.

$$\ln|3y+2| = x^3 - 12x + 3C$$

Now we use some logic in dealing with the constant C. Since C represents an arbitrary constant, $3C$ also represents an arbitrary constant. If we call the second arbitrary constant C_1, where $C_1 = 3C$, the equation becomes

$$\ln|3y+2| = x^3 - 12x + C_1.$$

Now exponentiate both sides of the equation (i.e., make each side of the equation the exponent for the base e).

$$e^{\ln|3y+2|} = e^{x^3-12x+C_1}$$
$$|3y+2| = e^{C_1}e^{x^3-12x}$$

Again define a new constant $C_2 = e^{C_1}$ (note that $C_2 > 0$):

$$|3y+2| = C_2e^{x^3-12x}.$$

Because of the absolute value on the left side of the equation, this corresponds to two separate equations:

$$3y + 2 = C_2e^{x^3-12x}$$

and

$$3y + 2 = -C_2e^{x^3-12x}.$$

The solution to either equation can be written in the form

$$y = \frac{-2 \pm C_2e^{x^3-12x}}{3}.$$

Since $C_2 > 0$, it does not matter whether we use plus or minus, so the constant can actually have either sign. Furthermore, the subscript on the constant C is entirely arbitrary, and can be dropped. Therefore the solution can be written as

$$y = \frac{-2 + Ce^{x^3-12x}}{3}, \text{ where } C = \pm C_2 \text{ or } C = 0.$$

Note that in writing a single general solution in this way, we are also allowing C to equal 0. This gives us the singular solution, $y = -\frac{2}{3}$, for the given differential equation. Check that this is indeed a solution of this differential equation!

5. No initial condition is imposed, so we are finished.

Exercise 8.3.1

Use the method of separation of variables to find a general solution to the differential equation

$$y' = 2xy + 3y - 4x - 6.$$

Hint

First factor the right-hand side of the equation by grouping, then use the five-step strategy of separation of variables.

Answer

$$y = 2 + Ce^{x^2+3x}$$

Example 8.3.2: Solving an Initial-Value Problem

Using the method of separation of variables, solve the initial-value problem

$$y' = (2x + 3)(y^2 - 4), \quad y(0) = -1.$$

Solution

Follow the five-step method of separation of variables.

1. In this example, $f(x) = 2x + 3$ and $g(y) = y^2 - 4$. Setting $g(y) = 0$ gives $y = \pm 2$ as constant solutions.

2. Divide both sides of the equation by $y^2 - 4$ and multiply by dx. This gives the equation

$$\frac{dy}{y^2 - 4} = (2x + 3)\, dx.$$

3. Next integrate both sides:

$$\int \frac{1}{y^2 - 4} dy = \int (2x + 3)\, dx. \tag{8.3.6}$$

To evaluate the left-hand side, use the method of partial fraction decomposition. This leads to the identity

$$\frac{1}{y^2 - 4} = \frac{1}{4}\left(\frac{1}{y - 2} - \frac{1}{y + 2}\right).$$

Then Equation 8.3.6 becomes

$$\frac{1}{4}\int \left(\frac{1}{y - 2} - \frac{1}{y + 2}\right) dy = \int (2x + 3)\, dx$$

$$\frac{1}{4}\left(\ln|y - 2| - \ln|y + 2|\right) = x^2 + 3x + C.$$

Multiplying both sides of this equation by 4 and replacing $4C$ with C_1 gives

$$\ln|y - 2| - \ln|y + 2| = 4x^2 + 12x + C_1$$

$$\ln\left|\frac{y - 2}{y + 2}\right| = 4x^2 + 12x + C_1.$$

4. It is possible to solve this equation for y. First exponentiate both sides of the equation and define $C_2 = e^{C_1}$:

$$\left|\frac{y - 2}{y + 2}\right| = C_2 e^{4x^2+12x}.$$

Next we can remove the absolute value and let a new constant C_3 be positive, negative, or zero, i.e., $C_3 = \pm C_2$ or $C_3 = 0$.

Then multiply both sides by $y + 2$.

$$y - 2 = C_3(y + 2)e^{4x^2+12x}$$

$$y - 2 = C_3 y e^{4x^2+12x} + 2C_3 e^{4x^2+12x}.$$

Now collect all terms involving y on one side of the equation, and solve for y:

$$y - C_3 y e^{4x^2+12x} = 2 + 2C_3 e^{4x^2+12x}$$

$$y\left(1 - C_3 e^{4x^2+12x}\right) = 2 + 2C_3 e^{4x^2+12x}$$

$$y = \frac{2 + 2C_3 e^{4x^2+12x}}{1 - C_3 e^{4x^2+12x}}.$$

5. To determine the value of C_3, substitute $x = 0$ and $y = -1$ into the general solution. Alternatively, we can put the same values into an earlier equation, namely the equation $\frac{y-2}{y+2} = C_3 e^{4x^2+12}$. This is much easier to solve for C_3:

$$\frac{y-2}{y+2} = C_3 e^{4x^2+12x}$$

$$\frac{-1-2}{-1+2} = C_3 e^{4(0)^2+12(0)}$$

$$C_3 = -3.$$

Therefore the solution to the initial-value problem is

$$y = \frac{2 - 6e^{4x^2+12x}}{1 + 3e^{4x^2+12x}}.$$

A graph of this solution appears in Figure 8.3.1.

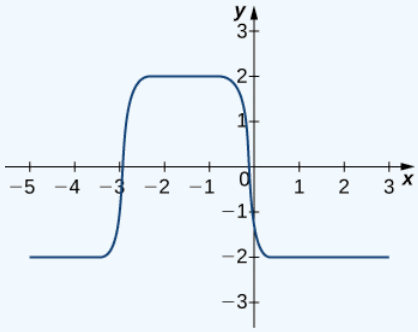


Figure 8.3.1: Graph of the solution to the initial-value problem $y' = (2x+3)(y^2-4), y(0) = -1$.

? Exercise 8.3.2

Find the solution to the initial-value problem

$$6y' = (2x+1)(y^2 - 2y - 8)$$

with $y(0) = -3$ using the method of separation of variables.

Hint

Follow the steps for separation of variables to solve the initial-value problem.

Answer

$$y = \frac{4 + 14e^{x^2+x}}{1 - 7e^{x^2+x}}$$

Applications of Separation of Variables

Many interesting problems can be described by separable equations. We illustrate two types of problems: solution concentrations and Newton's law of cooling.

Solution concentrations

Consider a tank being filled with a salt solution. We would like to determine the amount of salt present in the tank as a function of time. We can apply the process of separation of variables to solve this problem and similar problems involving *solution concentrations*.

Example 8.3.3: Determining Salt Concentration over Time

A tank containing 100 L of a brine solution initially has 4 kg of salt dissolved in the solution. At time $t = 0$, another brine solution flows into the tank at a rate of 2 L/min. This brine solution contains a concentration of 0.5 kg/L of salt. At the same time, a stopcock is opened at the bottom of the tank, allowing the combined solution to flow out at a rate of 2 L/min, so that the level of liquid in the tank remains constant (Figure 8.3.2). Find the amount of salt in the tank as a function of time (measured in minutes), and find the limiting amount of salt in the tank, assuming that the solution in the tank is well mixed at all times.

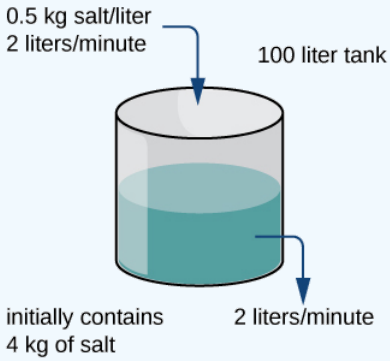


Figure 8.3.2: A brine tank with an initial amount of salt solution accepts an input flow and delivers an output flow. How does the amount of salt change with time?

Solution

First we define a function $u(t)$ that represents the amount of salt in kilograms in the tank as a function of time. Then $\frac{du}{dt}$ represents the rate at which the amount of salt in the tank changes as a function of time. Also, $u(0)$ represents the amount of salt in the tank at time $t = 0$, which is 4 kilograms.

The general setup for the differential equation we will solve is of the form

$$\frac{du}{dt} = \text{INFLOW RATE} - \text{OUTFLOW RATE}.$$

INFLOW RATE represents the rate at which salt enters the tank, and OUTFLOW RATE represents the rate at which salt leaves the tank. Because solution enters the tank at a rate of 2 L/min, and each liter of solution contains 0.5 kilogram of salt, every minute $2(0.5) = 1$ kilogram of salt enters the tank. Therefore INFLOW RATE = 1.

To calculate the rate at which salt leaves the tank, we need the concentration of salt in the tank at any point in time. Since the actual amount of salt varies over time, so does the concentration of salt. However, the volume of the solution remains fixed at 100 liters. The number of kilograms of salt in the tank at time t is equal to $u(t)$. Thus, the concentration of salt is $\frac{u(t)}{100}$ kg/L, and the solution leaves the tank at a rate of 2 L/min. Therefore salt leaves the tank at a rate of $\frac{u(t)}{100} \cdot 2 = \frac{u(t)}{50}$ kg/min, and OUTFLOW RATE is equal to $\frac{u(t)}{50}$. Therefore the differential equation becomes $\frac{du}{dt} = 1 - \frac{u}{50}$, and the initial condition is $u(0) = 4$. The initial-value problem to be solved is

$$\frac{du}{dt} = 1 - \frac{u}{50}, u(0) = 4.$$

The differential equation is a separable equation, so we can apply the five-step strategy for solution.

Step 1. Setting $1 - \frac{u}{50} = 0$ gives $u = 50$ as a constant solution. Since the initial amount of salt in the tank is 4 kilograms, this solution does not apply.

Step 2. Rewrite the equation as

$$\frac{du}{dt} = \frac{50-u}{50}.$$

Then multiply both sides by dt and divide both sides by $50-u$:

$$\frac{du}{50-u} = \frac{dt}{50}.$$

Step 3. Integrate both sides:

$$\int \frac{du}{50-u} = \int \frac{dt}{50}$$
$$-\ln|50-u| = \frac{t}{50} + C.$$

Step 4. Solve for $u(t)$:

$$\ln|50-u| = -\frac{t}{50} - C$$

$$e^{\ln|50-u|} = e^{-(t/50)-C}$$

$$|50-u| = C_1 e^{-t/50}, \text{ where } C_1 = e^{-C}.$$

Eliminate the absolute value by allowing the constant to be positive, negative, or zero, i.e., $C_1 = \pm e^{-C}$ or $C_1 = 0$:

$$50-u = C_1 e^{-t/50}.$$

Finally, solve for $u(t)$:

$$u(t) = 50 - C_1 e^{-t/50}.$$

Step 5. Solve for C_1:

$$u(0) = 50 - C_1 e^{-0/50}$$
$$4 = 50 - C_1$$
$$C_1 = 46.$$

The solution to the initial value problem is $u(t) = 50 - 46e^{-t/50}$. To find the limiting amount of salt in the tank, take the limit as t approaches infinity:

$$\lim_{t\to\infty} u(t) = 50 - 46e^{-t/50}$$
$$= 50 - 46(0) = 50.$$

Note that this was the constant solution to the differential equation. If the initial amount of salt in the tank is 50 kilograms, then it remains constant. If it starts at less than 50 kilograms, then it approaches 50 kilograms over time.

? Exercise 8.3.3

A tank contains 3 kilograms of salt dissolved in 75 liters of water. A salt solution of 0.4 kg salt/L is pumped into the tank at a rate of 6 L/min and is drained at the same rate. Solve for the salt concentration at time t. Assume the tank is well mixed at all times.

Hint

Follow the steps in Example $8.3.3$ and determine an expression for INFLOW and OUTFLOW. Formulate an initial-value problem, and then solve it.

Initial value problem:

$$\frac{du}{dt} = 2.4 - \frac{2u}{25},\ u(0) = 3$$

Answer

$$u(t) = 30 - 27e^{-t/50}$$

Newton's Law of Cooling

Newton's law of cooling states that the rate of change of an object's temperature is proportional to the difference between its own temperature and the ambient temperature (i.e., the temperature of its surroundings). If we let $T(t)$ represent the temperature of an object as a function of time, then $\frac{dT}{dt}$ represents the rate at which that temperature changes. The temperature of the object's surroundings can be represented by T_s. Then Newton's law of cooling can be written in the form

$$\frac{dT}{dt} = k(T(t) - T_s)$$

or simply

$$\frac{dT}{dt} = k(T - T_s).$$

The temperature of the object at the beginning of any experiment is the initial value for the initial-value problem. We call this temperature T_0. Therefore the initial-value problem that needs to be solved takes the form

$$\frac{dT}{dt} = k(T - T_s) \tag{8.3.7}$$

with $T(0) = T_0$, where k is a constant that needs to be either given or determined in the context of the problem. We use these equations in Example 8.3.4.

Example 8.3.4: Waiting for a Pizza to Cool

A pizza is removed from the oven after baking thoroughly, and the temperature of the oven is $350°F$. The temperature of the kitchen is $75°F$, and after 5 minutes the temperature of the pizza is $340°F$. We would like to wait until the temperature of the pizza reaches $300°F$ before cutting and serving it (Figure 8.3.3). How much longer will we have to wait?

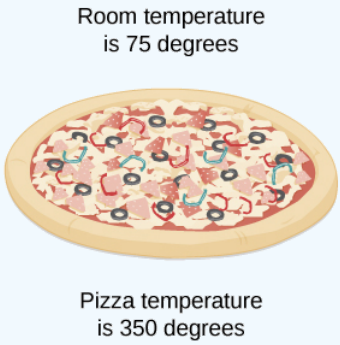


Figure 8.3.3: From Newton's law of cooling, if the pizza cools $10°F$ in 5 minutes, how long before it cools to $300°F$?

Solution

The ambient temperature (surrounding temperature) is $75°F$, so $T_s = 75$. The temperature of the pizza when it comes out of the oven is $350°F$, which is the initial temperature (i.e., initial value), so $T_0 = 350$. Therefore Equation 8.3.7 becomes

$$\frac{dT}{dt} = k(T - 75)$$

with $T(0) = 350$.

To solve the differential equation, we use the five-step technique for solving separable equations.

1. Setting the right-hand side equal to zero gives $T = 75$ as a constant solution. Since the pizza starts at $350°F$, this is not the solution we are seeking.

2. Rewrite the differential equation by multiplying both sides by dt and dividing both sides by $T - 75$:

$$\frac{dT}{T-75} = k\,dt.$$

3. Integrate both sides:

$$\int \frac{dT}{T-75} = \int k\,dt$$
$$\ln|T-75| = kt + C.$$

4. Solve for T by first exponentiating both sides:

$$\begin{aligned} e^{\ln|T-75|} &= e^{kt+C} \\ |T-75| &= C_1 e^{kt}, && \text{where } C_1 = e^C. \\ T-75 &= \pm C_1 e^{kt} \\ T-75 &= Ce^{kt}, && \text{where } C = \pm C_1 \text{ or } C = 0. \\ T(t) &= 75 + Ce^{kt}. \end{aligned}$$

5. Solve for C by using the initial condition $T(0) = 350$:

$$\begin{aligned} T(t) &= 75 + Ce^{kt} \\ T(0) &= 75 + Ce^{k(0)} \\ 350 &= 75 + C \\ C &= 275. \end{aligned}$$

Therefore the solution to the initial-value problem is

$$T(t) = 75 + 275e^{kt}.$$

To determine the value of k, we need to use the fact that after 5 minutes the temperature of the pizza is $340°F$. Therefore $T(5) = 340$. Substituting this information into the solution to the initial-value problem, we have

$$T(t) = 75 + 275e^{kt}$$

$$T(5) = 340 = 75 + 275e^{5k}$$

$$265 = 275e^{5k}$$

$$e^{5k} = \frac{53}{55}$$

$$\ln e^{5k} = \ln(\frac{53}{55})$$

$$5k = \ln(\frac{53}{55})$$

$$k = \frac{1}{5}\ln(\frac{53}{55}) \approx -0.007408.$$

So now we have $T(t) = 75 + 275e^{-0.007048t}$. When is the temperature $300°F$? Solving for t, we find

$$T(t) = 75 + 275e^{-0.007048t}$$

$$300 = 75 + 275e^{-0.007048t}$$

$$225 = 275e^{-0.007048t}$$

$$e^{-0.007048t} = \frac{9}{11}$$

$$\ln e^{-0.007048t} = \ln\frac{9}{11}$$

$$-0.007048t = \ln\frac{9}{11}$$

$$t = -\frac{1}{0.007048}\ln\frac{9}{11} \approx 28.5.$$

Therefore we need to wait an additional 23.5 minutes (after the temperature of the pizza reached $340°F$). That should be just enough time to finish this calculation.

? Exercise 8.3.4

A cake is removed from the oven after baking thoroughly, and the temperature of the oven is $450°F$. The temperature of the kitchen is $70°F$, and after 10 minutes the temperature of the cake is $430°F$.

a. Write the appropriate initial-value problem to describe this situation.
b. Solve the initial-value problem for $T(t)$.
c. How long will it take until the temperature of the cake is within $5°F$ of room temperature?

Hint

Determine the values of T_s and T_0 then use Equation 8.3.7.

Answer a

Initial-value problem

$$\frac{dT}{dt} = k(T - 70), \quad T(0) = 450$$

Answer b

$$T(t) = 70 + 380e^{kt}$$

Answer c

Approximately 114 minutes.

Key Concepts

- A separable differential equation is any equation that can be written in the form $y' = f(x)g(y)$.
- The method of separation of variables is used to find the general solution to a separable differential equation.

Key Equations

- **Separable differential equation**

  $y' = f(x)g(y)$

- **Solution concentration**

  $\frac{du}{dt} = \text{INFLOW RATE} - \text{OUTFLOW RATE}$

- **Newton's law of cooling**

  $\frac{dT}{dt} = k(T - T_s)$

Glossary

autonomous differential equation
an equation in which the right-hand side is a function of y alone

separable differential equation
any equation that can be written in the form $y' = f(x)g(y)$

separation of variables
a method used to solve a separable differential equation

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8.3E: Exercises for Section 8.3

In exercises 1 - 4, solve the following initial-value problems with the initial condition $y_0 = 0$ and graph the solution.

1) $\frac{dy}{dt} = y + 1$

Answer

$y = e^t - 1$

2) $\frac{dy}{dt} = y - 1$

3) $\frac{dy}{dt} = -y + 1$

Answer

$y = 1 - e^{-t}$

4) $\frac{dy}{dt} = -y - 1$

In exercises 5 - 14, find the general solution to the differential equation.

5) $x^2y' = (x + 1)y$

Answer

$y = Cxe^{-1/x}$

6) $y' = \tan(y)x$

7) $y' = 2xy^2$

Answer

$y = \frac{1}{C - x^2}$

8) $\frac{dy}{dt} = y\cos(3t + 2)$

9) $2x\frac{dy}{dx} = y^2$

Answer

$y = -\frac{2}{C + \ln|x|}$

10) $y' = e^y x^2$

11) $(1 + x)y' = (x + 2)(y - 1)$

Answer

$y = Ce^x(x + 1) + 1$

12) $\frac{dx}{dt} = 3t^2(x^2 + 4)$

13) $t\frac{dy}{dt} = \sqrt{1 - y^2}$

Answer

$y = \sin(\ln|t| + C)$

14) $y' = e^x e^y$

In exercises 15 - 24, find the solution to the initial-value problem.

15) $y' = e^{y-x}, \quad y(0) = 0$

Answer

$y = -\ln(e^{-x})$ which simplifies to $y = x$

16) $y' = y^2(x+1), \quad y(0) = 2$

17) $\dfrac{dy}{dx} = y^3 x e^{x^2}, \quad y(0) = 1$

Answer

$$y = \frac{1}{\sqrt{2 - e^{x^2}}}$$

18) $\dfrac{dy}{dt} = y^2 e^x \sin(3x), \quad y(0) = 1$

19) $y' = \dfrac{x}{\operatorname{sech}^2 y}, \quad y(0) = 0$

Answer

$$y = \tanh^{-1}\left(\frac{x^2}{2}\right)$$

20) $y' = 2xy(1+2y), \quad y(0) = -1$

21) $\dfrac{dx}{dt} = \ln(t)\sqrt{1-x^2}, \quad x(1) = 0$

Answer

$x = \sin(1 - t + t\ln t)$

22) $y' = 3x^2(y^2+4), \quad y(0) = 0$

23) $y' = e^y 5^x, \quad y(0) = \ln(\ln(5))$

Answer

$y = \ln(\ln(5)) - \ln(2 - 5^x)$

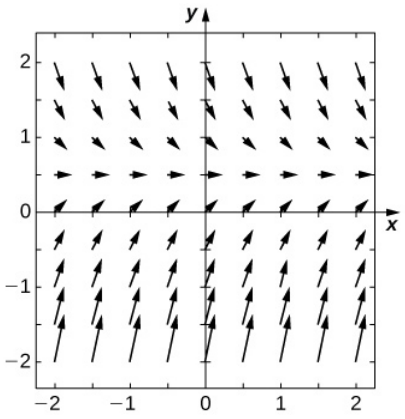
24) $y' = -2x\tan(y), \quad y(0) = \dfrac{\pi}{2}$

For problems 25 - 29, use a software program or your calculator to generate the directional fields. Solve explicitly and draw solution curves for several initial conditions. Are there some critical initial conditions that change the behavior of the solution?

25) [T] $y' = 1 - 2y$

Answer

$$y = Ce^{-2}x + \frac{1}{2}$$

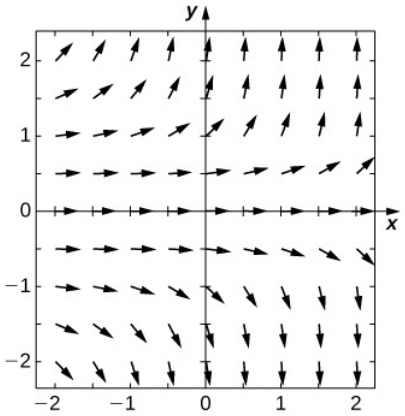


26) [T] $y' = y^2x^3$

27) [T] $y' = y^3e^x$

Answer

$$y = \frac{1}{\sqrt{2}\sqrt{C - e^x}}$$

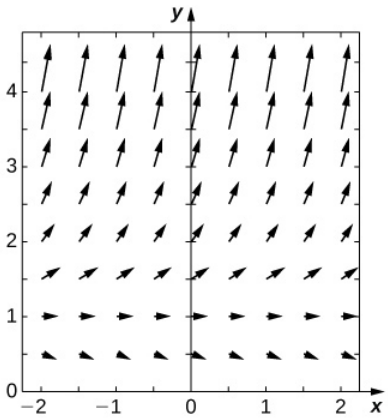


28) [T] $y' = e^y$

29) [T] $y' = y\ln(x)$

Answer

$$y = Ce^{-x}x^x$$



30) Most **drugs in the bloodstream** decay according to the equation $y' = cy$, where y is the concentration of the drug in the bloodstream. If the half-life of a drug is 2 hours, what fraction of the initial dose remains after 6 hours?

31) A drug is administered intravenously to a patient at a rate r mg/h and is cleared from the body at a rate proportional to the amount of drug still present in the body, d. Set up and solve the differential equation, assuming there is no drug initially present in the body.

Answer

$y = \frac{r}{d}(1 - e^{-dt})$

32) [T] How often should a drug be taken if its dose is 3 mg, it is cleared at a rate $c = 0.1$ mg/h, and 1 mg is required to be in the bloodstream at all times?

33) A tank contains 1 kilogram of salt dissolved in 100 liters of water. A salt solution of 0.1 kg salt/L is pumped into the tank at a rate of 2 L/min and is drained at the same rate. Solve for the salt concentration at time t. Assume the tank is well mixed.

Answer

Since the amount of salt at time t is given by the equation $A(t) = 10 - 9e^{-t/50}$, and the tank volume stays constant at 100 liters, the concentration at time t would be $c(t) = \frac{1}{100}\left(10 - 9e^{-t/50}\right)$.

34) A tank containing 10 kilograms of salt dissolved in 1000 liters of water has two salt solutions pumped in. The first solution of 0.2 kg salt/L is pumped in at a rate of 20 L/min and the second solution of 0.05 kg salt/L is pumped in at a rate of 5 L/min. The tank drains at 25 L/min. Assume the tank is well mixed. Solve for the salt concentration at time t.

35) [T] For the preceding problem, find how much salt is in the tank 1 hour after the process begins.

Answer

134.3 kilograms

36) Torricelli's law states that for a water tank with a hole in the bottom that has a cross-section of A and with a height of water h above the bottom of the tank, the rate of change of volume of water flowing from the tank is proportional to the square root of the height of water, according to $\frac{dV}{dt} = -A\sqrt{2gh}$, where g is the acceleration due to gravity. Note that $\frac{dV}{dt} = A\frac{dh}{dt}$. Solve the resulting initial-value problem for the height of the water, assuming a tank with a hole of radius 2 ft. The initial height of water is 100 ft.

37) For the preceding problem, determine how long it takes the tank to drain.

Answer

720 seconds

For problems 38 - 44, use Newton's law of cooling.

38) The liquid base of an ice cream has an initial temperature of $200°F$ before it is placed in a freezer with a constant temperature of $0°F$. After 1 hour, the temperature of the ice-cream base has decreased to $140°F$. Formulate and solve the initial-value problem to determine the temperature of the ice cream.

39) [T] The liquid base of an ice cream has an initial temperature of $210°F$ before it is placed in a freezer with a constant temperature of $20°F$. After 2 hours, the temperature of the ice-cream base has decreased to $170°F$. At what time will the ice cream be ready to eat? (Assume $30°F$ is the optimal eating temperature.)

Answer

24 hours 55 minutes

40) [T] You are organizing an ice cream social. The outside temperature is $80°F$ and the ice cream is at $10°F$. After 10 minutes, the ice cream temperature has risen by $10°F$. How much longer can you wait before the ice cream melts at $40°F$?

41) You have a cup of coffee at temperature $70°C$ and the ambient temperature in the room is $20°C$. Assuming a cooling rate k of 0.125, write and solve the differential equation to describe the temperature of the coffee with respect to time.

Answer

$T(t) = 20 + 50e^{-0.125t}$

42) [T] You have a cup of coffee at temperature $70°C$ that you put outside, where the ambient temperature is $0°C$. After 5 minutes, how much colder is the coffee?

43) You have a cup of coffee at temperature $70°C$ and you immediately pour in 1 part milk to 5 parts coffee. The milk is initially at temperature $1°C$. Write and solve the differential equation that governs the temperature of this coffee.

Answer

$T(t) = 20 + 38.5e^{-0.125t}$

44) You have a cup of coffee at temperature $70°C$, which you let cool 10 minutes before you pour in the same amount of milk at $1°C$ as in the preceding problem. How does the temperature compare to the previous cup after 10 minutes?

45) Solve the generic problem $y' = ay + b$ with initial condition $y(0) = c$.

Answer

$y = (c + ba)e^{ax} - \frac{b}{a}$

46) Prove the basic continual compounded interest equation. Assuming an initial deposit of P_0 and an interest rate of r, set up and solve an equation for continually compounded interest.

47) Assume an initial nutrient amount of I kilograms in a tank with L liters. Assume a concentration of c kg/L being pumped in at a rate of r L/min. The tank is well mixed and is drained at a rate of r L/min. Find the equation describing the amount of nutrient in the tank.

Answer

$y(t) = cL + (I - cL)e^{-rt/L}$

48) Leaves accumulate on the forest floor at a rate of 2 g/cm^2/yr and also decompose at a rate of 90% per year. Write a differential equation governing the number of grams of leaf litter per square centimeter of forest floor, assuming at time 0 there is no leaf litter on the ground. Does this amount approach a steady value? What is that value?

49) Leaves accumulate on the forest floor at a rate of 4 g/cm^2/yr. These leaves decompose at a rate of 10% per year. Write a differential equation governing the number of grams of leaf litter per square centimeter of forest floor. Does this amount approach a steady value? What is that value?

Answer

Differential Equation: $\frac{dy}{dt} = 4 - 0.1y$
Solution, the model for this situation: $y = 40(1 - e^{-0.1t})$,
Amount approaches a steady value of 40 g/cm^2

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8.4: The Logistic Equation

Learning Objectives

- Describe the concept of environmental carrying capacity in the logistic model of population growth.
- Draw a direction field for a logistic equation and interpret the solution curves.
- Solve a logistic equation and interpret the results.

Differential equations can be used to represent the size of a population as it varies over time. We saw this in an earlier chapter in the section on exponential growth and decay, which is the simplest model. A more realistic model includes other factors that affect the growth of the population. In this section, we study the logistic differential equation and see how it applies to the study of population dynamics in the context of biology.

Population Growth and Carrying Capacity

To model population growth using a differential equation, we first need to introduce some variables and relevant terms. The variable t. will represent time. The units of time can be hours, days, weeks, months, or even years. Any given problem must specify the units used in that particular problem. The variable P will represent population. Since the population varies over time, it is understood to be a function of time. Therefore we use the notation $P(t)$ for the population as a function of time. If $P(t)$ is a differentiable function, then the first derivative $\frac{dP}{dt}$ represents the instantaneous rate of change of the population as a function of time.

In Exponential Growth and Decay, we studied the exponential growth and decay of populations and radioactive substances. An example of an exponential growth function is $P(t) = P_0 e^{rt}$. In this function, $P(t)$ represents the population at time t, P_0 represents the **initial population** (population at time $t = 0$), and the constant $r > 0$ is called the **growth rate**. Figure 8.4.1 shows a graph of $P(t) = 100e^{0.03t}$. Here $P_0 = 100$ and $r = 0.03$.

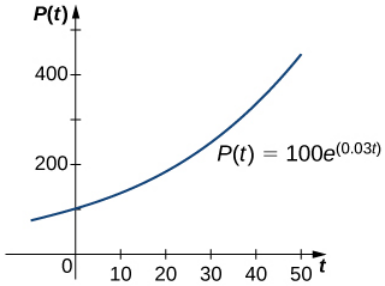


Figure 8.4.1: An exponential growth model of population.

We can verify that the function $P(t) = P_0 e^{rt}$ satisfies the initial-value problem

$$\frac{dP}{dt} = rP$$

with $P(0) = P_0$.

This differential equation has an interesting interpretation. The left-hand side represents the rate at which the population increases (or decreases). The right-hand side is equal to a positive constant multiplied by the current population. Therefore the differential equation states that the rate at which the population increases is proportional to the population at that point in time. Furthermore, it states that the constant of proportionality never changes.

One problem with this function is its prediction that as time goes on, the population grows without bound. This is unrealistic in a real-world setting. Various factors limit the rate of growth of a particular population, including birth rate, death rate, food supply, predators, and so on. The growth constant r usually takes into consideration the birth and death rates but none of the other factors, and it can be interpreted as a net (birth minus death) percent growth rate per unit time. A natural question to ask is whether the population growth rate stays constant, or whether it changes over time. Biologists have found that in many biological systems, the population grows until a certain steady-state population is reached. This possibility is not taken into account with exponential

growth. However, the concept of carrying capacity allows for the possibility that in a given area, only a certain number of a given organism or animal can thrive without running into resource issues.

Definition: Carrying Capacity

The *carrying capacity* of an organism in a given environment is defined to be the maximum population of that organism that the environment can sustain indefinitely.

We use the variable K to denote the carrying capacity. The growth rate is represented by the variable r. Using these variables, we can define the logistic differential equation.

Definition: Logistic Differential Equation

Let K represent the carrying capacity for a particular organism in a given environment, and let r be a real number that represents the growth rate. The function $P(t)$ represents the population of this organism as a function of time t, and the constant P_0 represents the initial population (population of the organism at time $t = 0$). Then the *logistic differential equation* is

$$\frac{dP}{dt} = rP\left(1 - \frac{P}{K}\right). \tag{8.4.1}$$

The logistic equation was first published by Pierre Verhulst in 1845. This differential equation can be coupled with the initial condition $P(0) = P_0$ to form an initial-value problem for $P(t)$.

Suppose that the initial population is small relative to the carrying capacity. Then $\frac{P}{K}$ is small, possibly close to zero. Thus, the quantity in parentheses on the right-hand side of Equation 8.4.1 is close to 1, and the right-hand side of this equation is close to rP. If $r > 0$, then the population grows rapidly, resembling exponential growth.

However, as the population grows, the ratio $\frac{P}{K}$ also grows, because K is constant. If the population remains below the carrying capacity, then $\frac{P}{K}$ is less than 1, so $1 - \frac{P}{K} > 0$. Therefore the right-hand side of Equation 8.4.1 is still positive, but the quantity in parentheses gets smaller, and the growth rate decreases as a result. If $P = K$ then the right-hand side is equal to zero, and the population does not change.

Now suppose that the population starts at a value higher than the carrying capacity. Then $\frac{P}{K} > 1$, and $1 - \frac{P}{K} < 0$. Then the right-hand side of Equation 8.4.1 is negative, and the population decreases. As long as $P > K$, the population decreases. It never actually reaches K because $\frac{dP}{dt}$ will get smaller and smaller, but the population approaches the carrying capacity as t approaches infinity. This analysis can be represented visually by way of a phase line. A **phase line** describes the general behavior of a solution to an autonomous differential equation, depending on the initial condition. For the case of a carrying capacity in the logistic equation, the phase line is as shown in Figure $8.4.2$.



Figure $8.4.2$: A phase line for the differential equation $\frac{dP}{dt} = rP(1 - \frac{P}{K})$.

This phase line shows that when P is less than zero or greater than K, the population decreases over time. When P is between 0 and K, the population increases over time.

Example 8.4.1: Examining the Carrying Capacity of a Deer Population

Let's consider the population of white-tailed deer (*Odocoileus virginianus*) in the state of Kentucky. The Kentucky Department of Fish and Wildlife Resources (KDFWR) sets guidelines for hunting and fishing in the state. Before the hunting season of 2004, it estimated a population of 900,000 deer. Johnson notes: "A deer population that has plenty to eat and is not hunted by humans or other predators will double every three years." (George Johnson, "The Problem of Exploding Deer Populations Has No Attractive Solutions," January 12,2001, accessed April 9, 2015)



Figure 8.4.3: (credit: modification of work by Rachel Kramer, Flickr)

This observation corresponds to a rate of increase $r = \dfrac{\ln(2)}{3} = 0.2311$, so the approximate growth rate is 23.11% per year. (This assumes that the population grows exponentially, which is reasonable—at least in the short term—with plentiful food supply and no predators.) The KDFWR also reports deer population densities for 32 counties in Kentucky, the average of which is approximately 27 deer per square mile. Suppose this is the deer density for the whole state (39,732 square miles). The carrying capacity K is 39,732 square miles times 27 deer per square mile, or 1,072,764 deer.

a. For this application, we have $P_0 = 900,000$, $K = 1,072,764$, and $r = 0.2311$. Substitute these values into Equation 8.4.1 and form the initial-value problem.
b. Solve the initial-value problem from part a.
c. According to this model, what will be the population in 3 years? Recall that the doubling time predicted by Johnson for the deer population was 3 years. How do these values compare?

Suppose the population managed to reach **1,200,000** What does the logistic equation predict will happen to the population in this scenario?

Solution

a. The initial value problem is

$$\frac{dP}{dt} = 0.2311P\left(1 - \frac{P}{1,072,764}\right), \quad P(0) = 900,000.$$

b. The logistic equation is an autonomous differential equation, so we can use the method of separation of variables.

Step 1: Setting the right-hand side equal to zero gives $P = 0$ and $P = 1,072,764$. This means that if the population starts at zero it will never change, and if it starts at the carrying capacity, it will never change.

Step 2: Rewrite the differential equation and multiply both sides by:

$$\frac{dP}{dt} = 0.2311P\left(\frac{1,072,764 - P}{1,072,764}\right)$$

$$dP = 0.2311P\left(\frac{1,072,764 - P}{1,072,764}\right)dt$$

$$\frac{dP}{P(1,072,764 - P)} = \frac{0.2311}{1,072,764}dt.$$

Step 3: Integrate both sides of the equation using partial fraction decomposition:

$$\int \frac{dP}{P(1,072,764-P)} = \int \frac{0.2311}{1,072,764}dt$$
$$\frac{1}{1,072,764}\int \left(\frac{1}{P}+\frac{1}{1,072,764-P}\right)dP = \frac{0.2311t}{1,072,764}+C$$
$$\frac{1}{1,072,764}\left(\ln|P|-\ln|1,072,764-P|\right) = \frac{0.2311t}{1,072,764}+C.$$

Step 4: Multiply both sides by 1,072,764 and use the quotient rule for logarithms:

$$\ln\left|\frac{P}{1,072,764-P}\right| = 0.2311t + C_1.$$

Here $C_1 = 1,072,764C$. Next exponentiate both sides and eliminate the absolute value:

$$e^{\ln\left|\frac{P}{1,072,764-P}\right|} = e^{0.2311t+C_1}$$
$$\left|\frac{P}{1,072,764-P}\right| = C_2e^{0.2311t}$$
$$\frac{P}{1,072,764-P} = C_2e^{0.2311t}.$$

Here $C_2 = e^{C_1}$ but after eliminating the absolute value, it can be negative as well. Now solve for:

$$P = C_2e^{0.2311t}(1,072,764-P)$$
$$P = 1,072,764C_2e^{0.2311t} - C_2Pe^{0.2311t}$$
$$P + C_2Pe^{0.2311t} = 1,072,764C_2e^{0.2311t}$$
$$P(1+C_2e^{0.2311t} = 1,072,764C_2e^{0.2311t}$$
$$P(t) = \frac{1,072,764C_2e^{0.2311t}}{1+C_2e^{0.2311t}}.$$

Step 5: To determine the value of C_2, it is actually easier to go back a couple of steps to where C_2 was defined. In particular, use the equation

$$\frac{P}{1,072,764-P} = C_2e^{0.2311t}.$$

The initial condition is $P(0) = 900,000$. Replace P with $900,000$ and t with zero:

$$\frac{P}{1,072,764-P} = C_2e^{0.2311t}$$
$$\frac{900,000}{1,072,764-900,000} = C_2e^{0.2311(0)}$$
$$\frac{900,000}{172,764} = C_2$$
$$C_2 = \frac{25,000}{4,799}$$
$$\approx 5.209.$$

Therefore

$$P(t) = \frac{1,072,764\left(\frac{25000}{4799}\right)e^{0.2311t}}{1+\left(\frac{25000}{4799}\right)e^{0.2311t}}$$
$$= \frac{1,072,764(25000)e^{0.2311t}}{4799+25000e^{0.2311t}.}$$

Dividing the numerator and denominator by **25,000** gives

$$P(t) = \frac{1,072,764e^{0.2311t}}{0.19196 + e^{0.2311t}}.$$

Figure 8.4.4 is a graph of this equation.

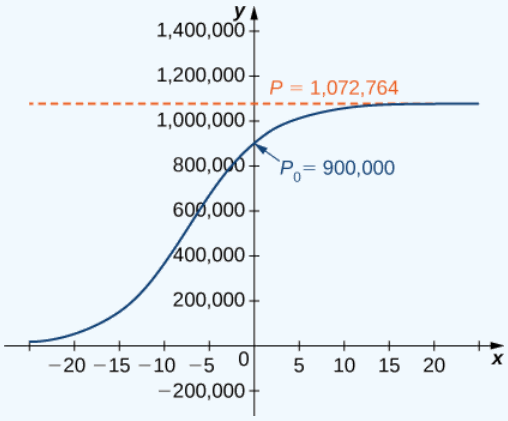


Figure 8.4.4: Logistic curve for the deer population with an initial population of **900,000** deer.

c. Using this model we can predict the population in **3** years.

$$P(3) = \frac{1,072,764e^{0.2311(3)}}{0.19196 + e^{0.2311(3)}} \approx 978,830\, deer$$

This is far short of twice the initial population of $900,000$. Remember that the doubling time is based on the assumption that the growth rate never changes, but the logistic model takes this possibility into account.

d. If the population reached **1,200,000** deer, then the new initial-value problem would be

$$\frac{dP}{dt} = 0.2311P\left(1 - \frac{P}{1,072,764}\right),\ P(0) = 1,200,000.$$

The general solution to the differential equation would remain the same.

$$P(t) = \frac{1,072,764C_2e^{0.2311t}}{1 + C_2e^{0.2311t}}$$

To determine the value of the constant, return to the equation

$$\frac{P}{1,072,764 - P} = C_2e^{0.2311t}.$$

Substituting the values $t = 0$ and $P = 1,200,000$, you get

$$C_2e^{0.2311(0)} = \frac{1,200,000}{1,072,764 - 1,200,000}$$

$$C_2 = -\frac{100,000}{10,603} \approx -9.431.$$

Therefore

$$P(t) = \frac{1,072,764C_2e^{0.2311t}}{1 + C_2e^{0.2311t}}$$

$$= \frac{1,072,764\left(-\frac{100,000}{10,603}\right)e^{0.2311t}}{1 + \left(-\frac{100,000}{10,603}\right)e^{0.2311t}}$$

$$= -\frac{107,276,400,000e^{0.2311t}}{100,000e^{0.2311t} - 10,603}$$

$$\approx \frac{10,117,551e^{0.2311t}}{9.43129e^{0.2311t} - 1}$$

This equation is graphed in Figure 8.4.5.

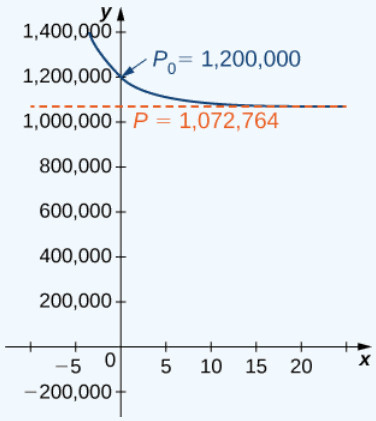


Figure 8.4.5: Logistic curve for the deer population with an initial population of **1,200,000** deer.

Solving the Logistic Differential Equation

The logistic differential equation is an autonomous differential equation, so we can use separation of variables to find the general solution, as we just did in Example 8.4.1.

Step 1: Setting the right-hand side equal to zero leads to $P = 0$ and $P = K$ as constant solutions. The first solution indicates that when there are no organisms present, the population will never grow. The second solution indicates that when the population starts at the carrying capacity, it will never change.

Step 2: Rewrite the differential equation in the form

$$\frac{dP}{dt} = \frac{rP(K-P)}{K}.$$

Then multiply both sides by dt and divide both sides by $P(K-P)$. This leads to

$$\frac{dP}{P(K-P)} = \frac{r}{K}dt.$$

Multiply both sides of the equation by K and integrate:

$$\int \frac{K}{P(K-P)}dP = \int r dt. \tag{8.4.2}$$

The left-hand side of this equation can be integrated using partial fraction decomposition. We leave it to you to verify that

$$\frac{K}{P(K-P)} = \frac{1}{P} + \frac{1}{K-P}.$$

Then the Equation 8.4.2 becomes

$$\int \frac{1}{P} + \frac{1}{K-P}dP = \int r dt$$

$$\ln|P| - \ln|K-P| = rt + C$$

$$\ln\left|\frac{P}{K-P}\right| = rt + C.$$

Now exponentiate both sides of the equation to eliminate the natural logarithm:

$$e^{\ln\left|\frac{P}{K-P}\right|} = e^{rt+C}$$

$$\left|\frac{P}{K-P}\right| = e^C e^{rt}.$$

We define $C_1 = e^c$ so that the equation becomes

$$\frac{P}{K-P} = C_1 e^{rt}. \tag{8.4.3}$$

To solve this equation for $P(t)$, first multiply both sides by $K-P$ and collect the terms containing P on the left-hand side of the equation:

$$\begin{aligned} P &= C_1 e^{rt}(K-P) \\ &= C_1 K e^{rt} - C_1 P e^{rt} \\ P + C_1 P e^{rt} &= C_1 K e^{rt}. \end{aligned}$$

Next, factor P from the left-hand side and divide both sides by the other factor:

$$\begin{aligned} P(1 + C_1 e^{rt}) &= C_1 K e^{rt} \\ P(t) &= \frac{C_1 K e^{rt}}{1 + C_1 e^{rt}}. \end{aligned}$$

The last step is to determine the value of C_1. The easiest way to do this is to substitute $t = 0$ and P_0 in place of P in Equation and solve for C_1:

$$\begin{aligned} \frac{P}{K-P} &= C_1 e^{rt} \\ \frac{P_0}{K-P_0} &= C_1 e^{r(0)} \\ C_1 &= \frac{P_0}{K-P_0}. \end{aligned}$$

Finally, substitute the expression for C_1 into Equation 8.4.3:

$$P(t) = \frac{C_1 K e^{rt}}{1 + C_1 e^{rt}} = \frac{\frac{P_0}{K-P_0} K e^{rt}}{1 + \frac{P_0}{K-P_0} e^{rt}}$$

Now multiply the numerator and denominator of the right-hand side by $(K-P_0)$ and simplify:

$$\begin{aligned} P(t) &= \frac{\frac{P_0}{K-P_0} K e^{rt}}{1 + \frac{P_0}{K-P_0} e^{rt}} \\ &= \frac{\frac{P_0}{K-P_0} K e^{rt}}{1 + \frac{P_0}{K-P_0} e^{rt}} \cdot \frac{K-P_0}{K-P_0} \\ &= \frac{P_0 K e^{rt}}{(K-P_0) + P_0 e^{rt}}. \end{aligned}$$

We state this result as a theorem.

Solution of the Logistic Differential Equation

Consider the logistic differential equation subject to an initial population of P_0 with carrying capacity K and growth rate r. The solution to the corresponding initial-value problem is given by

$$P(t) = \frac{P_0 K e^{rt}}{(K-P_0) + P_0 e^{rt}}$$

.

Now that we have the solution to the initial-value problem, we can choose values for P_0, r, and K and study the solution curve. For example, in Example we used the values $r = 0.2311, K = 1,072,764$, and an initial population of $900,000$ deer. This leads to the

solution

$$P(t) = \frac{P_0 K e^{rt}}{(K - P_0) + P_0 e^{rt}}$$
$$= \frac{900,000(1,072,764)e^{0.2311t}}{(1,072,764 - 900,000) + 900,000e^{0.2311t}}$$
$$= \frac{900,000(1,072,764)e^{0.2311t}}{172,764 + 900,000e^{0.2311t}}.$$

Dividing top and bottom by $900,000$ gives

$$P(t) = \frac{1,072,764e^{0.2311t}}{0.19196 + e^{0.2311t}}.$$

This is the same as the original solution. The graph of this solution is shown again in blue in Figure $8.4.6$, superimposed over the graph of the exponential growth model with initial population $900,000$ and growth rate 0.2311 (appearing in green). The red dashed line represents the carrying capacity, and is a horizontal asymptote for the solution to the logistic equation.

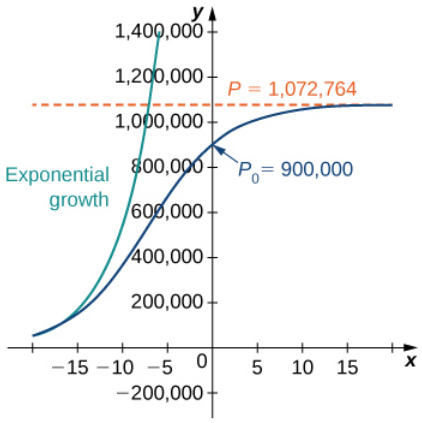


Figure $8.4.6$: A comparison of exponential versus logistic growth for the same initial population of $900,000$ organisms and growth rate of 23.11

Working under the assumption that the population grows according to the logistic differential equation, this graph predicts that approximately 20 years earlier (1984), the growth of the population was very close to exponential. The net growth rate at that time would have been around 23.1 per year. As time goes on, the two graphs separate. This happens because the population increases, and the logistic differential equation states that the growth rate decreases as the population increases. At the time the population was measured (2004), it was close to carrying capacity, and the population was starting to level off.

The solution to the logistic differential equation has a point of inflection. To find this point, set the second derivative equal to zero:

$$P(t) = \frac{P_0 K e^{rt}}{(K - P_0) + P_0 e^{rt}}$$
$$P'(t) = \frac{rP_0 K(K - P_0)e^{rt}}{((K - P_0) + P_0 e^{rt})^2}$$
$$P''(t) = \frac{r^2 P_0 K(K - P_0)^2 e^{rt} - r^2 P_0^2 K(K - P_0)e^{2rt}}{((K - P_0) + P_0 e^{rt})^3}$$
$$= \frac{r^2 P_0 K(K - P_0)e^{rt}((K - P_0) - P_0 e^{rt})}{((K - P_0) + P_0 e^{rt})^3}.$$

Setting the numerator equal to zero,

$$r^2 P_0 K(K - P_0)e^{rt}((K - P_0) - P_0 e^{rt}) = 0.$$

As long as $P_0 \neq K$, the entire quantity before and including $e^{rt}$is nonzero, so we can divide it out:

$$(K - P_0) - P_0 e^{rt} = 0.$$

Solving for t,

$$P_0 e^{rt} = K - P_0$$

$$e^{rt} = \frac{K - P_0}{P_0}$$

$$\ln e^{rt} = \ln \frac{K - P_0}{P_0}$$

$$rt = \ln \frac{K - P_0}{P_0}$$

$$t = \frac{1}{r} \ln \frac{K - P_0}{P_0}.$$

Notice that if $P_0 > K$, then this quantity is undefined, and the graph does not have a point of inflection. In the logistic graph, the point of inflection can be seen as the point where the graph changes from concave up to concave down. This is where the "leveling off" starts to occur, because the net growth rate becomes slower as the population starts to approach the carrying capacity.

Exercise 8.4.1

A population of rabbits in a meadow is observed to be 200 rabbits at time $t = 0$. After a month, the rabbit population is observed to have increased by 4. Using an initial population of 200 and a growth rate of 0.04, with a carrying capacity of 750 rabbits,

a. Write the logistic differential equation and initial condition for this model.
b. Draw a slope field for this logistic differential equation, and sketch the solution corresponding to an initial population of 200 rabbits.
c. Solve the initial-value problem for $P(t)$.
d. Use the solution to predict the population after 1 year.

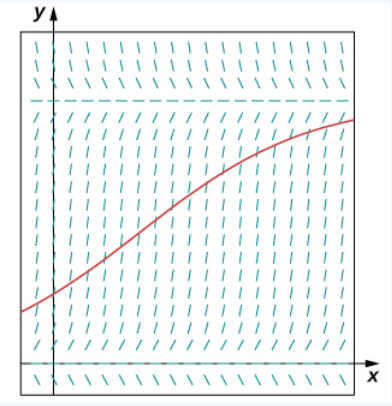
Hint

First determine the values of $r, K,$ and P_0. Then create the initial-value problem, draw the direction field, and solve the problem.

Answer

a. $\frac{dP}{dt} = 0.04(1 - \frac{P}{750}), P(0) = 200$

b.



c. $P(t) = \frac{3000e^{.04t}}{11 + 4e^{.04t}}$

d. After 12 months, the population will be $P(12) \approx 278$ rabbits.

Student Project: Logistic Equation with a Threshold Population

An improvement to the logistic model includes a **threshold population.** The threshold population is defined to be the minimum population that is necessary for the species to survive. We use the variable T to represent the threshold population. A differential equation that incorporates both the threshold population T and carrying capacity K is

$$\frac{dP}{dt} = -rP\left(1 - \frac{P}{K}\right)\left(1 - \frac{P}{T}\right)$$

where r represents the growth rate, as before.

1. The threshold population is useful to biologists and can be utilized to determine whether a given species should be placed on the endangered list. A group of Australian researchers say they have determined the threshold population for any species to survive: 5000 adults. (Catherine Clabby, "A Magic Number," American Scientist 98(1): 24, doi:10.1511/2010.82.24. accessed April 9, 2015, www.americanscientist.org/iss...a-magic-number). Therefore we use $T = 5000$ as the threshold population in this project. Suppose that the environmental carrying capacity in Montana for elk is $25,000$. Set up Equation using the carrying capacity of $25,000$ and threshold population of 5000. Assume an annual net growth rate of 18%.
2. Draw the direction field for the differential equation from step 1, along with several solutions for different initial populations. What are the constant solutions of the differential equation? What do these solutions correspond to in the original population model (i.e., in a biological context)?
3. What is the limiting population for each initial population you chose in step 2? (Hint: use the slope field to see what happens for various initial populations, i.e., look for the horizontal asymptotes of your solutions.)
4. This equation can be solved using the method of separation of variables. However, it is very difficult to get the solution as an explicit function of t. Using an initial population of $18,000$ elk, solve the initial-value problem and express the solution as an implicit function of t, or solve the general initial-value problem, finding a solution in terms of $r, K, T,$ and P_0.

Key Concepts

- When studying population functions, different assumptions—such as exponential growth, logistic growth, or threshold population—lead to different rates of growth.
- The logistic differential equation incorporates the concept of a carrying capacity. This value is a limiting value on the population for any given environment.
- The logistic differential equation can be solved for any positive growth rate, initial population, and carrying capacity.

Key Equations

- **Logistic differential equation and initial-value problem**

  $$\frac{dP}{dt} = rP\left(1 - \frac{P}{K}\right), \quad P(0) = P_0$$

- **Solution to the logistic differential equation/initial-value problem**

  $$P(t) = \frac{P_0 K e^{rt}}{(K - P_0) + P_0 e^{rt}}$$

- **Threshold population model**

  $$\frac{dP}{dt} = -rP\left(1 - \frac{P}{K}\right)\left(1 - \frac{P}{T}\right)$$

Glossary

carrying capacity

the maximum population of an organism that the environment can sustain indefinitely

growth rate

the constant $r > 0$ in the exponential growth function $P(t) = P_0 e^{rt}$

initial population

the population at time $t = 0$

logistic differential equation

a differential equation that incorporates the carrying capacity K and growth rate rr into a population model

phase line

a visual representation of the behavior of solutions to an autonomous differential equation subject to various initial conditions

threshold population

the minimum population that is necessary for a species to survive

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8.4E: Exercises for Section 8.4

Basic Logistic Model

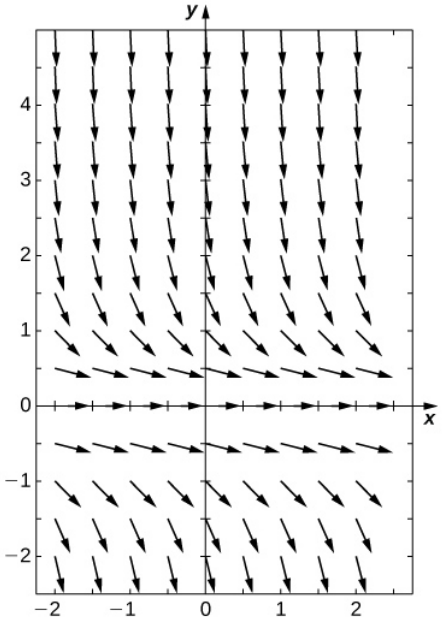
For problems 1 - 11, consider the logistic equation in the form $P' = CP - P^2$. Draw the directional field and find the stability of the equilibria.

1) $C = 3$

2) $C = 0$

Answer

$P = 0$ semi-stable



3) $C = -3$

4) Solve the logistic equation for $C = 10$ and an initial condition of $P(0) = 2$.

Answer

$$P = \frac{10e^{10x}}{e^{10x} + 4}$$

5) Solve the logistic equation for $C = -10$ and an initial condition of $P(0) = 2$.

6) A population of deer inside a park has a carrying capacity of 200 and a growth rate of 2. If the initial population is 50 deer, what is the population of deer at any given time?

Answer

$$P(t) = \frac{10000e^{0.02t}}{150 + 50e^{0.02t}}$$

7) A population of frogs in a pond has a growth rate of 5 If the initial population is 1000 frogs and the carrying capacity is 6000, what is the population of frogs at any given time?

8) [T] Bacteria grow at a rate of 20 per hour in a petri dish. If there is initially one bacterium and a carrying capacity of 1 million cells, how long does it take to reach $500,000$ cells?

Answer

69 hours 5 minutes

9) [T] Rabbits in a park have an initial population of 10 and grow at a rate of 4 per year. If the carrying capacity is 500, at what time does the population reach 100 rabbits?

10) [T] Two monkeys are placed on an island. After 5 years, there are 8 monkeys, and the estimated carrying capacity is 25 monkeys. When does the population of monkeys reach 16 monkeys?

Answer

8 years 11 months

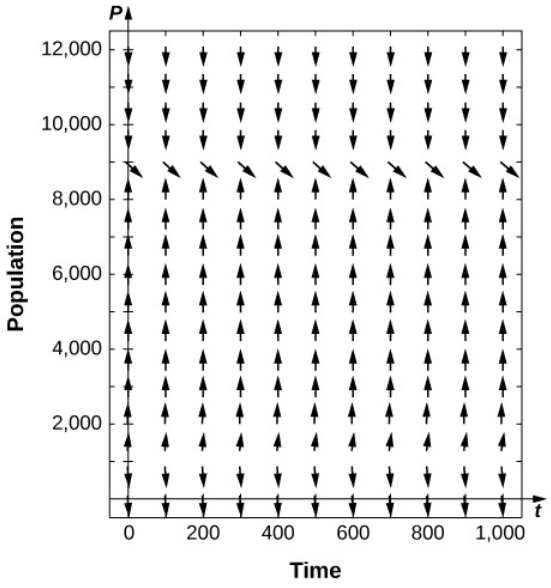
11) [T] A butterfly sanctuary is built that can hold 2000 butterflies, and 400 butterflies are initially moved in. If after 2 months there are now 800 butterflies, when does the population get to 1500 butterflies?

Logistic Population Model with Depletion

The following problems consider the logistic equation with an added term for depletion, either through death or emigration.

12) [T] The population of trout in a pond is given by $P' = 0.4P\left(1 - \dfrac{P}{10000}\right) - 400$, where 400 trout are caught per year. Use your calculator or computer software to draw a directional field and draw a few sample solutions. What do you expect for the behavior?

Answer

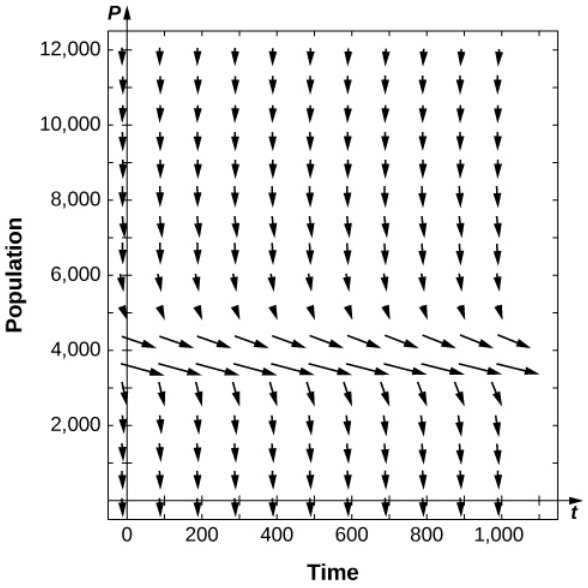


13) In the preceding problem, what are the stabilities of the equilibria $0 < P_1 < P_2$?

14) [T] For the preceding problem, use software to generate a directional field for the value $f = 400$. What are the stabilities of the equilibria?

Answer

P_1 semi-stable

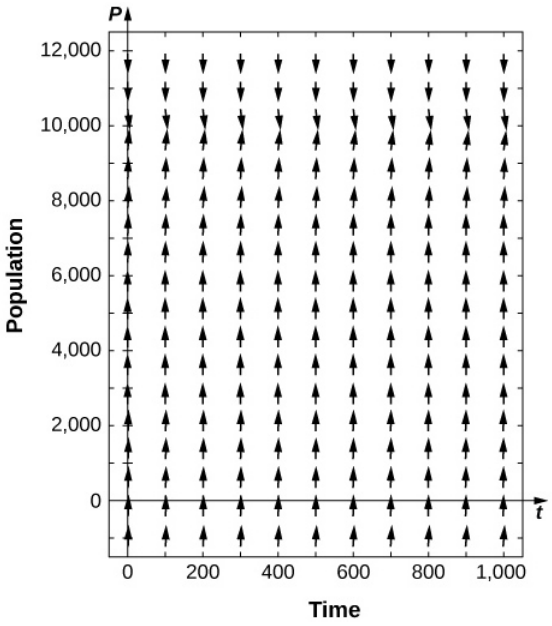


15) [T] For the preceding problems, use software to generate a directional field for the value $f = 600$. What are the stabilities of the equilibria?

16) [T] For the preceding problems, consider the case where a certain number of fish are added to the pond, or $f = -200$. What are the nonnegative equilibria and their stabilities?

Answer

$P_2 > 0$ stable



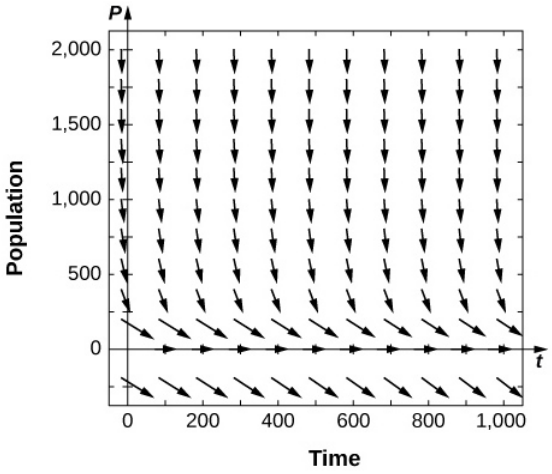
It is more likely that the amount of fishing is governed by the current number of fish present, so instead of a constant number of fish being caught, the rate is proportional to the current number of fish present, with proportionality constant k, as $P' = 0.4P\left(1 - \frac{P}{10000}\right) - kP$.

17) [T] For the previous fishing problem, draw a directional field assuming $k = 0.1$. Draw some solutions that exhibit this behavior. What are the equilibria and what are their stabilities?

18) [T] Use software or a calculator to draw directional fields for $k = 0.4$. What are the nonnegative equilibria and their stabilities?

Answer

$P_1 = 0$ is semi-stable



19) [T] Use software or a calculator to draw directional fields for $k = 0.6$. What are the equilibria and their stabilities?

20) Solve this equation, assuming a value of $k = 0.05$ and an initial condition of 2000 fish.

Answer

$$y = \frac{-20}{4 \times 10^{-6} - 0.002e^{0.01t}}$$

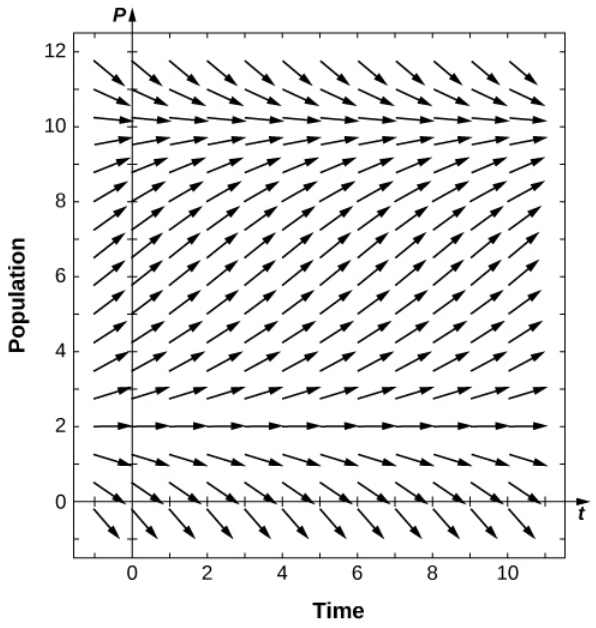
21) Solve this equation, assuming a value of $k = 0.05$ and an initial condition of 5000 fish.

Minimal Sustainable Population Thresholds

The following problems add in a minimal threshold value for the species to survive, T, which changes the differential equation to $P'(t) = rP\left(1 - \frac{P}{K}\right)\left(1 - \frac{T}{P}\right)$.

22) Draw the directional field of the threshold logistic equation, assuming $K = 10, r = 0.1, T = 2$. When does the population survive? When does it go extinct?

Answer



23) For the preceding problem, solve the logistic threshold equation, assuming the initial condition $P(0) = P_0$.

24) Bengal tigers in a conservation park have a carrying capacity of 100 and need a minimum of 10 to survive. If they grow in population at a rate of 1 per year, with an initial population of 15 tigers, solve for the number of tigers present.

Answer

$$P(t) = \frac{850 + 500e^{0.009t}}{85 + 5e^{0.009t}}$$

25) A forest containing ring-tailed lemurs in Madagascar has the potential to support 5000 individuals, and the lemur population grows at a rate of 5 per year. A minimum of 500 individuals is needed for the lemurs to survive. Given an initial population of 600 lemurs, solve for the population of lemurs.

26) The population of mountain lions in Northern Arizona has an estimated carrying capacity of 250 and grows at a rate of 0.25 per year and there must be 25 for the population to survive. With an initial population of 30 mountain lions, how many years will it take to get the mountain lions off the endangered species list (at least 100)?

Answer

13 years months

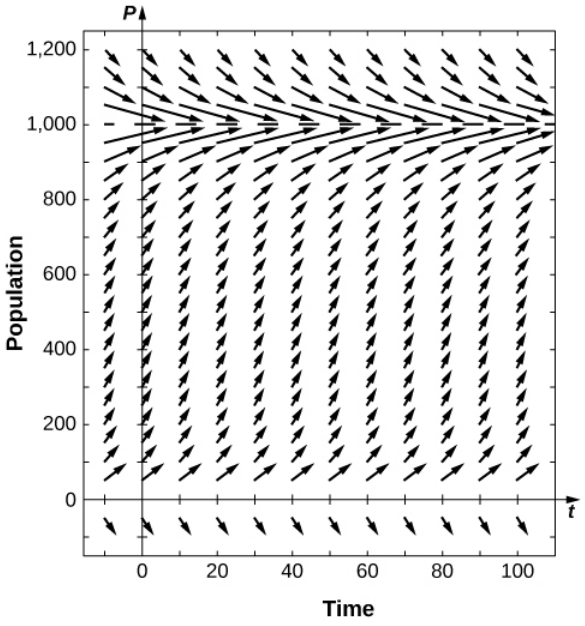
The Gompertz Equation

The following questions consider the Gompertz equation, a modification for logistic growth, which is often used for modeling cancer growth, specifically the number of tumor cells.

27) The Gompertz equation is given by $P(t)' = \alpha \ln\left(\frac{K}{P(t)}\right) P(t)$. Draw the directional fields for this equation assuming all parameters are positive, and given that $K = 1$.

28) Assume that for a population, $K = 1000$ and $\alpha = 0.05$. Draw the directional field associated with this differential equation and draw a few solutions. What is the behavior of the population?

Answer



29) Solve the Gompertz equation for generic α and K and $P(0) = P_0$.

30) [T] The Gompertz equation has been used to model tumor growth in the human body. Starting from one tumor cell on day 1 and assuming $\alpha = 0.1$ and a carrying capacity of 10 million cells, how long does it take to reach "detection" stage at 5 million cells?

Answer

31.465 days

31) [T] It is estimated that the world human population reached 3 billion people in 1959 and 6 billion in 1999. Assuming a carrying capacity of 16 billion humans, write and solve the differential equation for logistic growth, and determine what year the population reached 7 billion.

32) [T] It is estimated that the world human population reached 3 billion people in 1959 and 6 billion in 1999. Assuming a carrying capacity of 16 billion humans, write and solve the differential equation for Gompertz growth, and determine what year the population reached 7 billion. Was logistic growth or Gompertz growth more accurate, considering world population reached 7 billion on October $31, 2011$?

Answer

September 2008

33) Show that the population grows fastest when it reaches half the carrying capacity for the logistic equation $P' = rP\left(1 - \frac{P}{K}\right)$.

34) When does population increase the fastest in the threshold logistic equation $P'(t) = rP\left(1 - \frac{P}{K}\right)\left(1 - \frac{T}{P}\right)$?

Answer

$\frac{K+T}{2}$

35) When does population increase the fastest for the Gompertz equation $P(t)' = \alpha \ln\left(\frac{K}{P(t)}\right) P(t)$?

Below is a table of the populations of whooping cranes in the wild from 1940 to 2000. The population rebounded from near extinction after conservation efforts began. The following problems consider applying population models to fit the data. Assume a carrying capacity of $10,000$ cranes. Fit the data assuming years since 1940 (so your initial population at time 0 would be 22 cranes).

Year (years since conservation began)	Whooping Crane Population
1940(0)	22
1950(10)	31
1960(20)	36
1970(30)	57
1980(40)	91
1990(50)	159
2000(60)	256

Source: www.savingcranes.org/images/...wc_numbers.pdf

36) Find the equation and parameter r that best fit the data for the logistic equation.

Answer

$r = 0.0405$

37) Find the equation and parameters r and T that best fit the data for the threshold logistic equation.

38) Find the equation and parameter α that best fit the data for the Gompertz equation.

Answer

$\alpha = 0.0081$

39) Graph all three solutions and the data on the same graph. Which model appears to be most accurate?

40) Using the three equations found in the previous problems, estimate the population in 2010 (year 70 after conservation). The real population measured at that time was 437. Which model is most accurate?

Answer

Logistic: 361, Threshold: 436, Gompertz: 309.

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